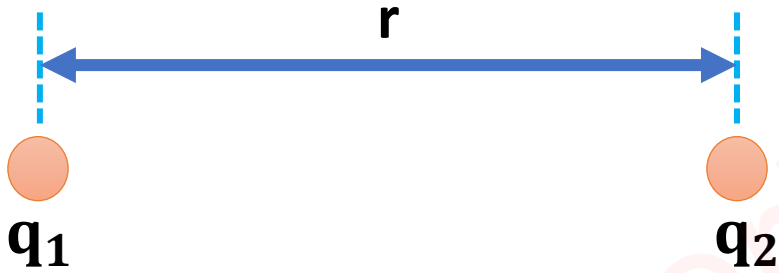


Coulomb's Law

Definition

- Force acting between two charges is directly proportional to charges and inversely proportional to the square of the distance between them.



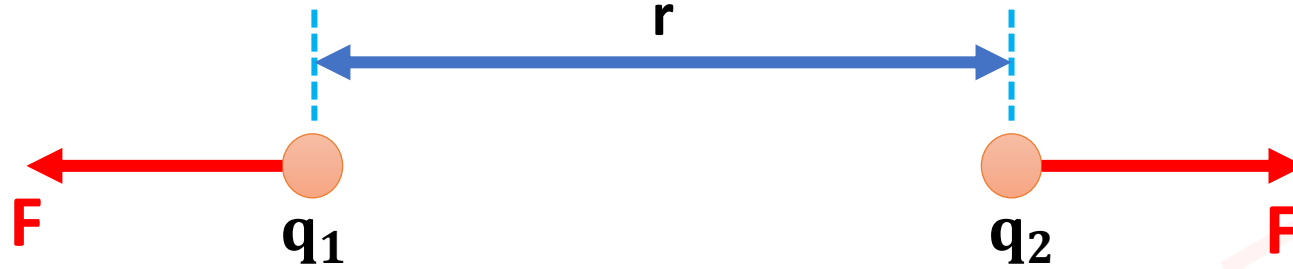
$$\Rightarrow F \propto \frac{q_1 q_2}{r^2}$$

$$\Rightarrow F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} = k \frac{q_1 q_2}{r^2}$$

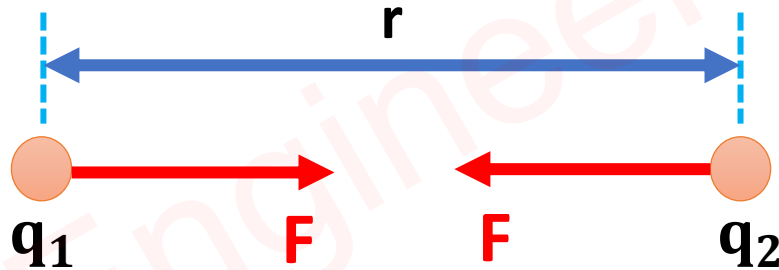
❖ Where, $k = 9 \times 10^9 \text{ SI}$

Force Direction using Coulomb's Law

- Case – I: Same polarity of charges ($q_1 q_2 > 0$)



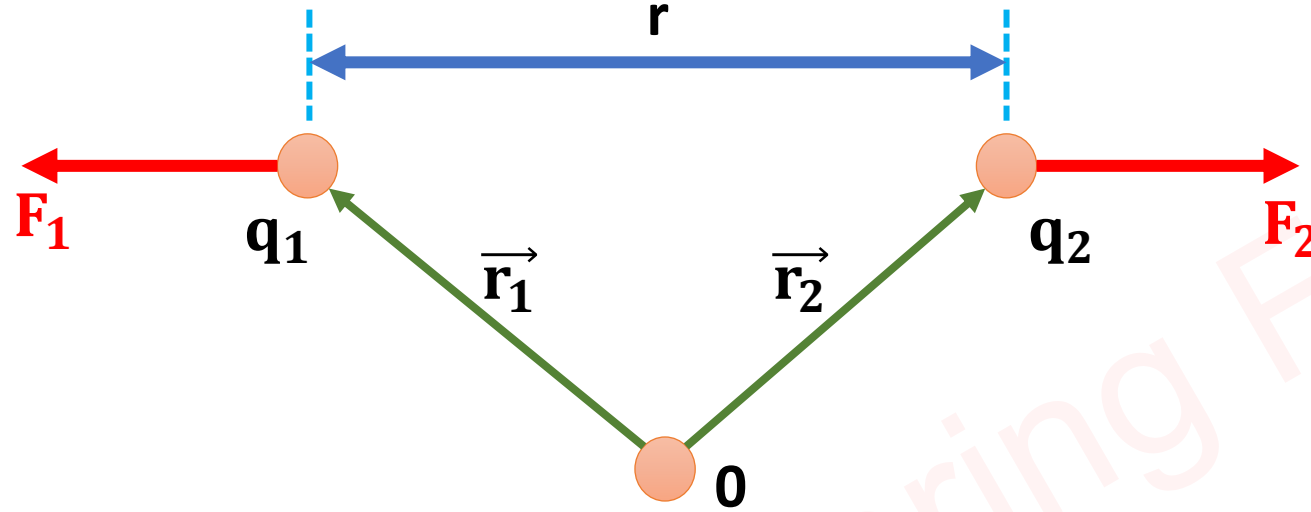
- Case – II: Opposite polarity of charges ($q_1 q_2 < 0$)



- Hence, with the same polarity of charges, there is a repulsive force between two charges, and with the opposite polarity of charges, there is an attractive force between two charges.

Force Calculation using Coulomb's Law

- Case – I: Same polarity of charges ($q_1 q_2 > 0$)



$$\Rightarrow \vec{F}_1 = k \frac{q_1 q_2}{r^2} \hat{r}_1$$

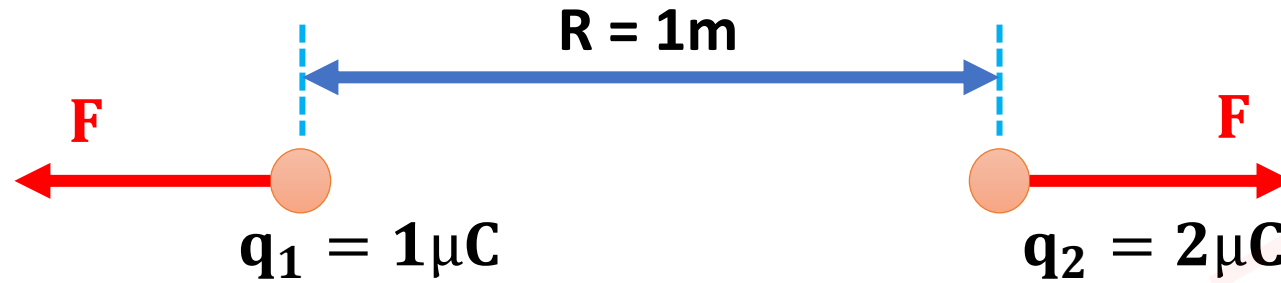
$$\Rightarrow \vec{F}_2 = k \frac{q_1 q_2}{r^2} \hat{r}_2$$

○ Where, $\hat{r}_1 = \frac{\vec{r}_1 - \vec{r}_2}{r}$

○ Where, $\hat{r}_2 = \frac{\vec{r}_2 - \vec{r}_1}{r}$

Example Based on Coulomb's Law

- Find the force acting on charges for given figure.

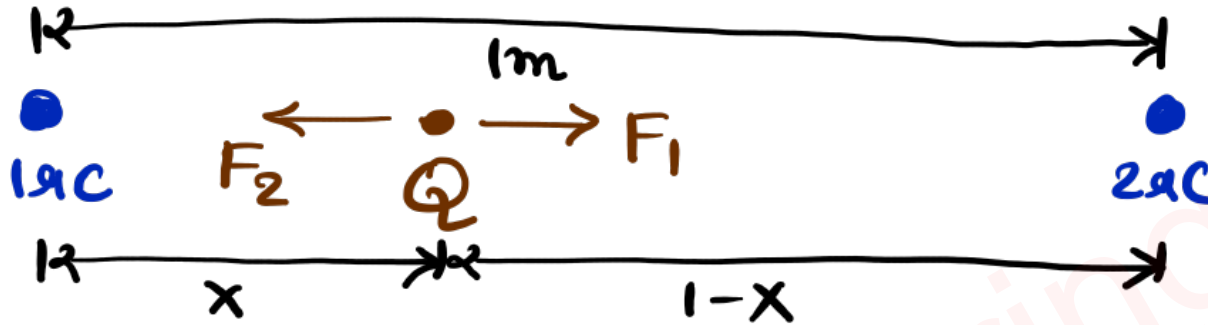


$$\Rightarrow F = k \frac{q_1 q_2}{r^2}$$

$$\Rightarrow F = 9 \times 10^9 \frac{(10^{-6} \times 2 \times 10^{-6})}{1^2} = 18 \times 10^{-3} \text{N}$$

Examples of Coulomb's Law

Question 1 – If two charges $1\mu\text{C}$ and $2\mu\text{C}$ are separated by 1m , then at what distance from $1\mu\text{C}$ Charge will the force on charge Q be zero?



$$\Rightarrow F_1 = F_2$$

$$\Rightarrow \frac{k(1)(Q)}{x^2} = \frac{k(2)(Q)}{(1-x)^2}$$

$$\Rightarrow (1-x)^2 = 2x^2$$

$$\Rightarrow (1-x) = \sqrt{2}x \quad \text{or} \quad \Rightarrow (1-x) = -\sqrt{2}x$$

$$\Rightarrow x = 0.41\text{m} \checkmark \quad \alpha$$

Question 2 – Point charges 1mC and -2mC are located at (3, 2, -1)m and (-1, -1, 4)m, respectively. Calculate the electric force on the 10nC charge located at (0, 3, 1)m.

(3, 2, -1)m
1mC
•

(-1, -1, 4)m
-2mC
•

(0, 3, 1)m
10nC
•
 $\vec{F}_1$
 $\vec{F}_2$

$$\rightarrow \vec{F}_1 = k \frac{q_1 q_2}{r_1^2} \hat{r}_1 = k \frac{q_1 q_2}{r_1^2} \left(\frac{\vec{r}_1}{r_1} \right)$$

$$\sim \vec{r}_1 = (-3, 1, 2)$$

$$r_1 = \sqrt{3^2 + 1^2 + 2^2} = \sqrt{14}$$

$$\rightarrow \vec{F}_1 = \frac{9 \times 10^9 \times 10^{-3} \times 10 \times 10^{-9}}{(\sqrt{14})^3} (-3, 1, 2)$$

$$= 1.71 \times 10^{-3} (-3, 1, 2) = (-5.13, 1.71, 3.42) \text{ mN}$$

$$\rightarrow \vec{F}_2 = k \frac{q_1 q_2}{r_2^2} \hat{r}_2 = k \frac{q_1 q_2}{r_2^2} \frac{\vec{r}_2}{r_2}$$

$$\rightarrow \vec{r}_2 = (-1, -4, 3)$$

$$r_2 = \sqrt{1^2 + 4^2 + 3^2} = \sqrt{26}$$

$$\rightarrow \vec{F}_2 = \frac{9 \times 10^9 \times 10 \times 10^{-9} \times 2 \times 10^{-3}}{(\sqrt{26})^3} (-1, -4, 3)$$

$$= 1.35 \times 10^{-3} (-1, -4, 3)$$

$$= (-1.35, -5.4, 4.05) \text{ mN}$$

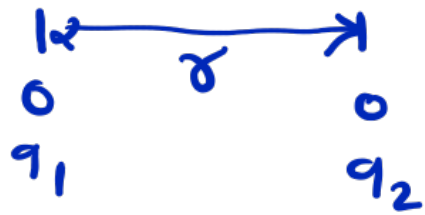
$$\sim \vec{F} = \vec{F}_1 + \vec{F}_2$$

$$\boxed{\vec{F} = (-6.48, -3.69, 7.47) \text{ mN}}$$

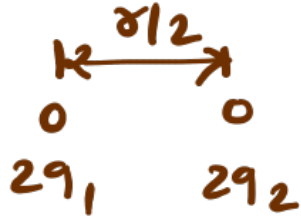
Examples of Coulomb's Law

Question 3 – The force acting between two point charges kept at a certain distance is ϕ . Now magnitudes of charges are doubled and distance between them is halved, the force acting between them is

- a. ϕ
- b. 4ϕ
- c. 8ϕ
- d. ✓ 16ϕ



$$\rightarrow \phi = k \frac{q_1 q_2}{r^2}$$



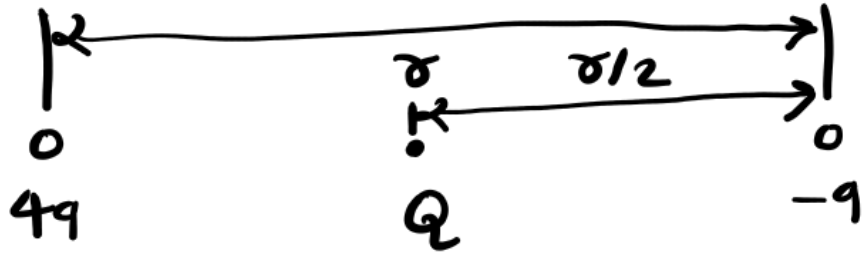
$$\rightarrow \phi' = k \frac{(2q_1)(2q_2)}{(r/2)^2}$$

$$= 16 \left(k \frac{q_1 q_2}{r^2} \right)$$

$$= 16\phi$$

Question 4 – The distance between two point charges $4q$ and $-q$ is r . A third charge Q is placed at their midpoint. The resultant force acting on $-q$ is zero then $Q = \dots\dots\dots$

- a. ☒ $-q$
- b. ☐ q
- c. ☐ $-4q$
- d. ☐ $4q$



$$\therefore F = 0 = F_Q + F_{4q}$$

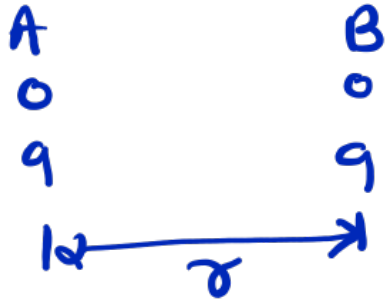
$$\Rightarrow 0 = \cancel{k} \frac{(Q)(-1)}{(\cancel{r}/2)^2} + \cancel{k} \frac{(4q)(-1)}{\cancel{r}^2}$$

$$\Rightarrow 0 = 4Q + 4q$$

$$\Rightarrow Q = -q$$

Question 5 – Two identical metal spheres A and B carry the same charge q . when the two spheres are at a distance r from each other, the force acting between them is F . Another identical uncharged sphere C is first brought in contact with sphere A, After that it is touched by sphere B and then separated from it. Now the force acting between A and B at the same distance is

- a. F
- b. $2F$
- ☒ c. $3F/8$
- d. $F/4$



$$\rightarrow F = k \frac{q^2}{r^2}$$

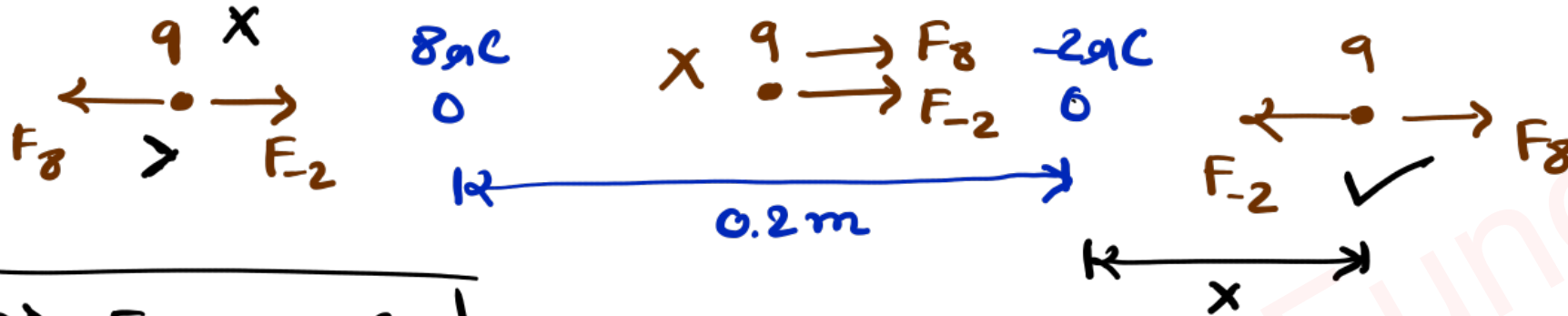


$$\rightarrow F' = k \frac{(q/2)(3q/4)}{r^2}$$

$$= \frac{3}{8} \left(k \frac{q^2}{r^2} \right)$$

$$= \frac{3}{8} F$$

Question 6 – Two Electric charges having magnitude $8\mu\text{C}$ and $-2\mu\text{C}$ are separated by 20cm . Where should a third charge be placed so that the resultant force acting on it is zero?



$$\Rightarrow F = k \frac{q_1 q_2}{r^2}$$

$$\Rightarrow 0 = F_{-2} + F_8$$

$$\Rightarrow 0 = \frac{k (q) (-2\mu)}{x^2} + \frac{k (q) (8\mu)}{(x+0.2)^2}$$

$$\Rightarrow \frac{2}{x^2} = \frac{8}{(x+0.2)^2}$$

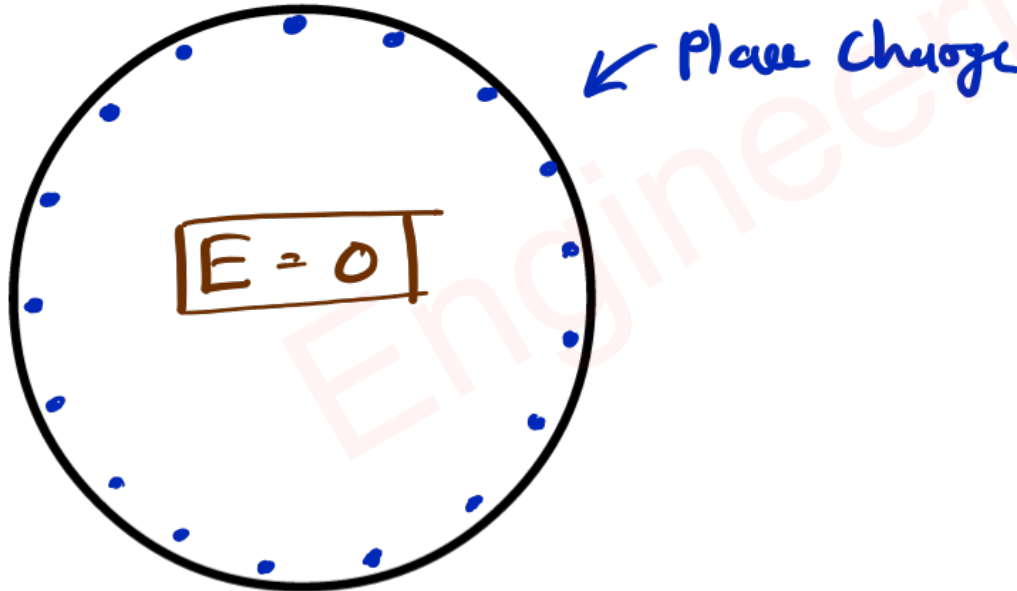
$$\Rightarrow (x+0.2)^2 = 4x^2$$

$$\Rightarrow 2x = x + 0.2 \Rightarrow \boxed{x = 0.2\text{m}}$$

Examples of Electric Charge

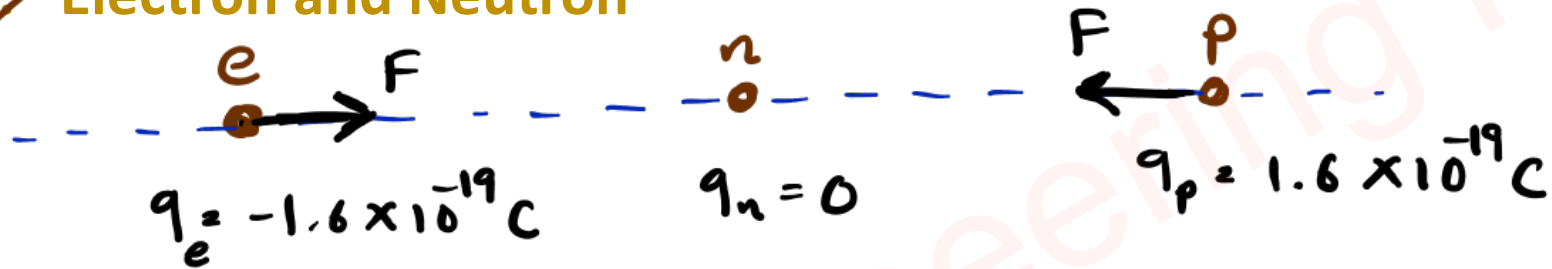
Question 1 – When a charge is given to a conductor.

- a ✓ It distributes uniformly all over the surface
- b. It distributes uniformly all over the volume
- c. It distributes on the surface, inversely proportional to the radius of curvature
- d. It stays where it was placed



Question 2 – Consider an electron, a neutron, and a proton initially at rest and placed along a straight line such that the neutron is exactly at the center of the line joining the electron and proton. At $t=0$, the particles are released but are constrained to move along the same straight line. Which of these will collide first?

- a. The particle will never collide
- b. All will collide together
- c. Proton and Neutron
- ✓ d. Electron and Neutron



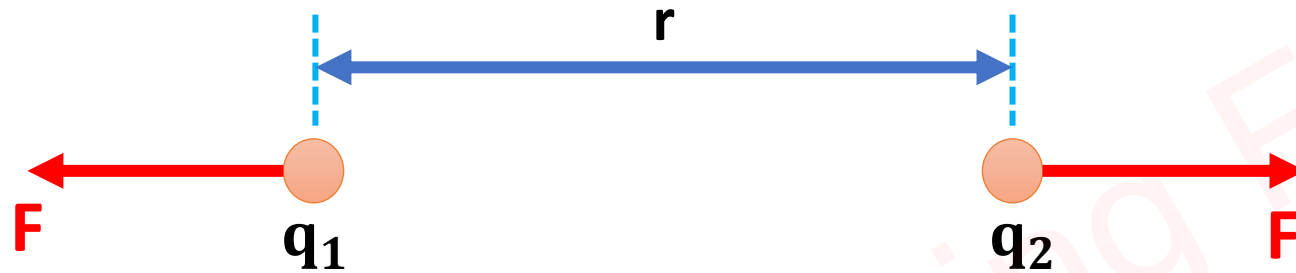
$$\Rightarrow F = ma \Rightarrow a = F/m$$

$$\begin{aligned} \Rightarrow m_e &= 9.1 \times 10^{-31} \text{ kg} \\ m_p &= 1.67 \times 10^{-27} \text{ kg} \end{aligned}$$

Electric Field due to Point Charge

Basics of Electric Field

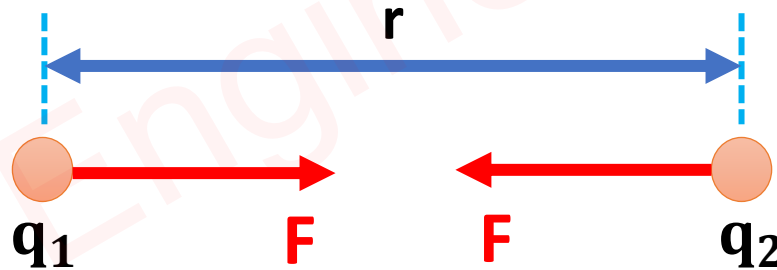
- Case – I: Same polarity of charges ($q_1 q_2 > 0$)



$$\Rightarrow F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} = k \frac{q_1 q_2}{r^2}$$

❖ Where, $k = 9 \times 10^9 \text{ SI}$

- Case – II: Opposite polarity of charges ($q_1 q_2 < 0$)



- This force on charge happens due to Electric Field.

Electric Field due to Point Charge

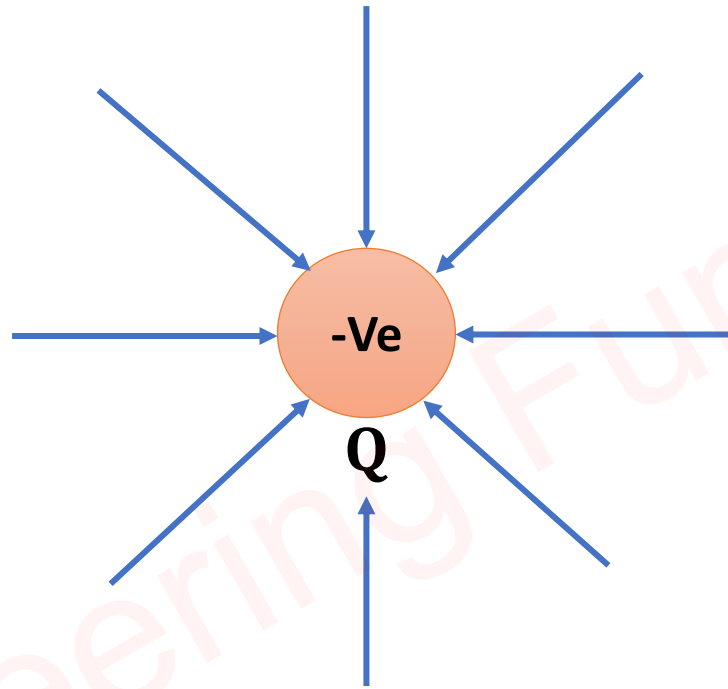
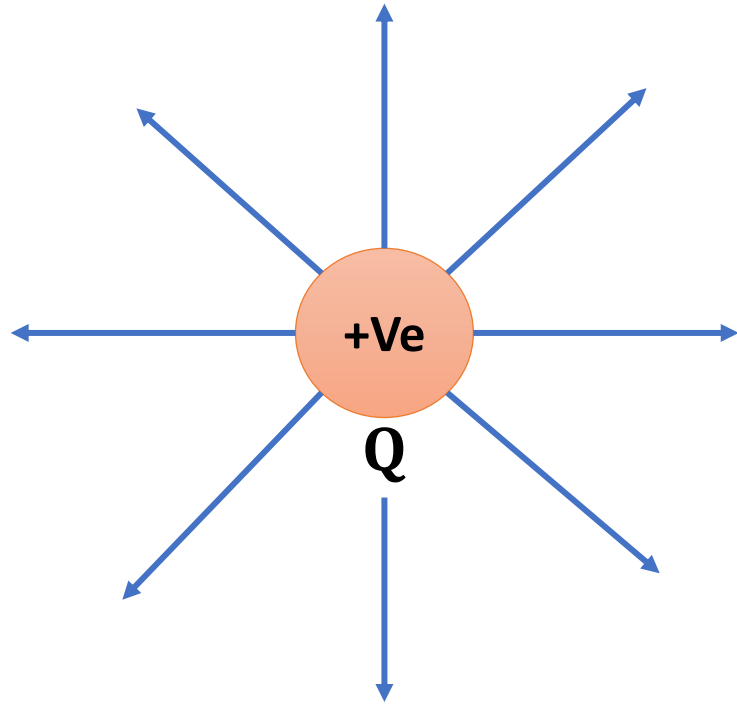
- Electric Field due to Point Charge Q is amount of Force acting on 1C Charge.



$$\Rightarrow E = \frac{F}{q} = \frac{1}{4\pi\epsilon_0} \frac{Q(1C)}{r^2} = k \frac{Q}{r^2}$$

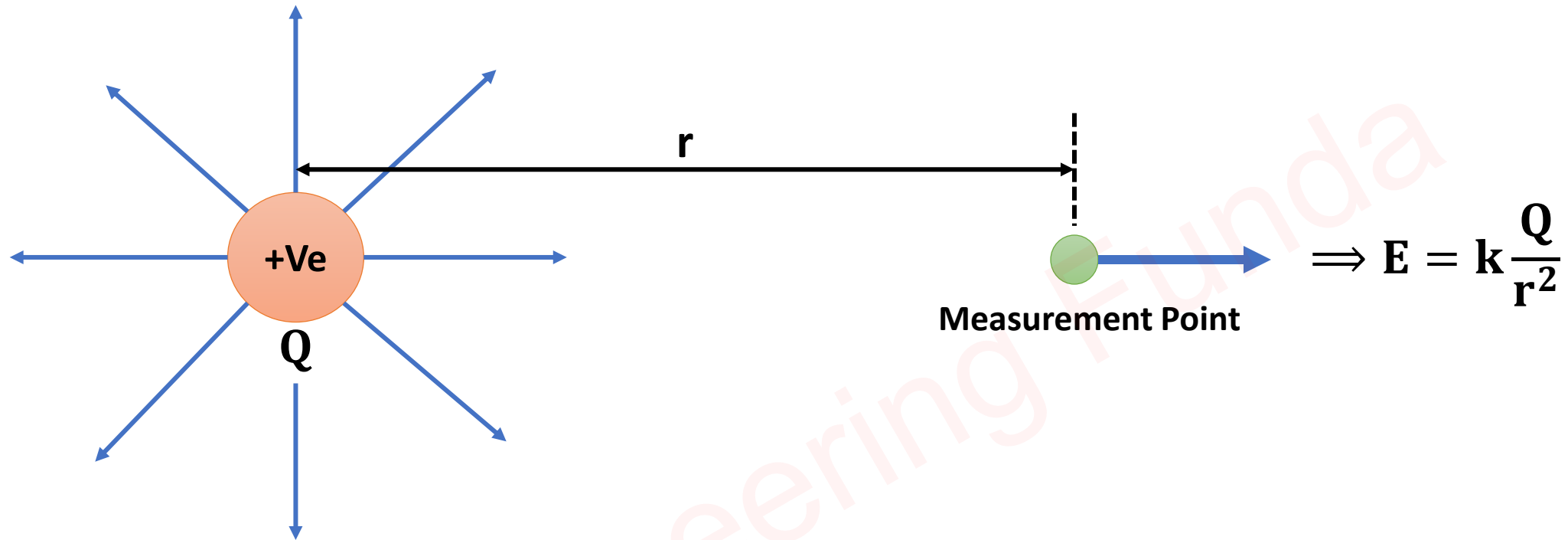
❖ Where, $k = 9 \times 10^9 \text{ SI}$

Direction of Electric Field due to Point Charge



- Electric Field is outward from the Positive charge.
- Electric Field is inward to the Negative charge.

Calculation of Electric Field due to Point Charge

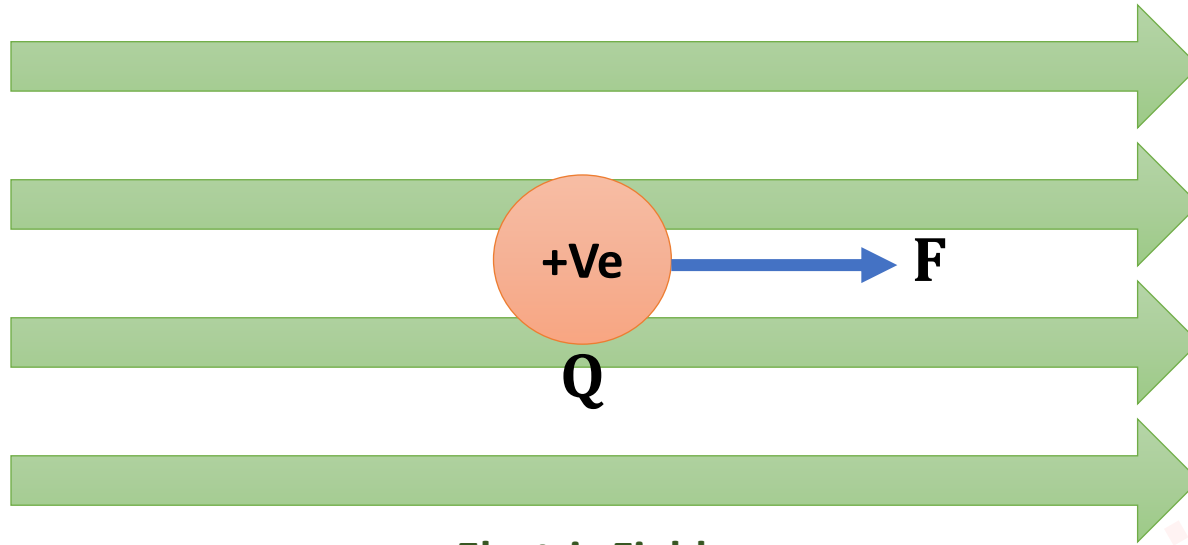


- Electric Field due to the point charge

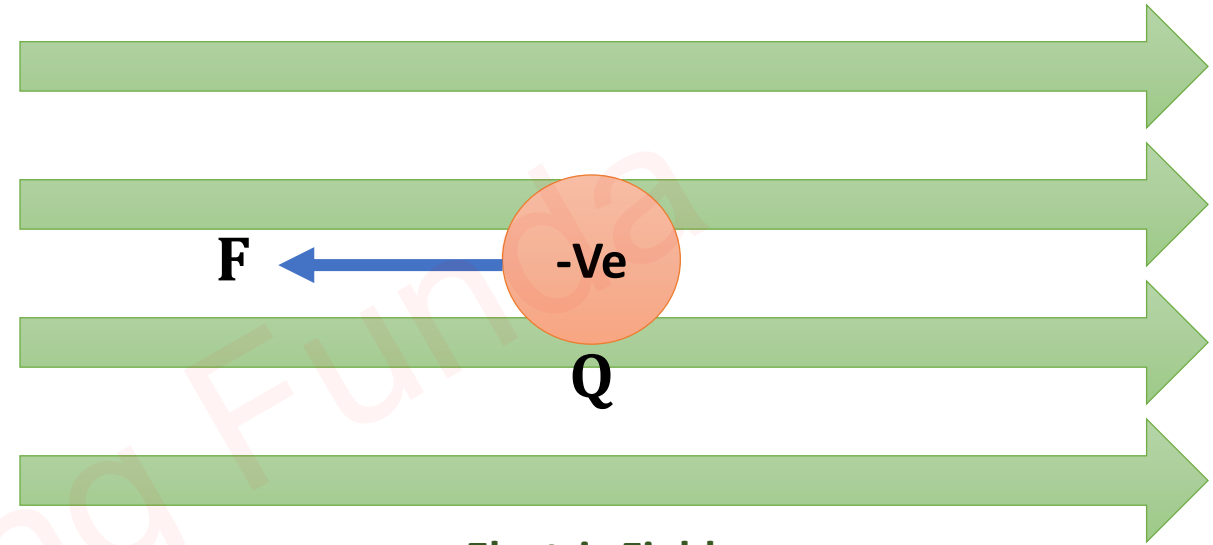
$$\Rightarrow E = k \frac{Q}{r^2}$$

$$\Rightarrow \vec{E} = k \frac{Q}{r^2} \hat{r} = k \frac{Q}{r^2} \frac{\vec{r}}{r} = k \frac{Q}{r^3} \vec{r}$$

Direction of Force under Electric Field on Point Charge



Electric Field



Electric Field

- Force on Positive charge is in the direction of Electric Field.
- Force on Negative charge is in Opposite direction of Electric Field.

Examples of Electric Field due to Point Charge

Question 1 – Find the Electric Field $\vec{E}$ at (0, 3, 4)m due to a point charge $Q = 0.5\mu\text{C}$ placed at the origin.

$$Q = 0.5\mu\text{C}$$

(0, 0, 0)

(0, 3, 4) $\vec{E}$

$$\Rightarrow \vec{E} = k \frac{q}{r^2} \hat{r} = k \frac{q}{r^2} \left(\frac{\vec{r}}{r} \right) = k \frac{q}{r^3} \vec{r}$$

$$\Rightarrow \vec{r} = (0, 3, 4)$$

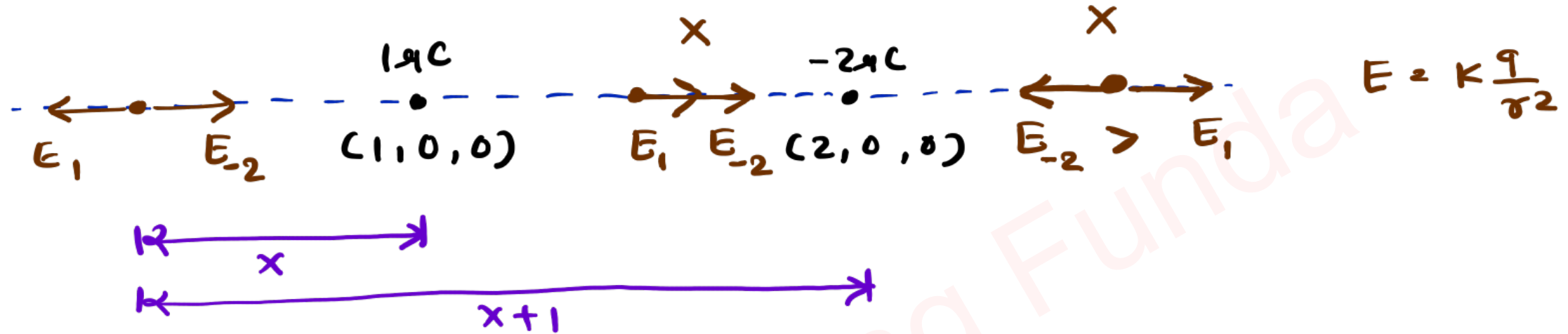
$$r = \sqrt{0^2 + 3^2 + 4^2} = 5$$

$$\Rightarrow \vec{E} = 9 \times 10^9 \times \frac{0.5 \times 10^{-6}}{5^2} (0, 3, 4)$$

$$= 36 (0, 3, 4)$$

$$= (0, 108, 144) \text{ N/C or V/m}$$

Question 2 – Find the position at which the Electric field is zero. If $1\mu\text{C}$ is located at $(1, 0, 0)\text{m}$ and $-2\mu\text{C}$ is located at $(2, 0, 0)\text{m}$.



$$\Rightarrow E_1 = E_2$$

$$\Rightarrow \cancel{k} \frac{q_1}{x^2} = \cancel{k} \frac{q_2}{(x+1)^2}$$

$$\Rightarrow \frac{1 \times 10^{-6}}{x^2} = \frac{2 \times 10^{-6}}{(x+1)^2}$$

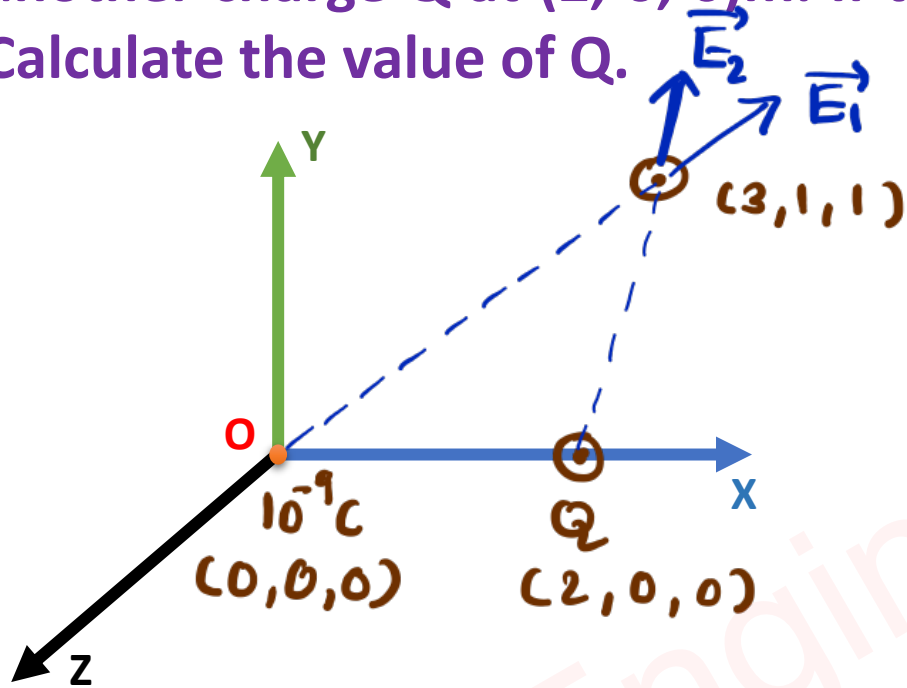
$$\Rightarrow (x+1)^2 = 2x^2$$

$$\Rightarrow x+1 = \sqrt{2}x \Rightarrow x = 2.41 \text{ m}$$

$$\rightarrow (-1.41, 0, 0)$$

Examples of Electric Field due to Point Charge

Question 3 – A Charge 10^{-9}C is located at the origin of a cartesian coordinate system and another charge Q at $(2, 0, 0)\text{m}$. If the x-component of Electric Field at $(3, 1, 1)\text{m}$ is zero, then Calculate the value of Q .



$$\rightarrow \vec{E}_2 = k \frac{q_2}{r_2^2} \hat{r}_2 = k \frac{q_2}{r_2^3} \vec{r}_2$$

$$\vec{r}_2 = (3, 1, 1) - (2, 0, 0) = (1, 1, 1)$$

$$r_2 = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\begin{aligned} \rightarrow \vec{E}_2 &= 9 \times 10^9 \times \frac{Q}{(\sqrt{3})^3} (1, 1, 1) \\ &= 1.73 \times 10^9 \times Q (1, 1, 1) \end{aligned}$$

$$\rightarrow \vec{E}_1 = k \frac{q_1}{r_1^2} \hat{r}_1 = k \frac{q_1}{r_1^3} \vec{r}_1$$

$$\vec{r}_1 = (3, 1, 1)$$

$$r_1 = \sqrt{3^2 + 1^2 + 1^2} = \sqrt{11}$$

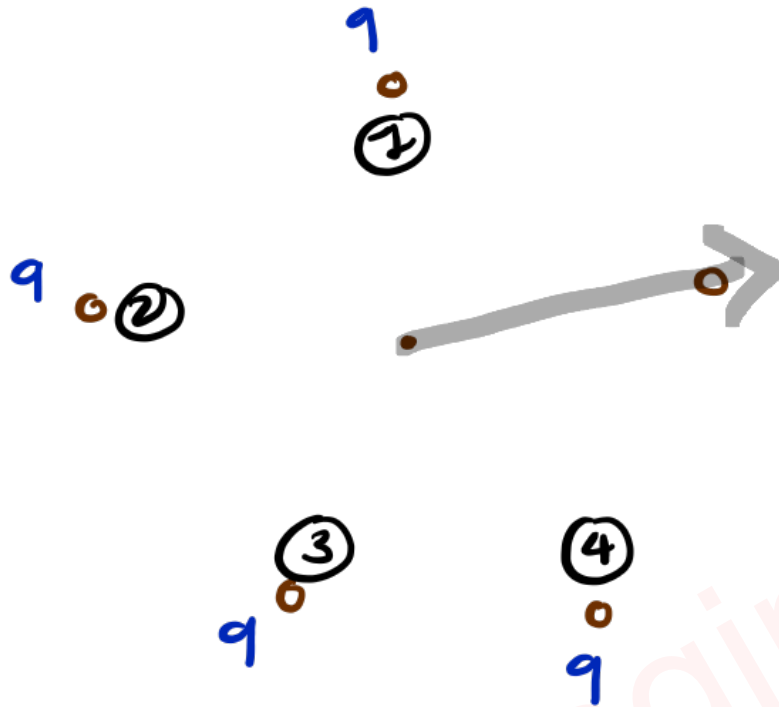
$$\rightarrow \vec{E}_1 = \frac{9 \times 10^9 \times 10^{-9}}{(\sqrt{11})^3} (3, 1, 1)$$

$$= 0.2466 (3, 1, 1) \text{ V/m}$$

$$\rightarrow E_x = 0 = 1.73 \times 10^9 \times Q + 0.2466 \times 3$$

$$\Rightarrow Q = - \frac{0.2466 \times 3}{1.73 \times 10^9} = -0.427 \times 10^{-9} \text{ C}$$

Question 4 – Four particles, each having a charge q , are placed on the four vertices of a regular pentagon. The distance of each corner from the center is a . Find the electric field at the center of the pentagon.



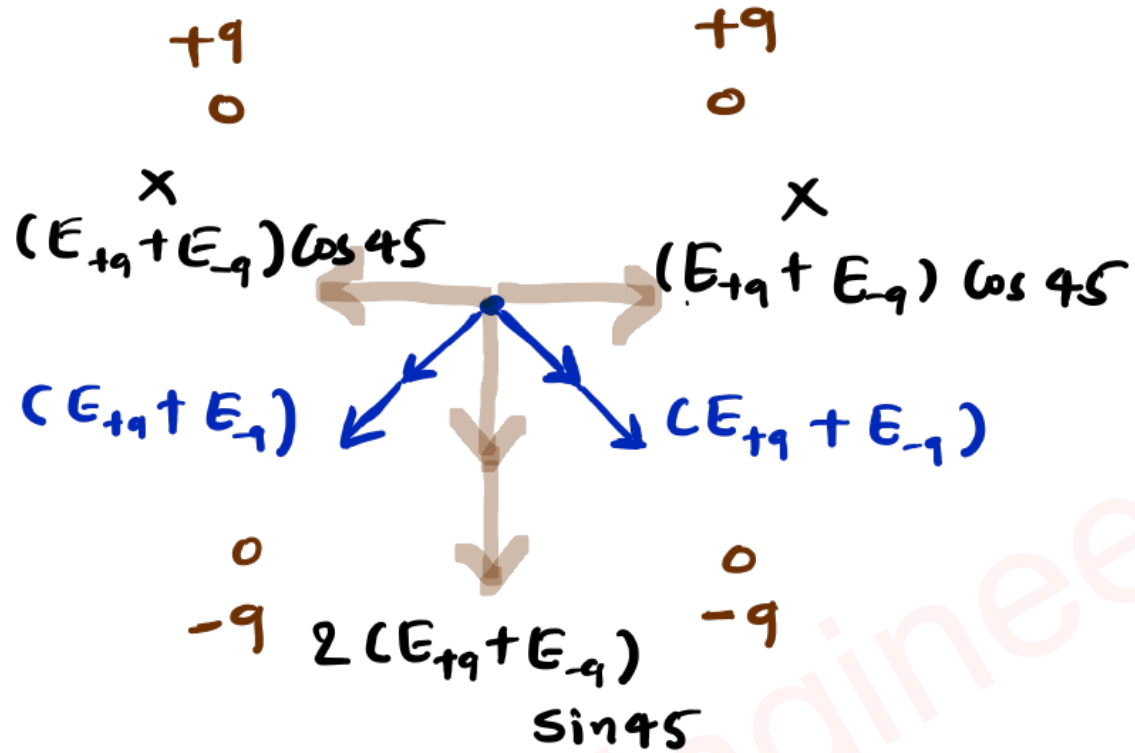
→ If 5 charges are placed on each vertices then $\vec{E} = 0$

$$\Rightarrow \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4 + \vec{E}_5 = 0$$

$$\Rightarrow \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4 = -\vec{E}_5 = \ominus k \frac{q}{a^2}$$

-Ve sign shows E field in outward direction.

Question 5 – Four electric charges $+q$, $+q$, $-q$, and $-q$ are placed on a square's vertices A, B, C, and D. The length of the square is a , calculate the intensity of the resultant electric field at the center.

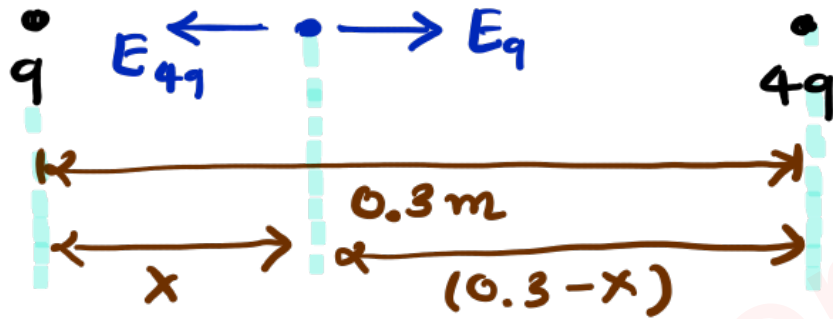


$$\begin{aligned}
 \rightarrow E &= 2(E_{+q} + E_{-q}) \sin 45 \\
 &= 4E \sin 45 \\
 &= 4 \left(k \frac{q}{r^2} \right) \sin 45 \\
 &= 4 \left(k \frac{q}{(a/\sqrt{2})^2} \right) \left(\frac{1}{\sqrt{2}} \right)
 \end{aligned}$$

$$E = 4\sqrt{2} k \frac{q}{a^2}$$

Examples of Electric Field due to Point Charge

Question 6 – Two point charges of q and $4q$ are kept 30cm apart. At a distance ^{10 cm} q from, on the straight line joining them, the intensity of the electric field is zero.



$$\Rightarrow E_{4q} = E_q$$

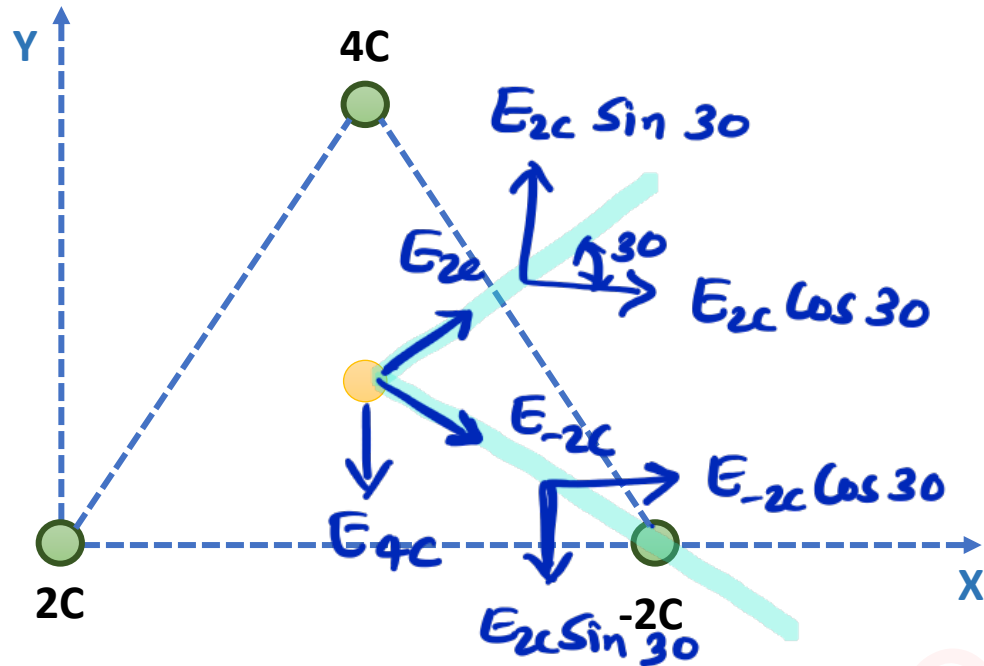
$$\Rightarrow \cancel{4} \frac{(4\cancel{q})}{(0.3-x)^2} = \cancel{q} \frac{q}{x^2}$$

$$\Rightarrow 4x^2 = (0.3-x)^2$$

$$\Rightarrow 2x = 0.3 - x$$

$$\Rightarrow \boxed{x = 0.1\text{ m}} = 10\text{ cm}$$

Question 7 – Find the electric field at the center of the equilateral triangle on the side, $a=1\text{m}$.



$$\begin{aligned}
 \vec{E}_{4C} &= -\hat{j} \left(k \frac{q}{r^2} \right) \\
 &= -\hat{j} \left(9 \times 10^9 \times \frac{4}{\left(\frac{1}{\sqrt{3}}\right)^2} \right) \\
 &= -\hat{j} \, 9 \times 3 \times 4 \times 10^9 \\
 &= -\hat{j} \, 108 \times 10^9 \text{ V/m}
 \end{aligned}$$

$$\begin{aligned}
 \vec{E}_{2C} &= \hat{i} E_{2C} \cos 30 - \hat{j} E_{2C} \sin 30 \\
 &= \hat{i} \left(9 \times 10^9 \times \frac{2}{\left(\frac{1}{\sqrt{3}}\right)^2} \times \frac{\sqrt{3}}{2} \right) - \\
 &\quad \hat{j} \left(9 \times 10^9 \times \frac{2}{\left(\frac{1}{\sqrt{3}}\right)^2} \times \frac{1}{2} \right) \\
 &= \hat{i} (46.76 \times 10^9) - \hat{j} (27 \times 10^9) \text{ V/m}
 \end{aligned}$$

$$\begin{aligned}
 \vec{E} &= \vec{E}_{4C} + \vec{E}_{2C} + \vec{E}_{-2C} \\
 &= 2 \times 46.76 \times 10^9 \hat{i} - 108 \times 10^9 \hat{j} \\
 &= 93.52 \times 10^9 \hat{i} - 108 \times 10^9 \hat{j}
 \end{aligned}$$

$$\Rightarrow \vec{E}_{2c} = \hat{i} E_{2c} \cos 30 + \hat{j} E_{2c} \sin 30$$

$$= \hat{i} \left(9 \times 10^9 \times \frac{2}{(\sqrt{3})^2} \times \frac{\sqrt{3}}{2} \right) +$$

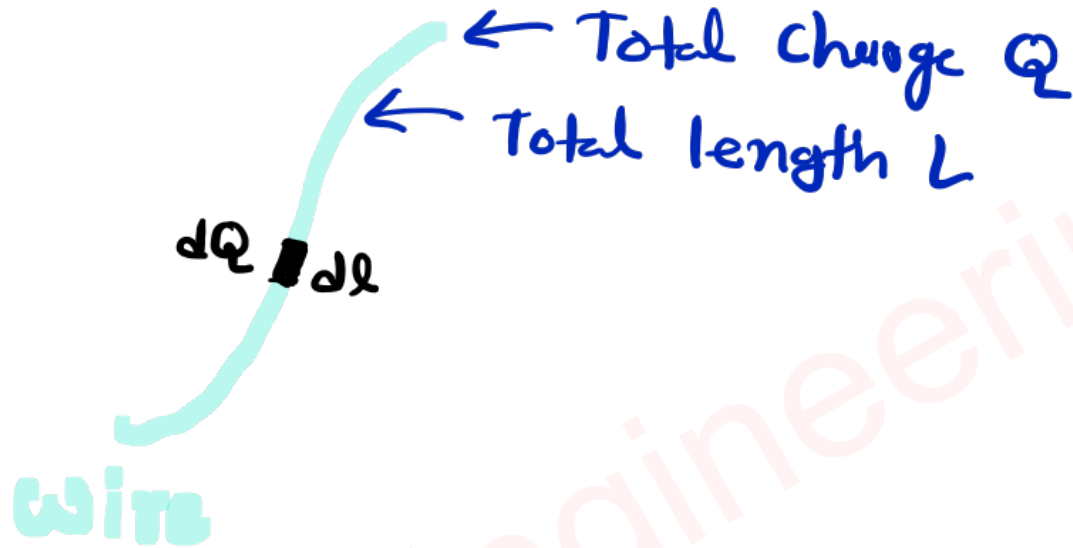
$$\hat{j} \left(9 \times 10^9 \times \frac{2}{(\sqrt{3})^2} \times \frac{1}{2} \right)$$

$$= \hat{i} (46.76 \times 10^9) + \hat{j} (27 \times 10^9) \text{ V/m}$$

Charge Density

Line Charge Density

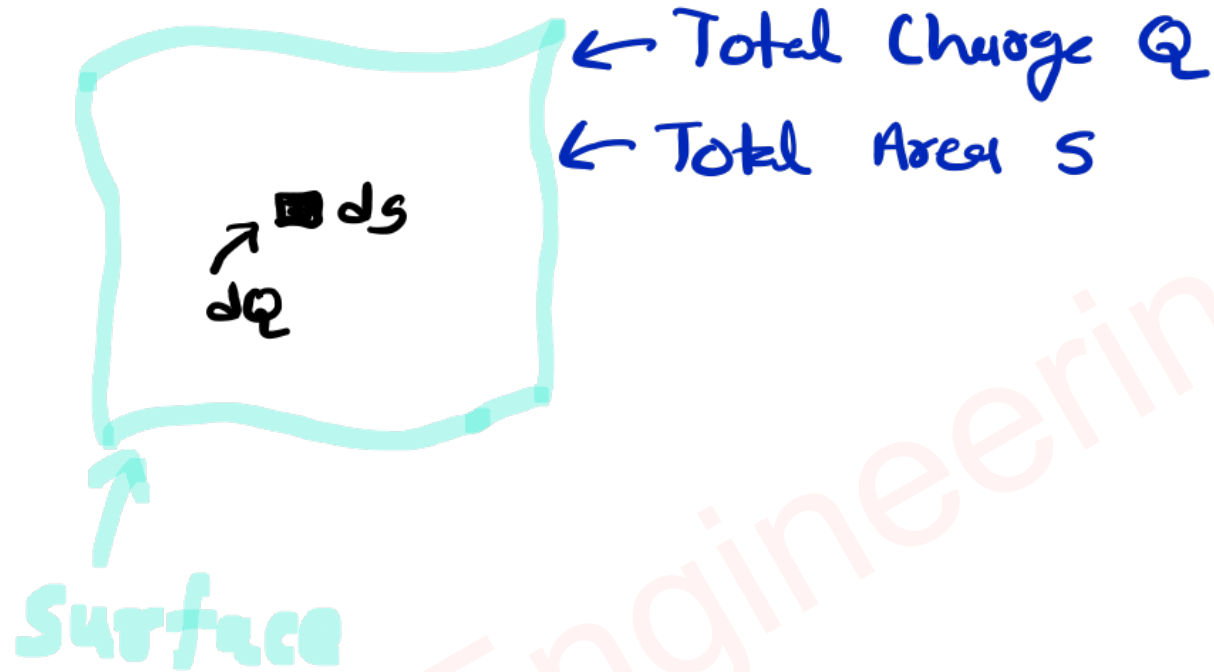
- It is the Charge per unit Length.
- Example: Wire, Thin Rod etc.



$$\leadsto \lambda_L = \frac{Q}{L} = \frac{dQ}{dl} \rightarrow dQ = \lambda_L dl \\ \Rightarrow Q = \int \lambda_L dl$$

Surface Charge Density

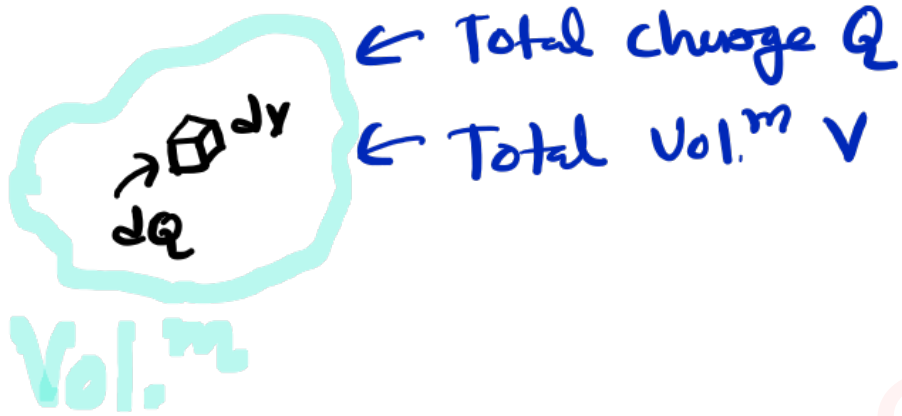
- It is the Charge per unit Area.
- Example: Paper Foil, Capacitor with parallel plates etc.



$$\rightarrow j_s = \frac{Q}{S} = \frac{dQ}{ds} \rightarrow dQ = j_s ds$$
$$\Rightarrow Q = \int j_s ds$$

Volume Charge Density

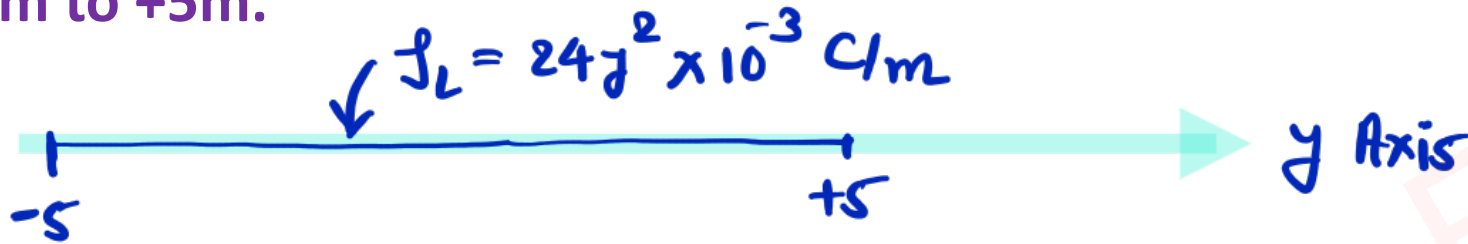
- It is the Charge per unit Volume.
- Example: Metallic Body with 3D.



$$\begin{aligned} \leadsto f_v &= \frac{Q}{V} = \frac{dQ}{dV} \rightarrow dQ = f_v dV \\ &\Rightarrow Q = \int f_v dV \end{aligned}$$

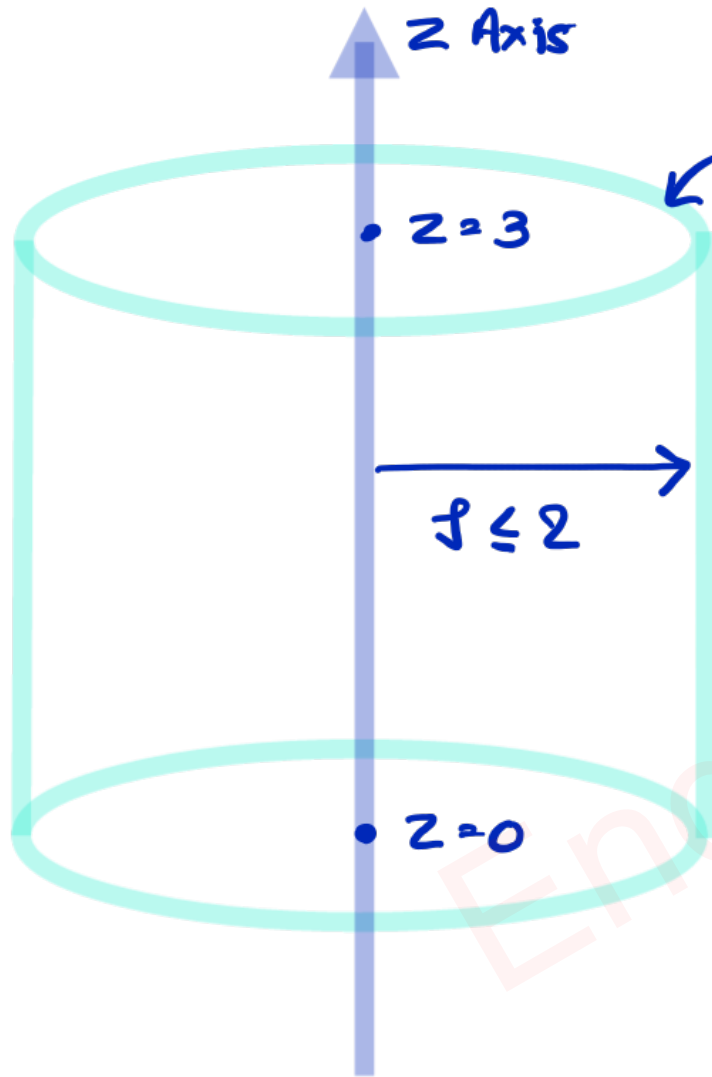
Examples of Charge Density

Question 1 – If the line charge density $\rho_L = 24y^2$ mC/m. Find the total charge on Y axis from -5m to +5m.



$$\begin{aligned} \Rightarrow Q &= \int \rho_L \, dl \\ &= \int_{-5}^5 24y^2 \times 10^{-3} \, dy \\ &= 24 \times 10^{-3} \left[\frac{y^3}{3} \right]_{-5}^5 \\ &= \frac{8}{3} \times \frac{250}{3} \times 10^{-3} \\ &= 2 \text{ C} \end{aligned}$$

Question 2 – Find the total charge contained in the cylindrical volume defined by $\rho \leq 2, 0 \leq z \leq 3, \rho_V = 20\rho z \text{ mC/m}^3$



$$\rho_V = 20 \rho z \times 10^{-3} \text{ C/m}^3$$

$$\Rightarrow Q = \int \rho_V dV$$

$$= \int_0^3 \int_0^{2\pi} \int_0^2 20 \times 10^{-3} \times \rho z \left(\rho d\rho d\phi dz \right)$$

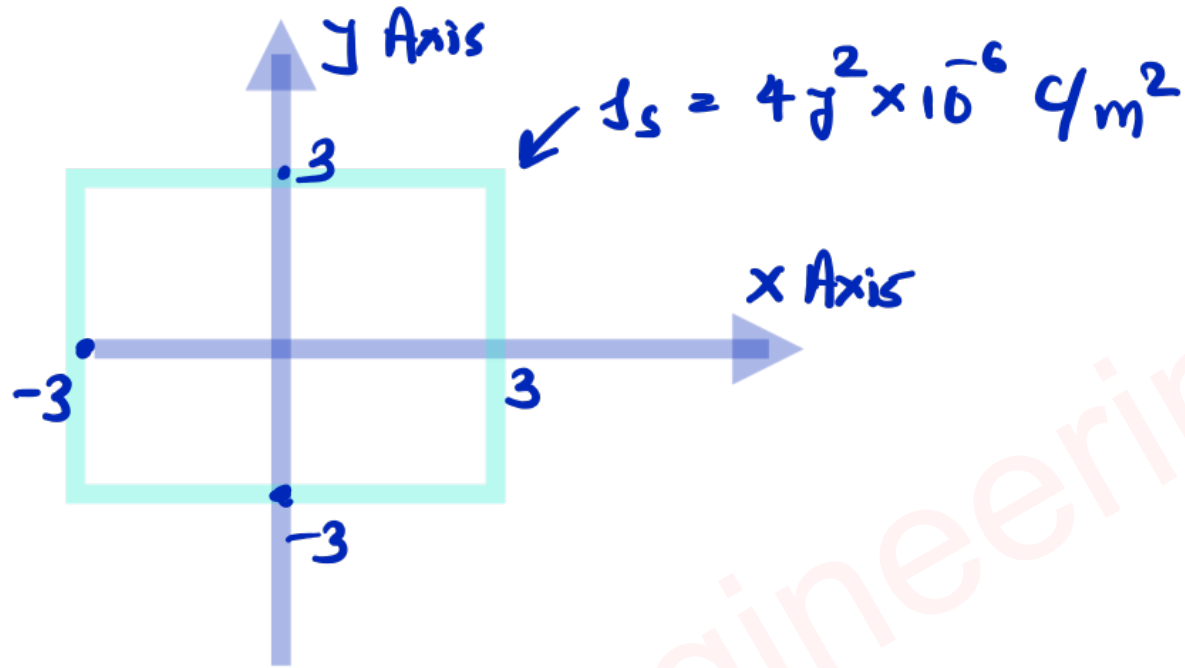
$$= 20 \times 10^{-3} \times \left[\frac{\rho^3}{3} \right]_0^2 \left[\phi \right]_0^{2\pi} \left[\frac{z^2}{2} \right]_0^3$$

$$= 20 \times 10^{-3} \times \frac{8}{3} \times 2\pi \times \frac{9}{2}$$

$$= 1.508 \text{ C}$$

Question 3 – A square plate residing in a XY plane is situated in the space defined by $-3 \leq x \leq 3$ and $-3 \leq y \leq 3$

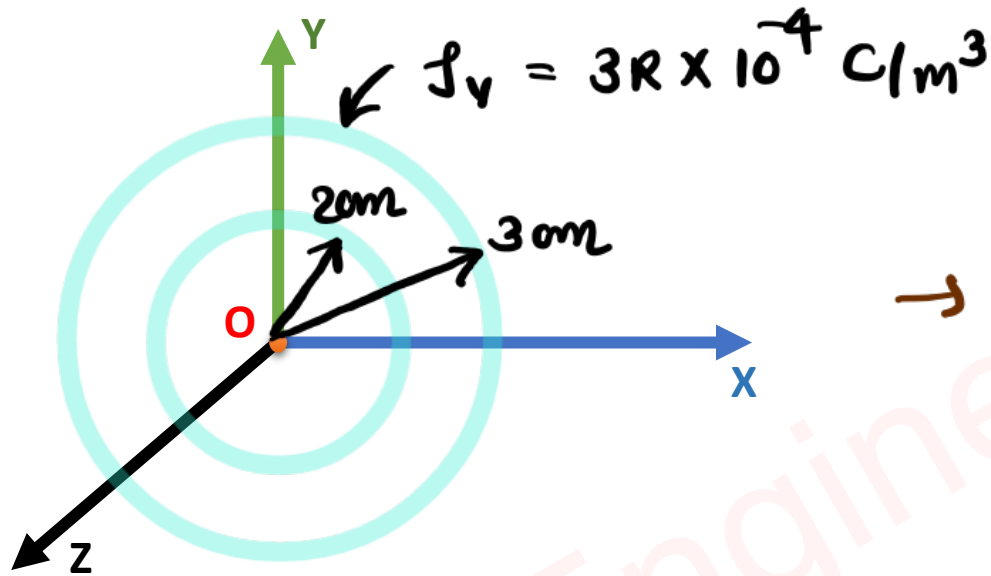
Find the total charge on the plate if surface charge density is given by $\rho_s = 4y^2 \mu\text{C}/\text{m}^2$.



$$\begin{aligned} Q &= \int \rho_s \, ds \\ &= \int_{-3}^3 \int_{-3}^3 4y^2 \times 10^{-6} \, dx \, dy \\ &= 4 \times 10^{-6} \left[\frac{y^3}{3} \right]_{-3}^3 \left[x \right]_{-3}^3 \\ &= 4 \times 10^{-6} \times 18 \times 6 \\ &= 432 \mu\text{C} \\ &= 0.432 \text{ mC} \end{aligned}$$

Examples of Charge Density

Question 4 – A thick spherical shell centered at the origin extends between $R=2\text{cm}$ and $R=3\text{cm}$. If the volume charge density is $\rho_V = 3R \times 10^{-4} \text{ C/m}^3$. Find the total charge contained in the shell.



$$\begin{aligned} \rightarrow R &\in (2, 3) \text{ cm} \\ \theta &\in (0, \pi) \\ \phi &\in (0, 2\pi) \end{aligned}$$

$$\begin{aligned} \rightarrow Q &= \int \rho_V dV \\ &= \int_2^3 \int_0^{2\pi} \int_0^\pi 3R \times 10^{-4} (R^2 \sin \theta d\theta d\phi dR) \\ &= 3 \times 10^{-4} \left[\frac{R^4}{4} \right]_{2 \times 10^{-2}}^{3 \times 10^{-2}} [-\cos \theta]_0^\pi [\phi]_0^{2\pi} \\ &= 3 \times 10^{-4} \times \frac{65}{4} \times 10^{-8} \times 2 \times 2\pi \\ &= 0.612 \text{ nC} \end{aligned}$$

Question 5 – The volume charge density $\rho_V = \rho_0 e^{-|x|-|y|-|z|}$ C/m³ exists over all free space. Calculate the total charge present.

$$\Rightarrow x \in (-\infty, \infty)$$

$$y \in (-\infty, \infty)$$

$$z \in (-\infty, \infty)$$

$$\Rightarrow Q = \int \rho_V dV$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \rho_0 e^{-|x|-|y|-|z|} dx dy dz$$

$$= \int_{-\infty}^{\infty} \rho_0 e^{-|x|} dx \int_{-\infty}^{\infty} e^{-|y|} dy \int_{-\infty}^{\infty} e^{-|z|} dz$$

$$= \rho_0 \left[2 \int_0^{\infty} e^{-x} dx \right] \left[2 \int_0^{\infty} e^{-y} dy \right] \left[2 \int_0^{\infty} e^{-z} dz \right]$$

$$= 8 \rho_0 \left[\frac{e^{-x}}{-1} \right]_0^{\infty} \left[\frac{e^{-y}}{-1} \right]_0^{\infty} \left[\frac{e^{-z}}{-1} \right]_0^{\infty}$$

$$= 8 \rho_0$$

Question 6 – The Spherical volume charge is given by

$$\rho_V = \rho_{V0} \left(1 - \frac{r^2}{a^2}\right) \text{ C/m}^3, r < a$$

$$\rho_V = 0 \text{ C/m}^3, r > a$$

$$\left| \begin{array}{l} \rightarrow \theta \in (0, \pi) \\ \phi \in (0, 2\pi) \\ r \in (0, a) \end{array} \right.$$

Calculate the total charge Q.

$$\rightarrow Q = \int \rho_V dV$$

$$= \int_0^a \int_0^{2\pi} \int_0^\pi \rho_{V0} \left(1 - \frac{r^2}{a^2}\right) (r^2 \sin \theta dr d\phi d\theta)$$

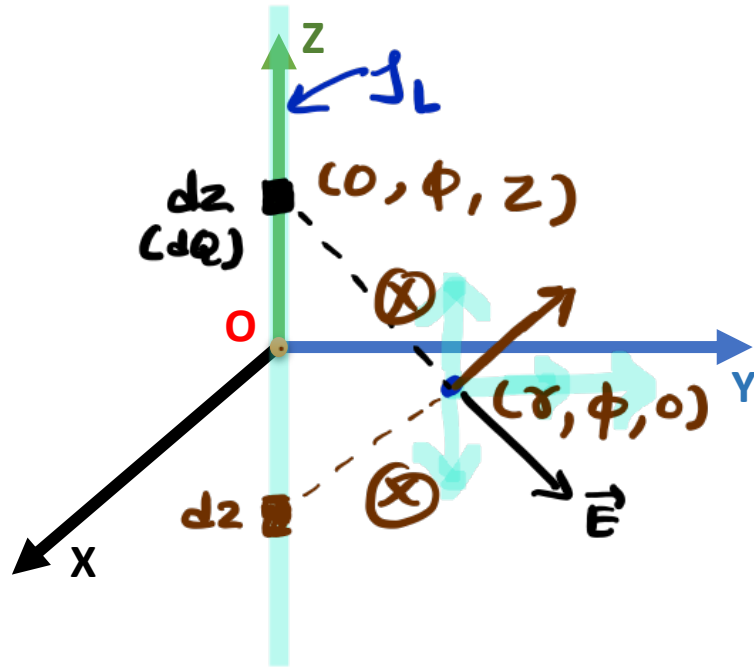
$$= \rho_{V0} [-\cos \theta]_0^\pi [\phi]_0^{2\pi} \left[\frac{r^3}{3} - \frac{r^5}{5a^2}\right]_0^a$$

$$= \rho_{V0} [2][2\pi] \left[\frac{a^3}{3} - \frac{a^5}{5a^2}\right]$$

$$= \rho_{V0} a^3 [4\pi] \left[\frac{2}{15}\right]$$

$$Q = \frac{8\pi}{15} a^3 \rho_{V0}$$

Electric Field due to Line Charge



→ Electric field due to point charge

$$\Rightarrow \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^3} \vec{r}$$

→ For dz length $\vec{E}$ field is

$$\Rightarrow d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^3} \vec{r}$$

$$\Rightarrow \vec{r} = (r, \phi, 0) - (0, \phi, z) = (r, 0, -z)$$

$$r = \sqrt{r^2 + z^2}$$

$$dq = \lambda_L dz$$

→ Hence, Electric field is

$$\Rightarrow d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{(\lambda_L dz)}{(r^2 + z^2)^{3/2}} (r, 0, -z)$$

~) Here, z direction has no Electric field.

$$\Rightarrow d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{\rho_L dz}{(r^2 + z^2)^{3/2}} (r, 0, 0)$$

$$\Rightarrow \vec{E} = \int_{-\infty}^{\infty} \frac{1}{4\pi\epsilon_0} \frac{\rho_L dz}{(r^2 + z^2)^{3/2}} (r \hat{a}_r)$$

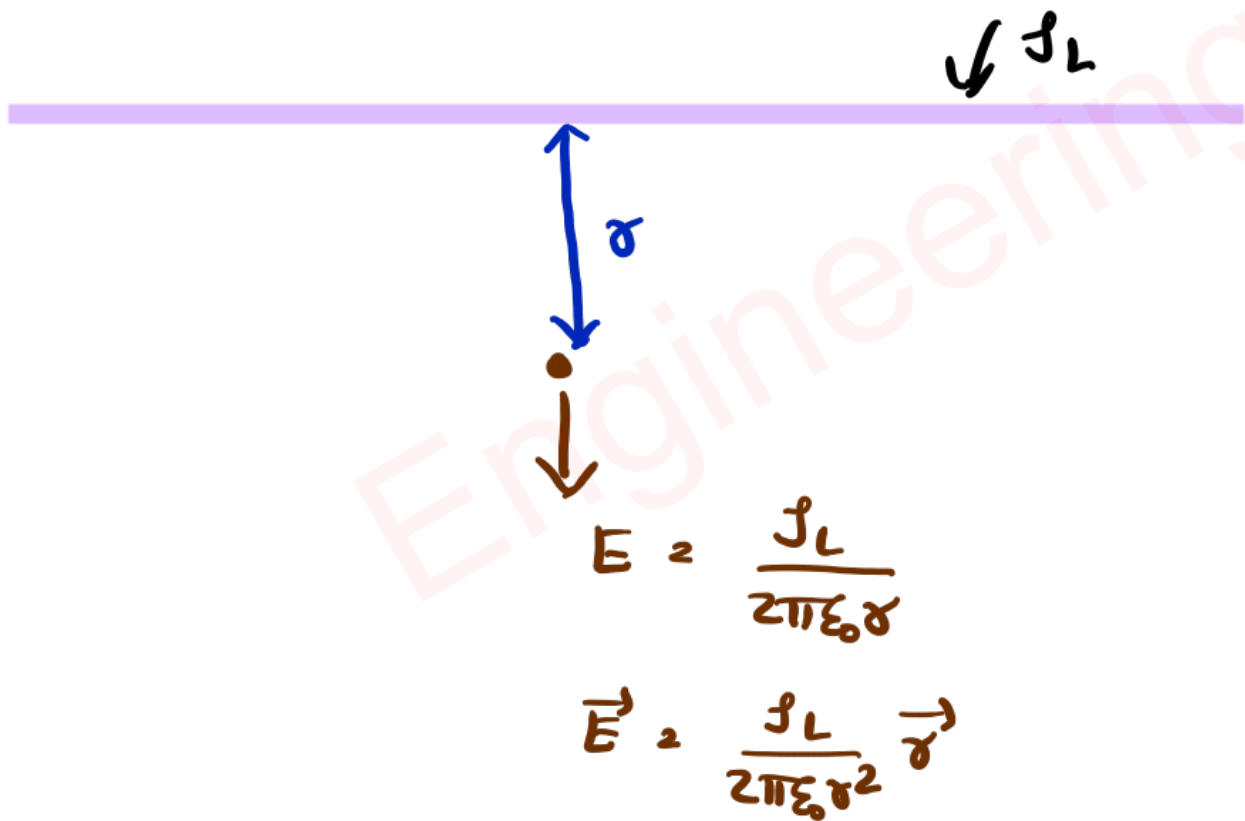
$$= \frac{\rho_L}{4\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{r dz}{(r^2 + z^2)^{3/2}} \hat{a}_r$$

$$\sim) \begin{array}{l} z = r \tan \theta \\ dz = r \sec^2 \theta d\theta \end{array} \quad \left| \begin{array}{l} z \in (-\infty, \infty) \\ \theta \in (-\pi/2, \pi/2) \end{array} \right.$$

$$\Rightarrow \vec{E} = \frac{\rho_L}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{\cancel{r}^2 \cancel{\sec^2 \theta}}{\cancel{r}^2 \cancel{\sec^2 \theta}} d\theta = \frac{\rho_L}{4\pi\epsilon_0 r} [\sin \theta]_{-\pi/2}^{\pi/2} \hat{a}_r$$

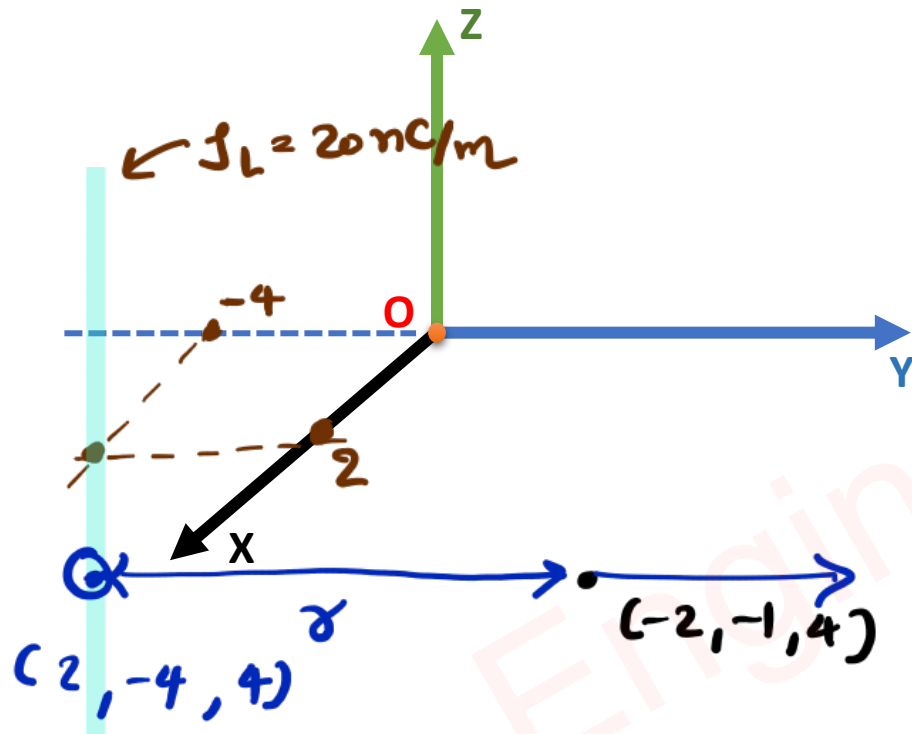
$$\Rightarrow \vec{E} = \frac{J_L}{2\pi\epsilon_0 r} \hat{a}_r = \frac{J_L}{2\pi\epsilon_0 r^2} \vec{r} = \frac{2k J_L}{r^2} \vec{r}$$

$$\left[k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI} \right]$$



Examples of Electric Field due to Line Charge

Question 1 – The line charge density $\rho_L = 20 \text{ nC/m}$ is placed at $x = 2 \text{ m}$, $y = -4 \text{ m}$. Find the Electric Field at $(-2, -1, 4) \text{ m}$ due to line charge.



$$\vec{r} = (-2, -1, 4) - (2, -4, 4) \\ = (-4, 3, 0)$$

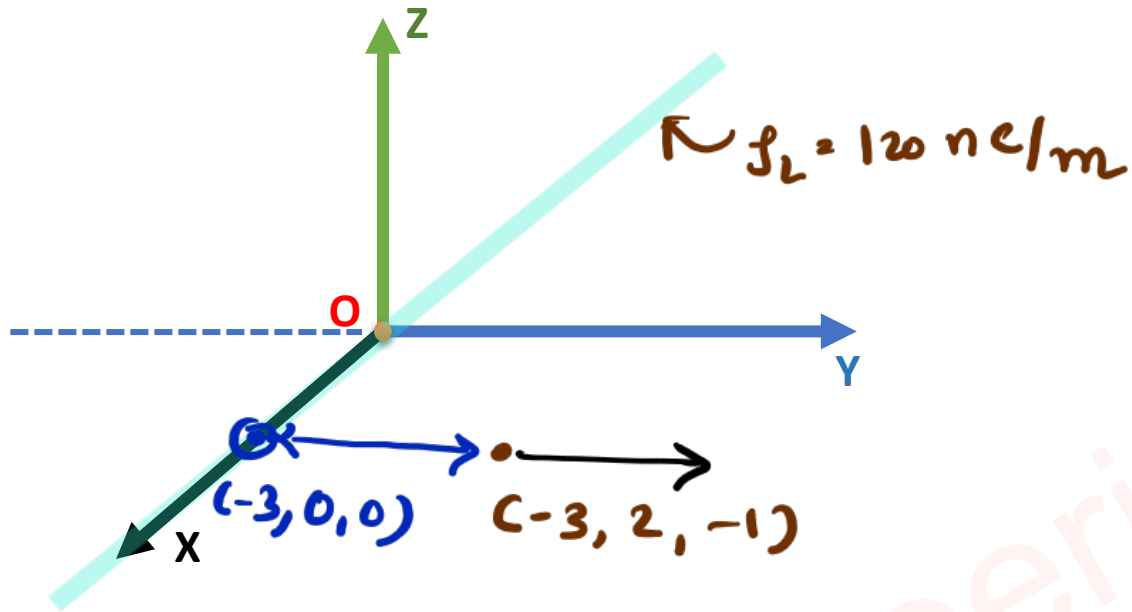
$$r = \sqrt{4^2 + 3^2} = 5$$

$$\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 r} \hat{r} = \frac{\rho_L}{2\pi\epsilon_0 r^2} \vec{r} \\ = \frac{20 \times 10^{-9} \times 18 \times 10^9}{25} (-4, 3, 0)$$

$$= 14.4 (-4, 3, 0)$$

$$= (-57.6, 43.2, 0) \text{ V/m or N/C}$$

Question 2 – The line charge density $\rho_l = 120 \text{ nC/m}$ is placed x axis. Find the Electric Field at $(-3, 2, -1) \text{ m}$ due to line charge.



$$\vec{r} = (-3, 2, -1) - (-3, 0, 0) = (0, 2, -1)$$

$$r = \sqrt{0^2 + 2^2 + 1^2} = \sqrt{5}$$

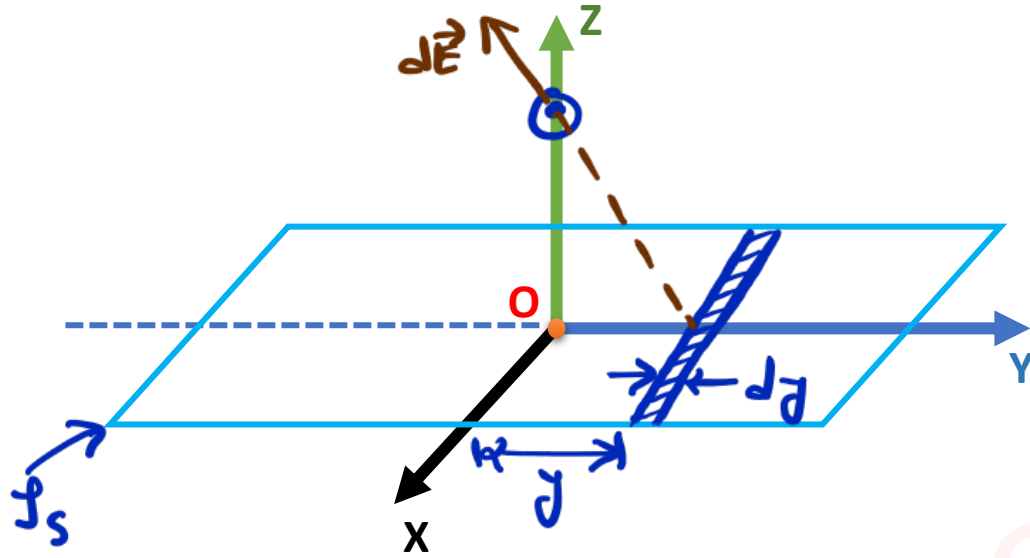
$$\vec{E} = \frac{\rho_l}{2\pi\epsilon_0} \frac{\vec{r}}{r^2} = \frac{\rho_l}{2\pi\epsilon_0 r^2} \vec{r}$$

$$= \frac{120 \times 10^{-9} \times 18 \times 10^9}{5} \times (0, 2, -1)$$

$$= 432 (0, 2, -1)$$

$$= (0, 864, -432) \text{ V/m or N/C}$$

Electric Field due to Surface Charge



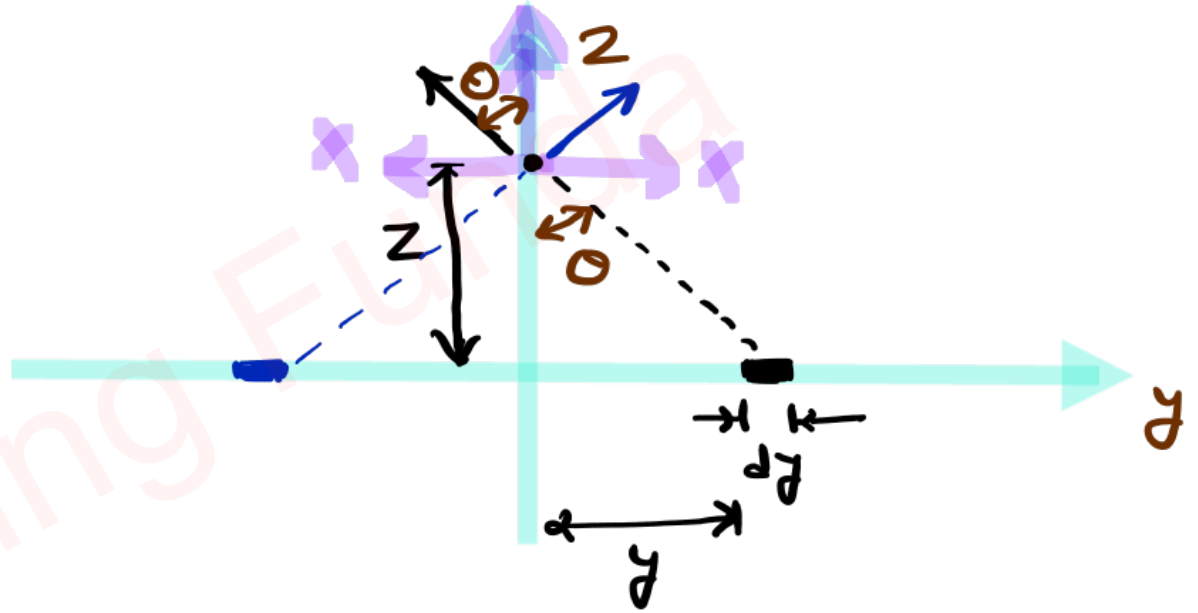
→ Electric field due to line charge.

$$\Rightarrow d\vec{E} = \frac{\lambda L}{2\pi\epsilon_0 R} \hat{q}_r$$

$$\Rightarrow \lambda L = \sigma_s dl$$

$$R = \sqrt{y^2 + z^2}$$

$$\hat{q}_r = \hat{i}_z \cos \theta = \frac{z}{\sqrt{y^2 + z^2}} \hat{i}_z$$



$$\Rightarrow d\vec{E} = \frac{I_s dy}{2\pi\epsilon_0 \sqrt{y^2+z^2}} \times \frac{z}{\sqrt{y^2+z^2}} \hat{i}_z$$

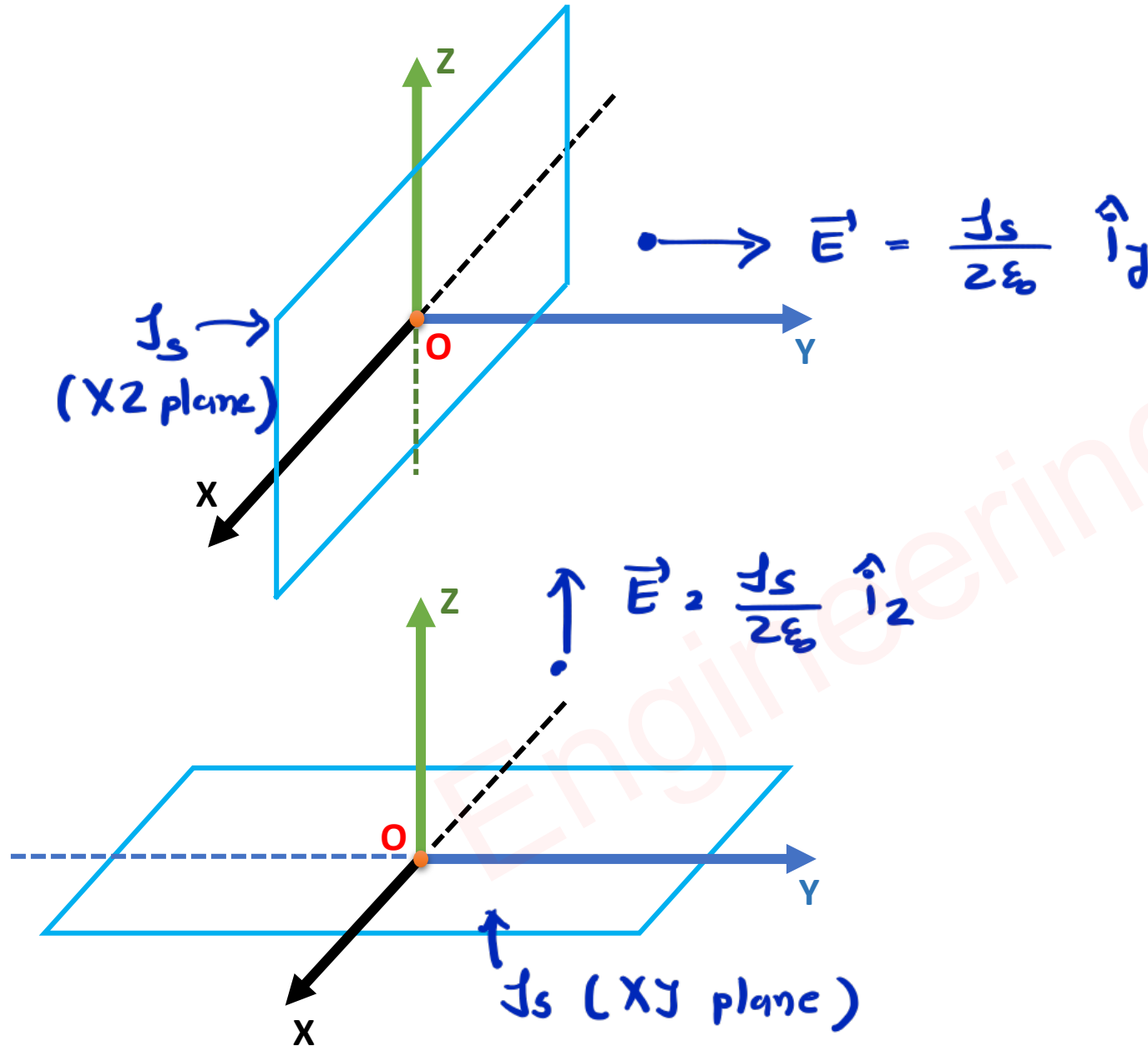
$$\Rightarrow \vec{E} = \int d\vec{E} = \int_{-\infty}^{\infty} \frac{z I_s dy}{2\pi\epsilon_0 (y^2+z^2)} \hat{i}_z$$

$$= \frac{I_s}{2\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{z dy}{y^2+z^2} \hat{i}_z$$

$$= \frac{I_s}{2\pi\epsilon_0} \left[\tan^{-1}(y/z) \right]_{-\infty}^{\infty} \hat{i}_z$$

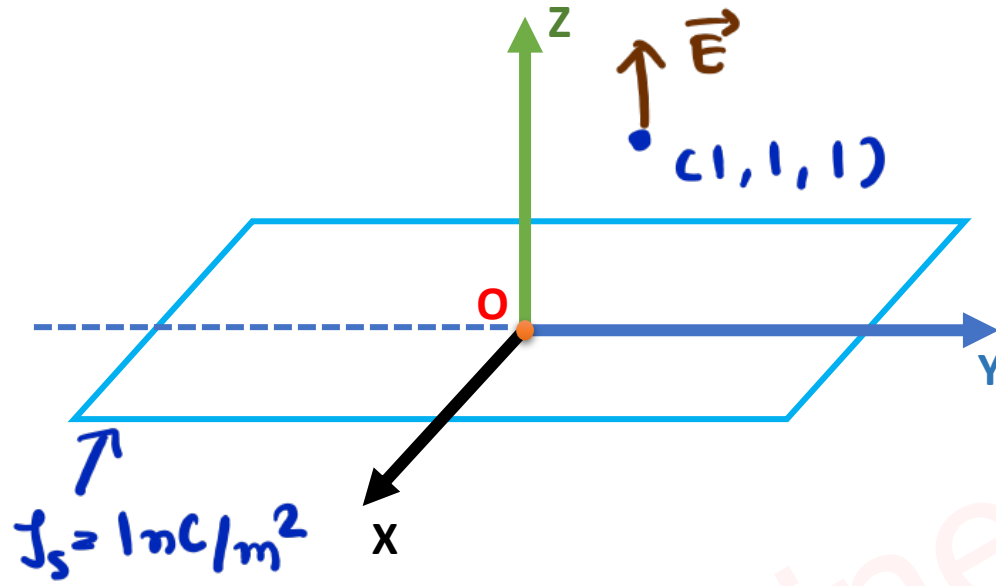
$$\Rightarrow \vec{E} = \frac{I_s}{2\epsilon_0} \hat{i}_z$$

Direction of Electric Field due to Surface Charge



Example of Electric Field due to Surface Charge

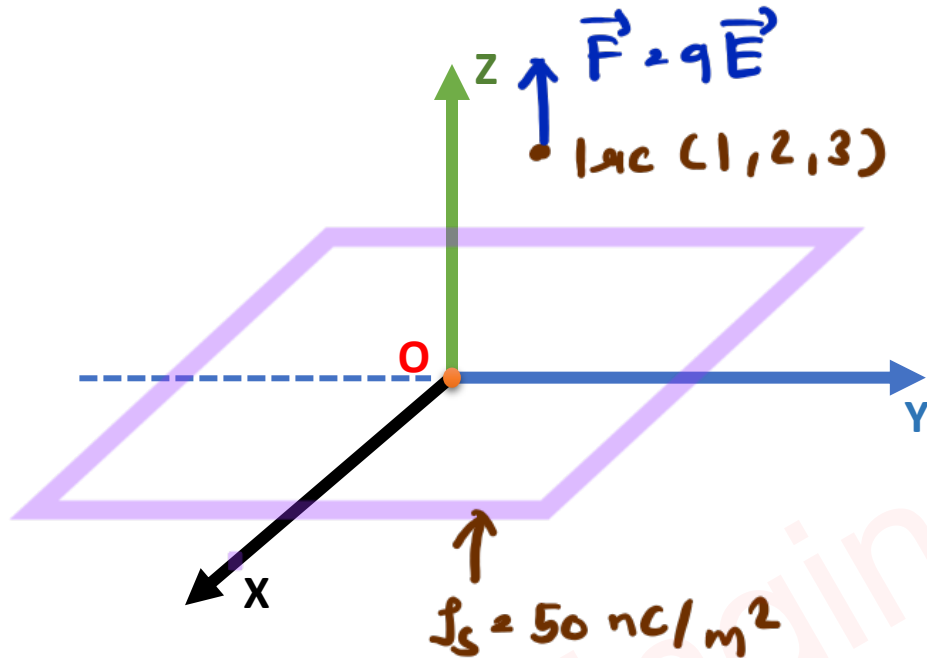
Question – If the surface charge density $\rho_s = 1 \text{ nC/m}^2$ is placed on XY plane. Find the Electric Field at (1, 1, 1)m due to Surface charge.



$$\begin{aligned}\vec{E} &= \frac{\rho_s}{2\epsilon_0} \hat{i}_z \\ &= \frac{1 \times 10^{-9}}{2 \times 8.854 \times 10^{-12}} \hat{i}_z \\ &= 56.47 \hat{i}_z \text{ V/m or N/C}\end{aligned}$$

Examples of Electric Field due to Surface Charge

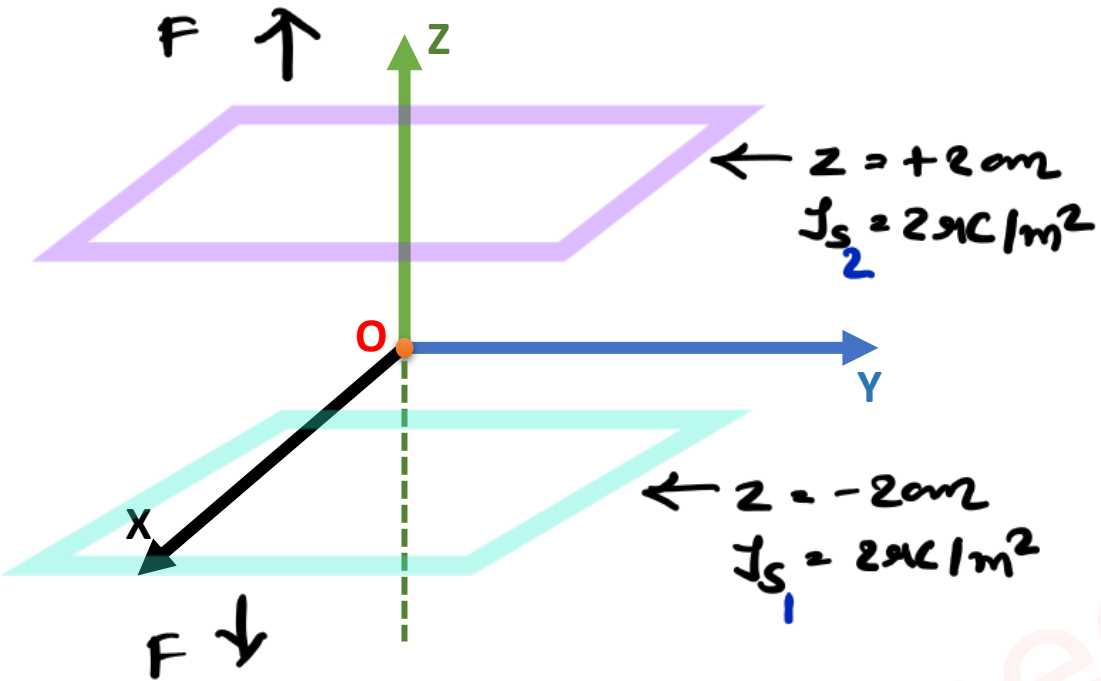
Question 1 – The surface charge density $\rho_s = 50 \text{ nC/m}^2$ is placed XY plane. Find the Electric force on $1 \mu\text{C}$ charge located at $(1, 2, 3) \text{ m}$.



$$\begin{aligned}\rightarrow \vec{E} &= \frac{\rho_s}{2\epsilon_0} \hat{i}_z \\ &= \frac{50 \times 10^{-9}}{2 \times 8.854 \times 10^{-12}} \hat{i}_z \\ &= 2.82 \times 10^3 \hat{i}_z \text{ V/m or F/C}\end{aligned}$$

$$\begin{aligned}\rightarrow \vec{F} &= q\vec{E} \\ &= 10^{-6} \times 2.82 \times 10^3 \hat{i}_z \\ &= 2.82 \times 10^{-3} \hat{i}_z \text{ N}\end{aligned}$$

Question 2 – The surface charge density $\rho_s = 2\mu\text{C}/\text{m}^2$ is placed at $z = \pm 2\text{cm}$. Find the Electric force per unit area exerted on each other.



As both surfaces have +ve charge, they will experience repulsive force.

$$\Rightarrow \frac{F}{A} = \frac{qE}{A}$$

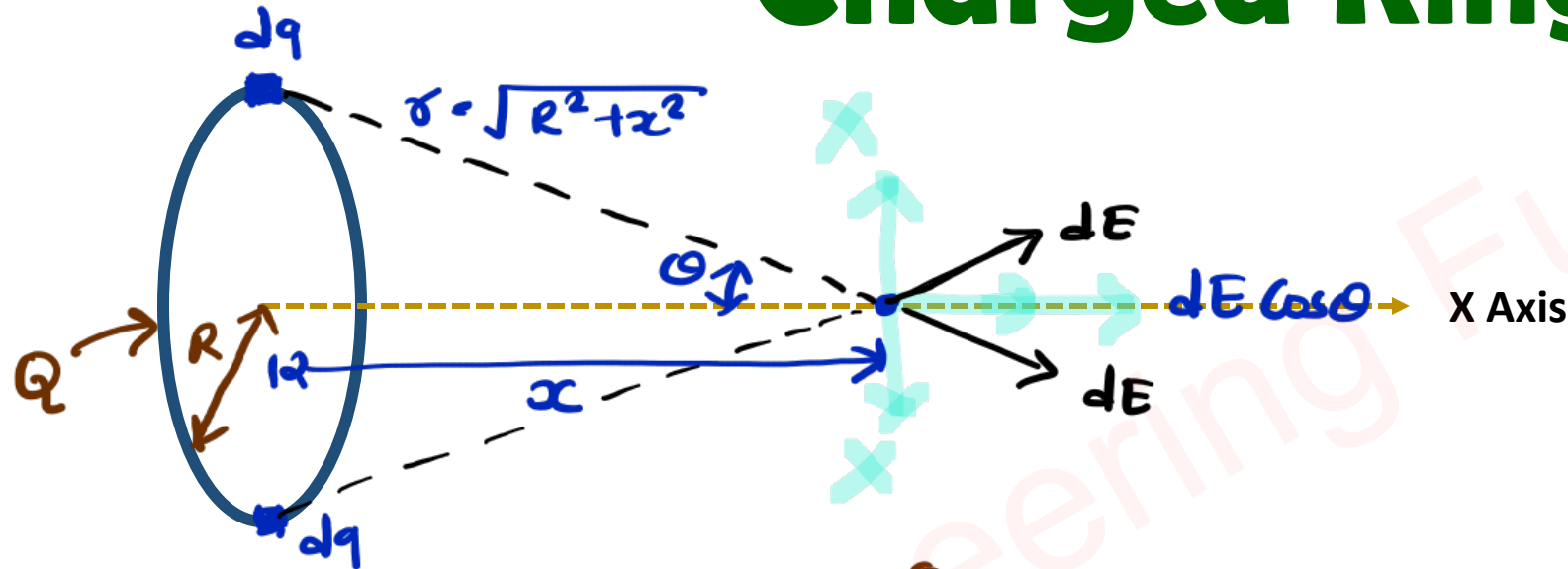
$$\Rightarrow \frac{F}{A} = \rho_{s2} \left(\frac{\rho_{s1}}{2\epsilon_0} \right)$$

$$\Rightarrow \frac{F}{A} = \frac{\rho_{s1} \rho_{s2}}{2\epsilon_0}$$

$$\Rightarrow \frac{F}{A} = \frac{4 \times 10^{-12}}{2 \times 8.854 \times 10^{-12}}$$

$$\Rightarrow \frac{F}{A} = 0.225 \text{ N/m}^2$$

Electric Field on the Axis of Uniformly Charged Ring



$$\Rightarrow \cos \theta = \frac{x}{\sqrt{R^2 + x^2}}$$

$$\begin{aligned} \Rightarrow \hat{r} &= \cos \theta \hat{i}_x \\ &= \frac{x}{\sqrt{R^2 + x^2}} \hat{i}_x \end{aligned}$$

→ line charge density $\lambda_L = \frac{Q}{2\pi R}$

→ Electric field due to point charge dq .

$$dE = k \frac{dq}{r^2} \hat{r}$$

→ Net Electric field

$$\vec{E}_{\text{net}} = \int d\vec{E}$$

$$= K \frac{Q}{r^2} \left(\frac{x}{\sqrt{R^2 + x^2}} \right) \hat{i}_x$$

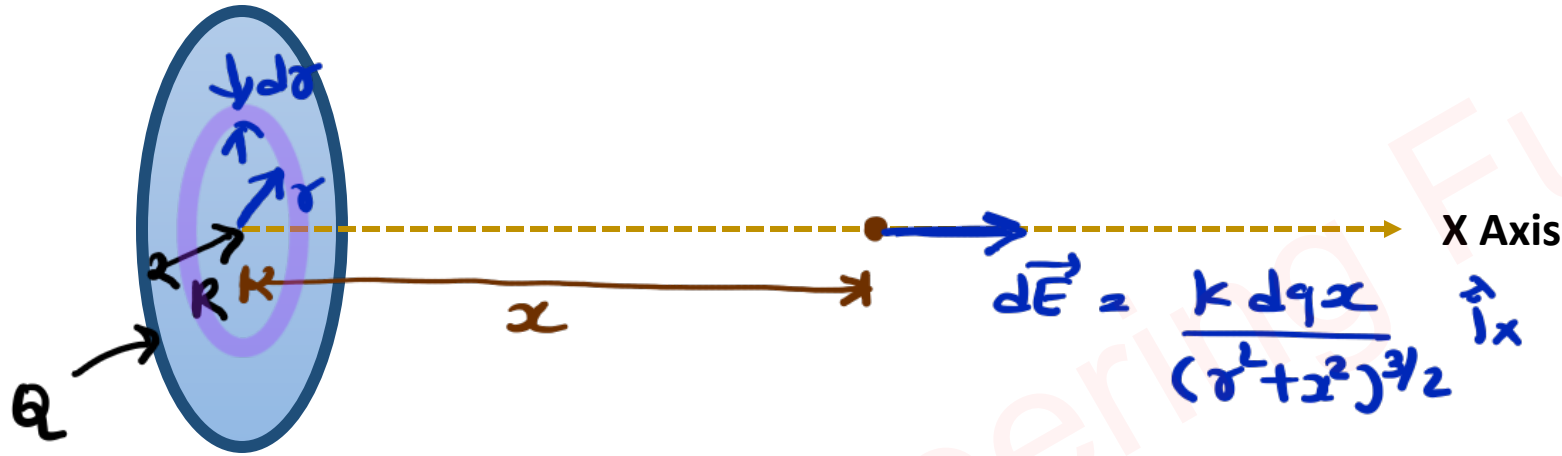
$$= K \frac{Qx}{(R^2 + x^2)^{3/2}} \hat{i}_x$$

$$= K \frac{\int_L (2\pi R) x}{(R^2 + x^2)^{3/2}} \hat{i}_x$$

$$= \left(\frac{1}{4\pi\epsilon_0} \right) \frac{\int_L (2\pi R) x}{(R^2 + x^2)^{3/2}} \hat{i}_x$$

$$= \frac{\int_L R x}{2\epsilon_0 (R^2 + x^2)^{3/2}} \hat{i}_x$$

Electric Field on the Axis of Uniformly Charged Disc



$$\sim \text{Surface charge } \sigma_s = \frac{Q}{\pi R^2}$$

$$\sim dq = \sigma_s (2\pi r dr)$$

$$\sim d\vec{E} = \frac{k x (2\pi r dr) \sigma_s}{(r^2 + x^2)^{3/2}} = \frac{1}{2 \cancel{4} \pi \epsilon_0} \frac{x (\cancel{2\pi} r dr) \sigma_s}{(r^2 + x^2)^{3/2}} = \frac{x \sigma_s r dr}{2 \epsilon_0 (r^2 + x^2)^{3/2}} \hat{i}_x$$

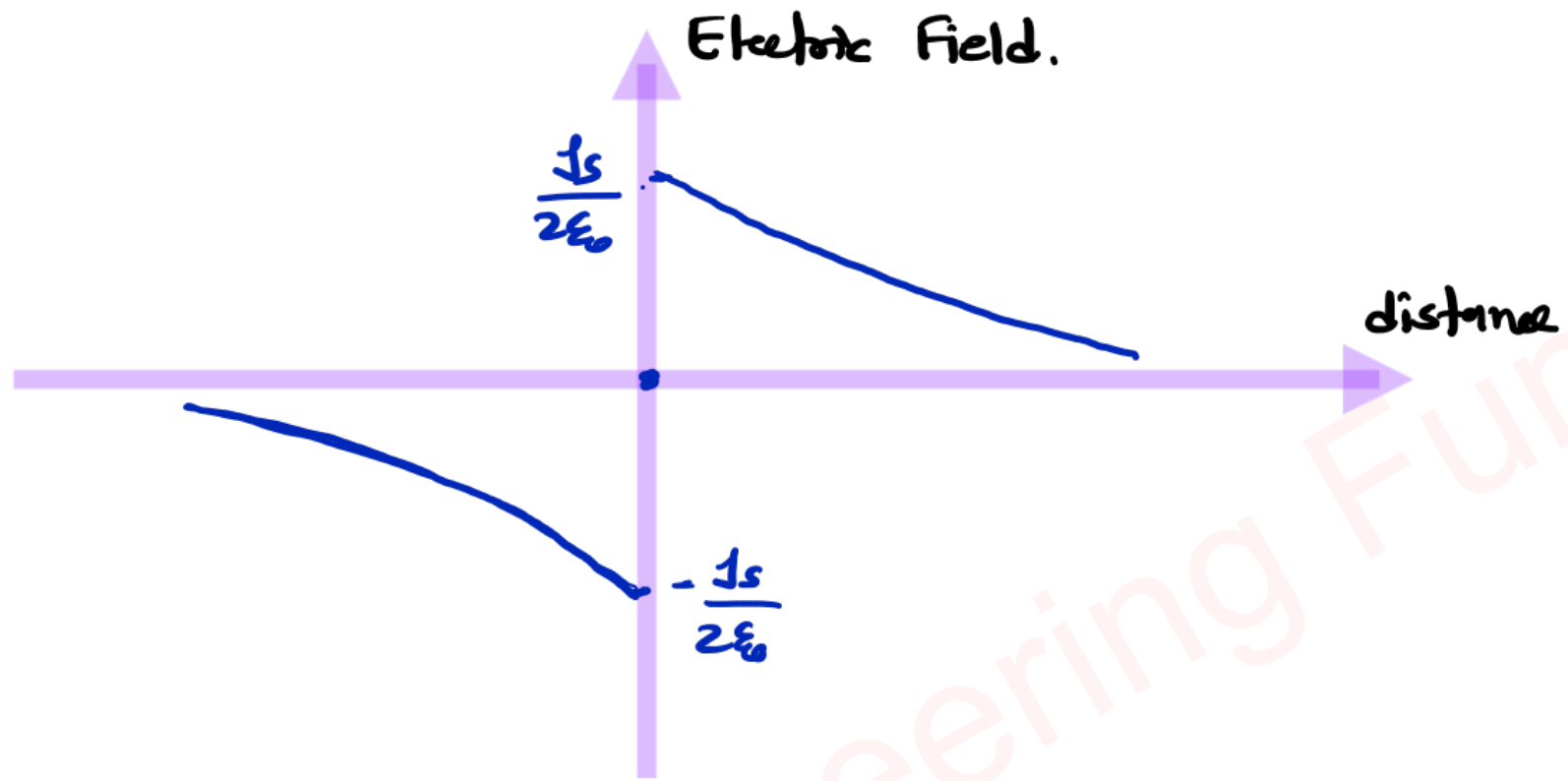
→ For Net Electric field $r \in (0, R)$

$$\Rightarrow \vec{E} = \int_0^R \frac{x \cdot \sigma \cdot d\sigma}{2\epsilon_0 (r^2 + x^2)^{3/2}} \hat{i}_x$$
$$= \frac{x \cdot \sigma}{2\epsilon_0} \int_0^R \frac{\sigma \cdot d\sigma}{(r^2 + x^2)^{3/2}} \hat{i}_x$$

$$\rightarrow \text{Let } x^2 + r^2 = t^2 \Rightarrow 2r \cdot dr = 2t \cdot dt \quad \left| \begin{array}{l} r \in (0, R) \\ t \in (x, \sqrt{x^2 + R^2}) \end{array} \right.$$
$$\Rightarrow r \cdot dr = t \cdot dt$$

$$\rightarrow \int_0^R \frac{r \cdot dr}{(r^2 + x^2)^{3/2}} = \int_x^{\sqrt{x^2 + R^2}} \frac{t \cdot dt}{t^3} = \int_x^{\sqrt{x^2 + R^2}} t^{-2} \cdot dt = \left[\frac{t^{-1}}{-1} \right]_x^{\sqrt{x^2 + R^2}} = \left[\frac{1}{x} - \frac{1}{\sqrt{x^2 + R^2}} \right]$$

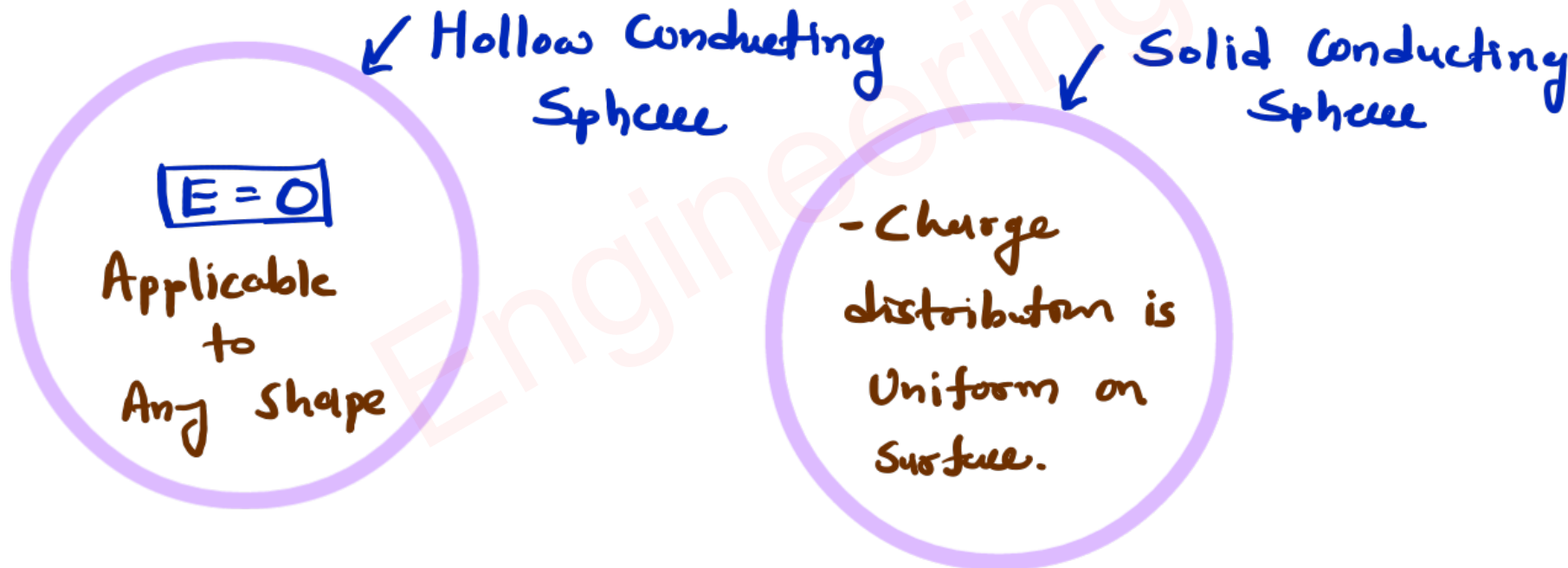
$$\Rightarrow \vec{E} = \frac{x \cdot \sigma}{2\epsilon_0} \left[\frac{1}{x} - \frac{1}{\sqrt{x^2 + R^2}} \right] = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{x^2 + R^2}} \right] \hat{i}_x$$



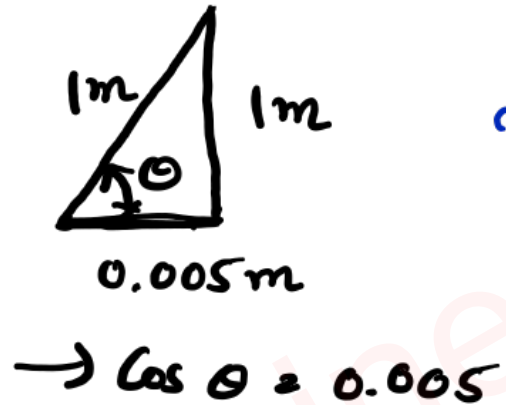
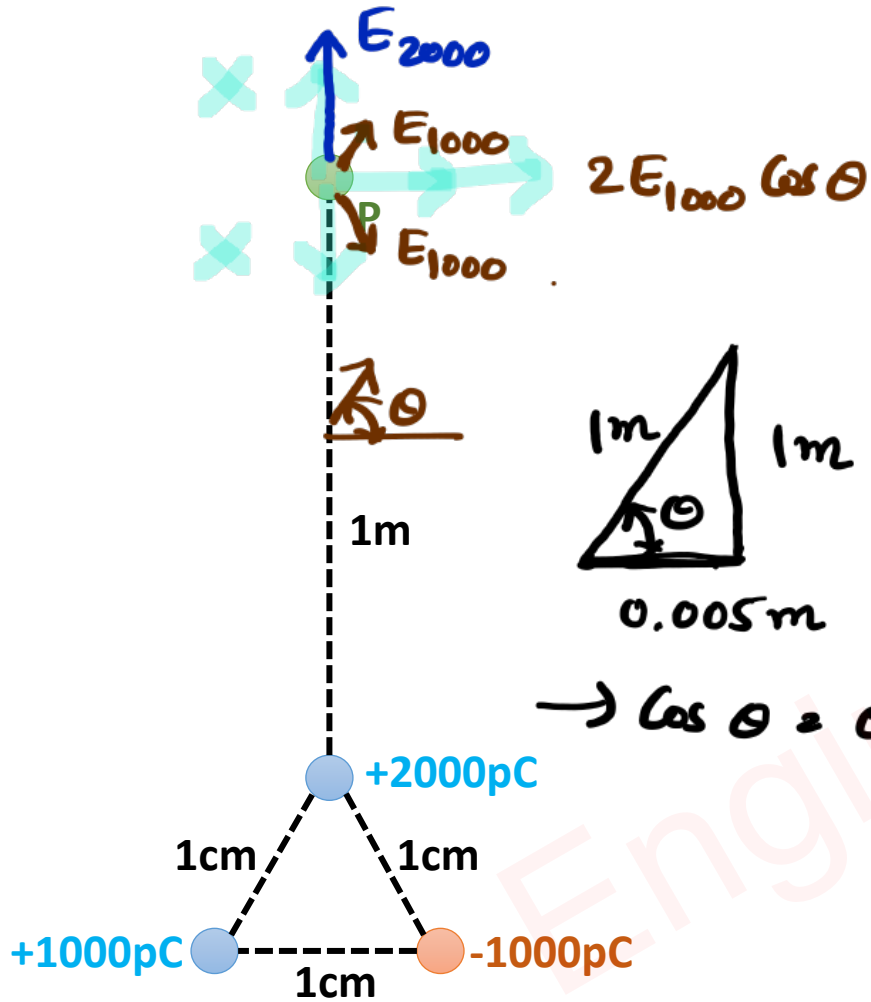
Examples of Electric Field

Question 1 – Inside a hollow conducting sphere

- ✓ a. Electric Field is zero
- b. Electric Field is non-zero constant
- c. Electric Field changes with the magnitude of the charge given to the conductor
- d. Electric Field changes with distance from the centre of the sphere



Question 2 – Determine the electric field intensity at the point p for the arrangement shown in the figure.



$$\begin{aligned} \Rightarrow E_x &= 2 E_{1000} \cos \theta \\ &= 2 \left(k \frac{q}{r^2} \right) \cos \theta \\ &= 2 \times 9 \times 10^9 \times \frac{1000 \times 10^{-12}}{1^2} \times 0.005 \\ &= 0.09 \hat{i}_x \end{aligned}$$

$$\begin{aligned} \Rightarrow E_y &= E_{2000} = k \frac{q}{r^2} \\ &= 9 \times 10^9 \times \frac{2000 \times 10^{-12}}{1^2} \\ &= 18 \hat{i}_y \end{aligned}$$

$$\begin{aligned} \Rightarrow \vec{E} &= E_x \hat{i}_x + E_y \hat{i}_y \\ &= (0.09 \hat{i}_x + 18 \hat{i}_y) \text{ N/C, V/m} \end{aligned}$$

Question 3 – The electric field $\vec{E}$ at the point $(1, 1, 0)$ due to a point charge of $1\mu\text{C}$ located at $(-1, 1, 1)$ is

$$\begin{array}{c} \bullet \\ 1\mu\text{C} \\ (-1, 1, 1) \end{array}$$

$$\begin{array}{c} \bullet \rightarrow \vec{E} \\ (1, 1, 0) \end{array}$$

$$\begin{aligned} \rightarrow \vec{r} &= (1, 1, 0) - (-1, 1, 1) \\ &= (2, 0, -1) \end{aligned}$$

$$\rightarrow r = \sqrt{2^2 + 0^2 + 1^2} = \sqrt{5}$$

$$\therefore \vec{E} = k \frac{q}{r^2} \hat{r} = k \frac{q}{r^3} \vec{r}$$

$$= \frac{9 \times 10^9 \times 10^{-6}}{(\sqrt{5})^3} (2, 0, -1)$$

$$= 804 (2, 0, -1)$$

$$= (1608, 0, -804) \text{ N/C or V/m}$$

Question 4 – If the $\vec{E}$ is the electric field intensity, $\nabla \cdot (\nabla \times \vec{E})$ is equal to

- a. $\vec{E}$
- b. $|\vec{E}|$
- c. Null vector = (0, 0, 0)
- ✓ d. Zero = 0

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0$$

$$\nabla \times (\nabla F) = (0, 0, 0)$$

Electric Flux Density

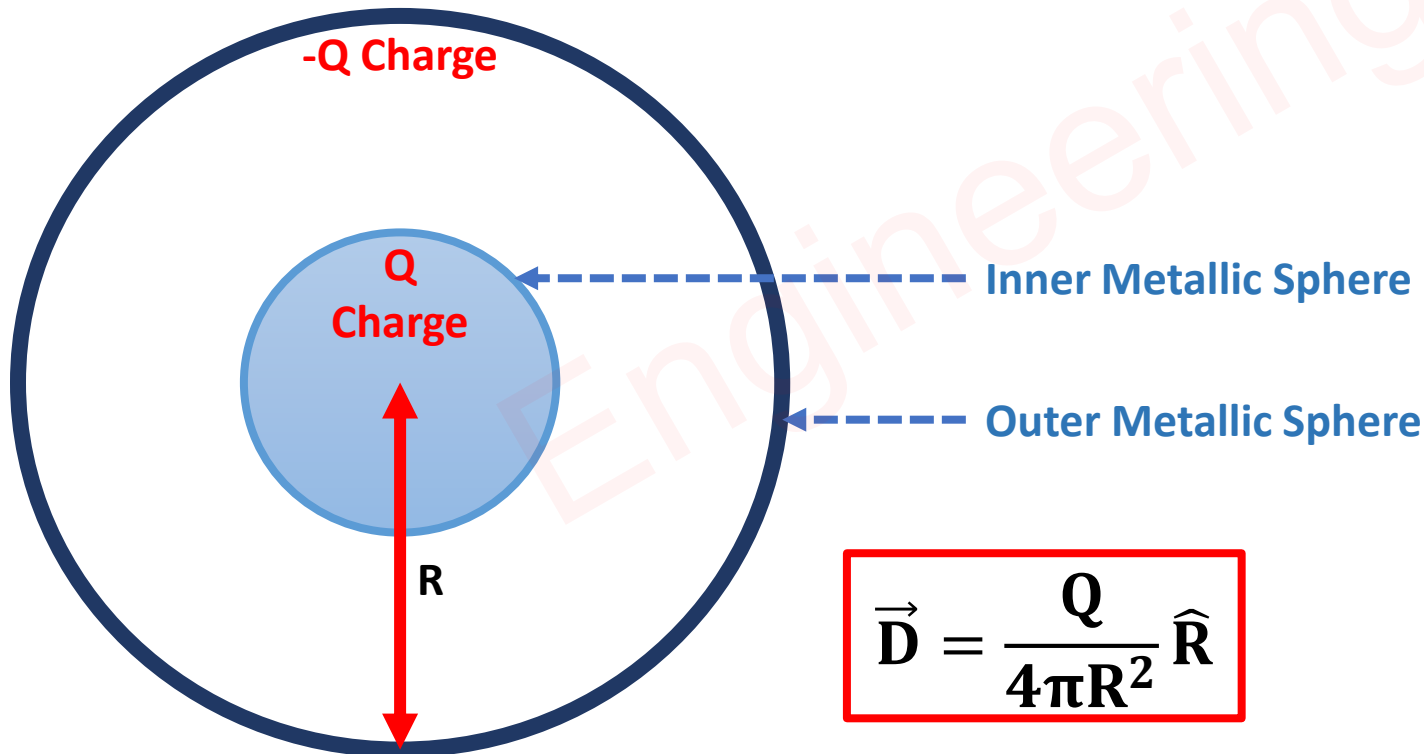
Outlines

- Faraday's Experiment to understand Electric Flux Density
- Measurement of Electric Flux Density
- Relation between Electric Flux Density and Electric Field

Engineering Funda

Faraday's Experiment to Understand Electric Flux Density

- Two Metallic Spheres are separated by Dielectric Material (Air).
- Q Charge is displaced in the Inner Metallic Sphere.
- After that, Faradays Observed that the $-Q$ charge was on the Outer Metallic Sphere.
- **Electric Flux Density $\vec{D}$** is the charge displaced per unit surface area.
- Direction of Electric Flux Density $\vec{D}$ is in Radial Direction.
- Unit of Electric Flux Density $\vec{D}$ is C/m^2 .



$$\vec{D} = \frac{Q}{4\pi R^2} \hat{R}$$

Relation between Electric Flux Density & Electric Field

- Electric Flux Density $\vec{D}$ is given by:

$$\vec{D} = \frac{Q}{4\pi R^2} \hat{R}$$

- Electric Field $\vec{E}$ is given by:

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 R^2} \hat{R}$$

- Hence, Relation between Electric Flux Density $\vec{D}$ and Electric Field $\vec{E}$ is given by:

$$\vec{D} = \epsilon_0 \vec{E}$$

- Hence, Generalized Relation between Electric Flux Density $\vec{D}$ and Electric Field $\vec{E}$ is given by:

$$\vec{D} = \epsilon_r \epsilon_0 \vec{E}$$

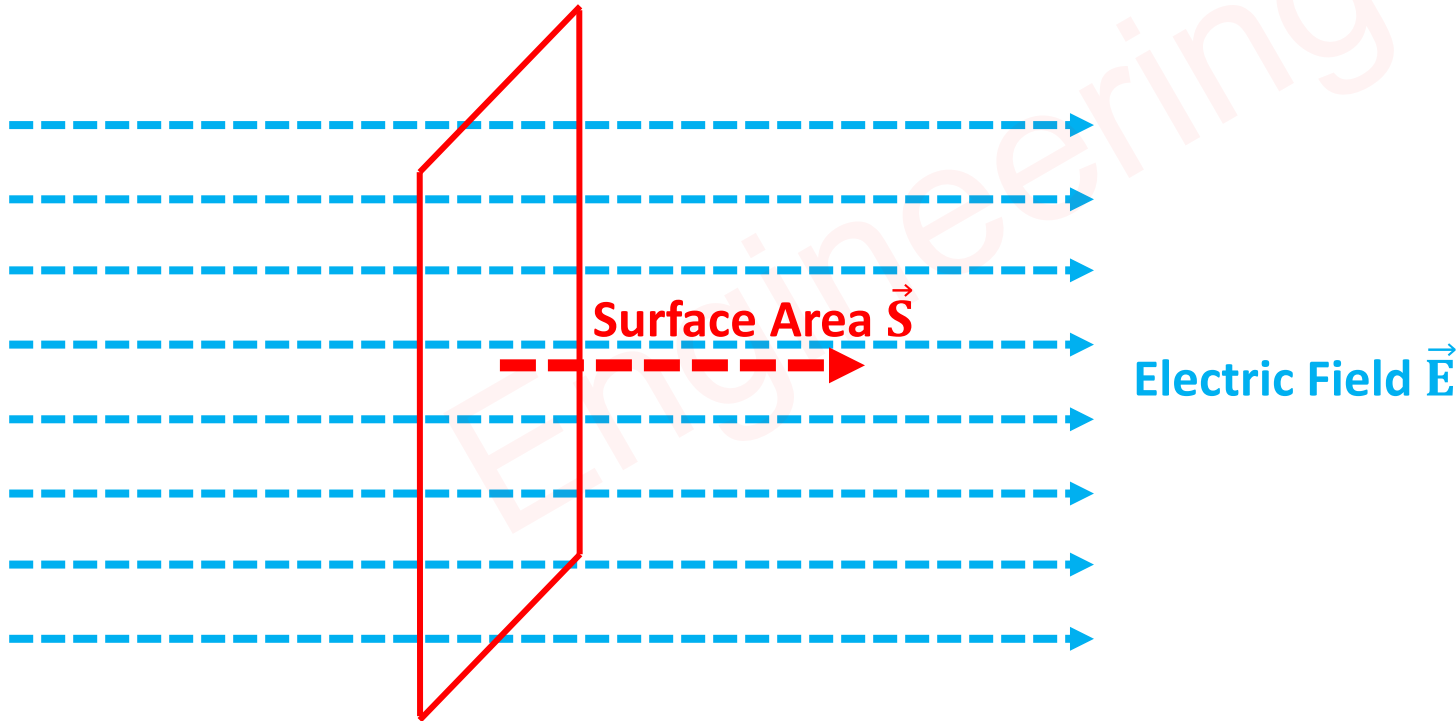
Electric Flux

Basics of Electric Flux

- Electric Field passing through any surface is Electric Flux ϕ .

$$\phi = \vec{E} \cdot \vec{S} = ES \cos \theta$$

- Unit of Electric Flux ϕ is Weber (Wb) or Vm.



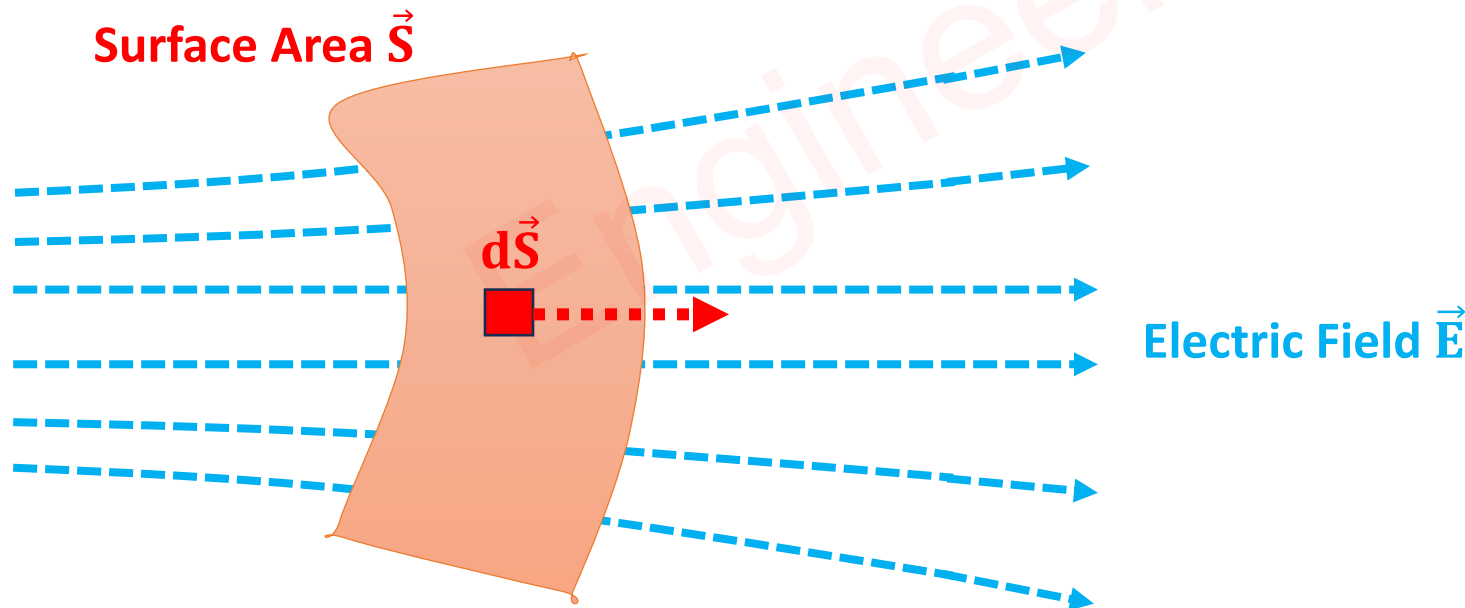
Calculation of Electric Flux

- For a Calculation of Electric Flux ϕ , Non-Uniform Electric Field $\vec{E}$ and uneven surface is considered.
- For a small differential area $d\vec{S}$, differential flux $d\phi$ is given by:

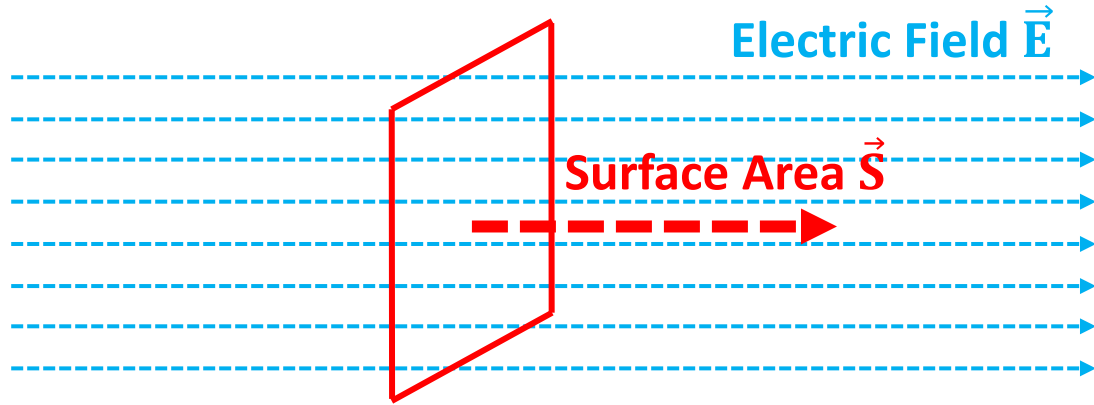
$$d\phi = \vec{E} \cdot d\vec{S}$$

- Hence total Flux ϕ is given by:

$$\phi = \int d\phi = \int \vec{E} \cdot d\vec{S}$$

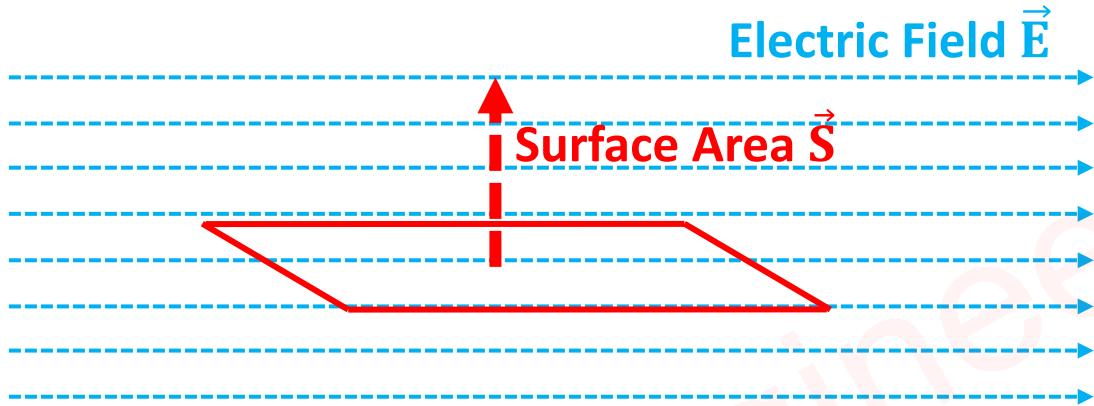


Case Studies of Electric Flux



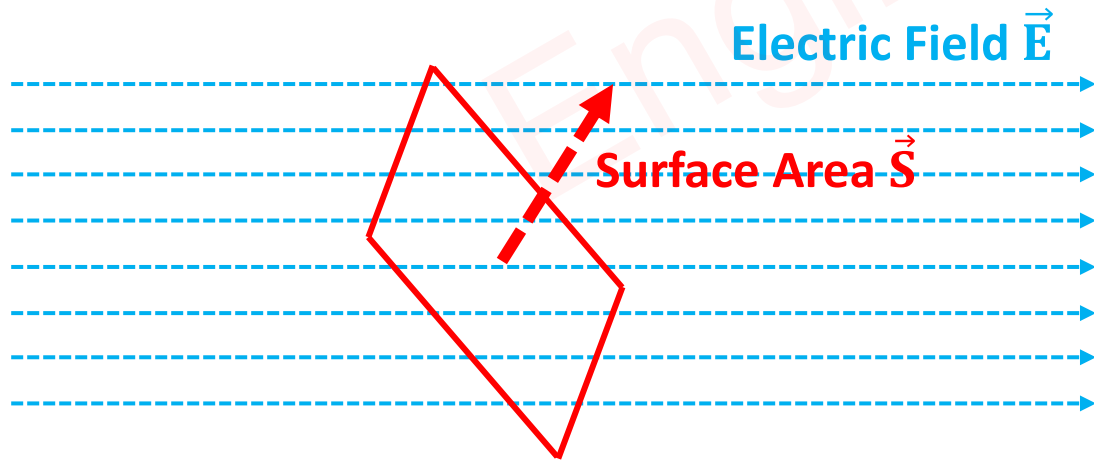
- Angle between $\vec{E}$ and $\vec{S}$ is 0° .

$$\phi = \vec{E} \cdot \vec{S} = ES \cos \theta = ES$$



- Angle between $\vec{E}$ and $\vec{S}$ is 90° .

$$\phi = \vec{E} \cdot \vec{S} = ES \cos \theta = 0$$



- Angle between $\vec{E}$ and $\vec{S}$ is θ .

$$\phi = \vec{E} \cdot \vec{S} = ES \cos \theta$$

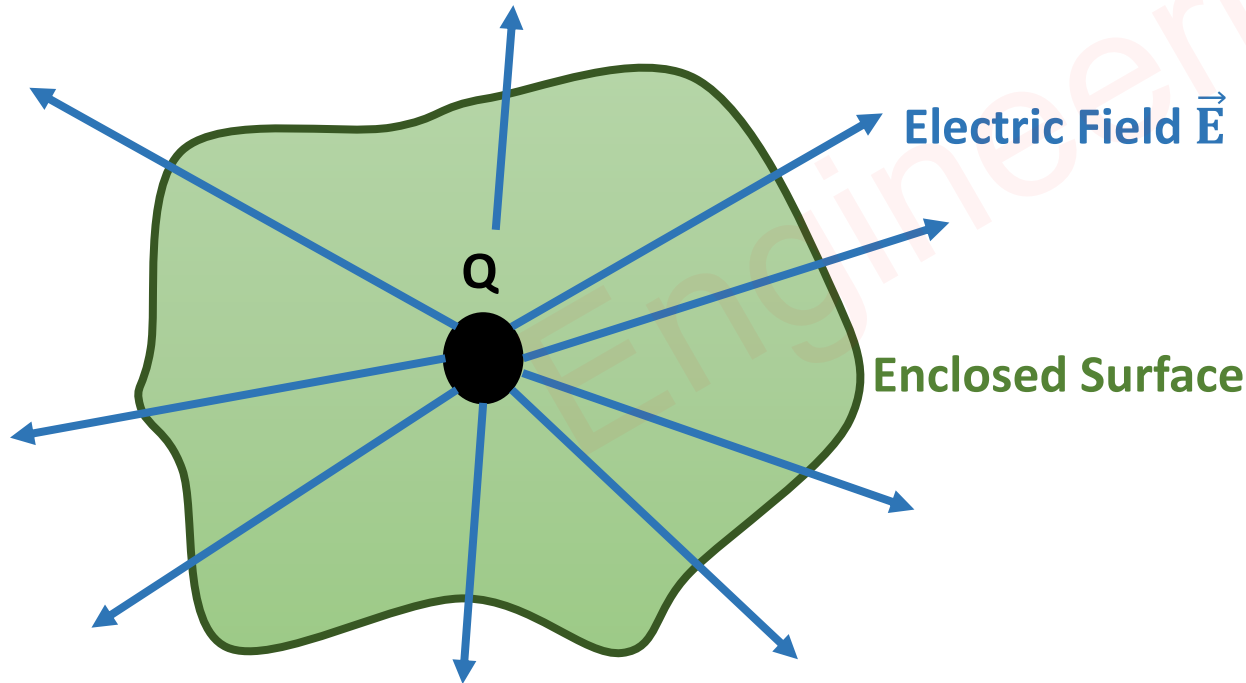
Gauss's Law for Electric Field

Statement

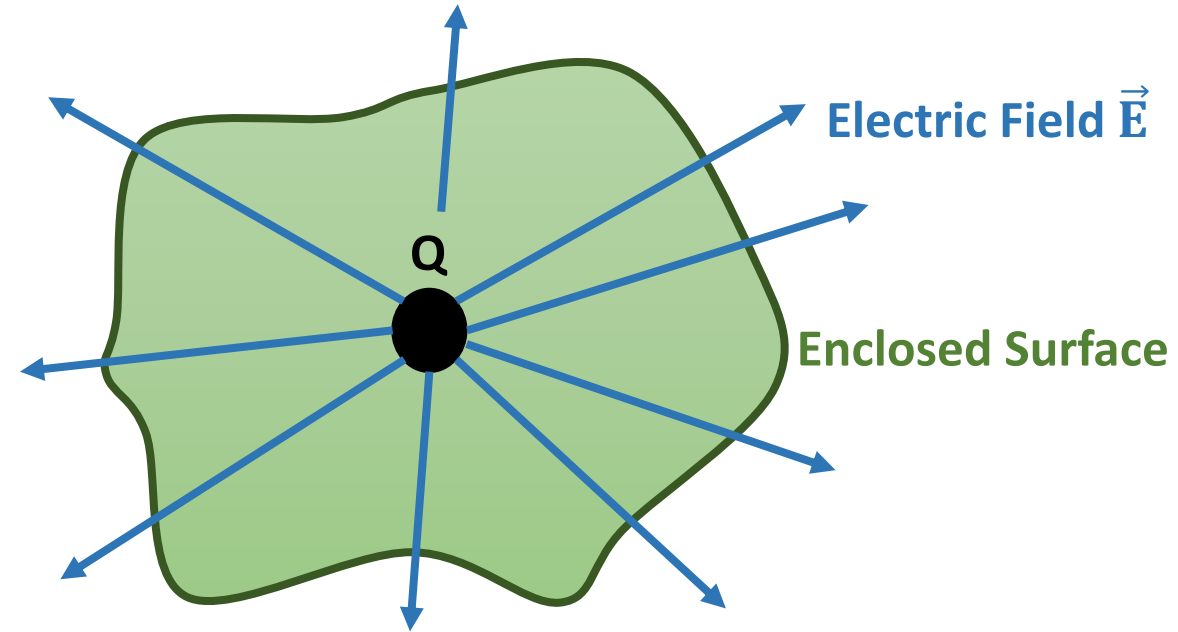
- It states that the total Electric Flux passing through any closed surface is equal to the enclosed charge divided by ϵ_0 .

$$\Phi = \int \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0}$$

$$\Phi = \int \vec{D} \cdot d\vec{S} = Q$$



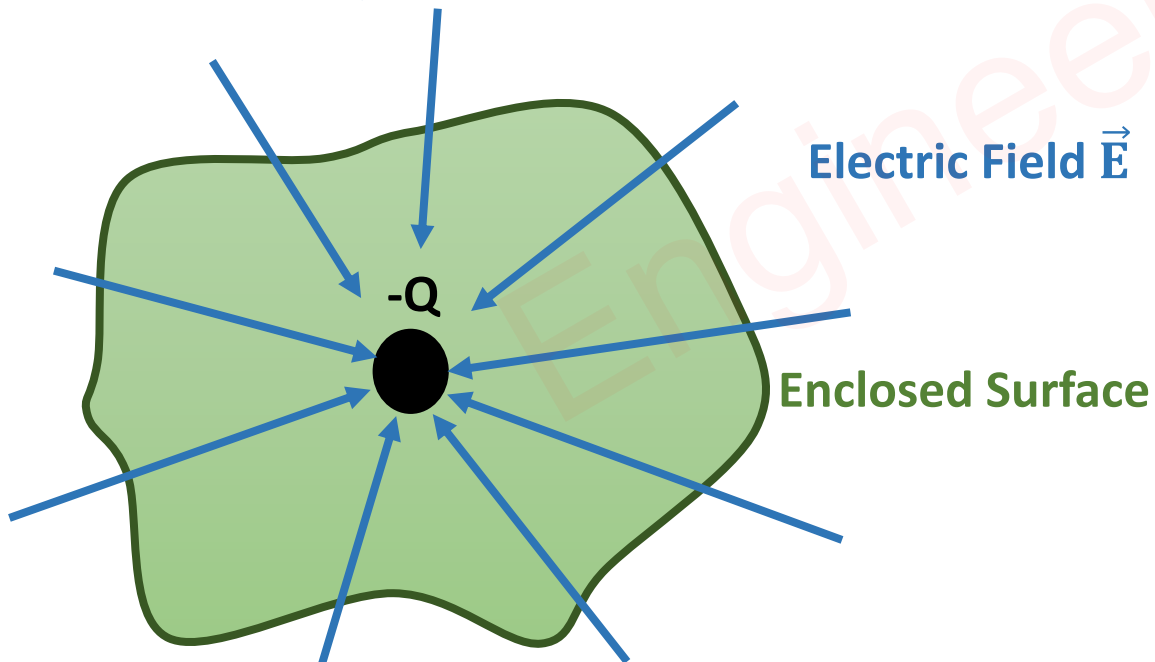
Case Studies of Gauss's Law of Electric Field



- If the charge is positive, then Electric Flux leaves the enclosed surface.

$$\Phi = \int \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0}$$

$$\Phi = \int \vec{D} \cdot d\vec{S} = Q$$

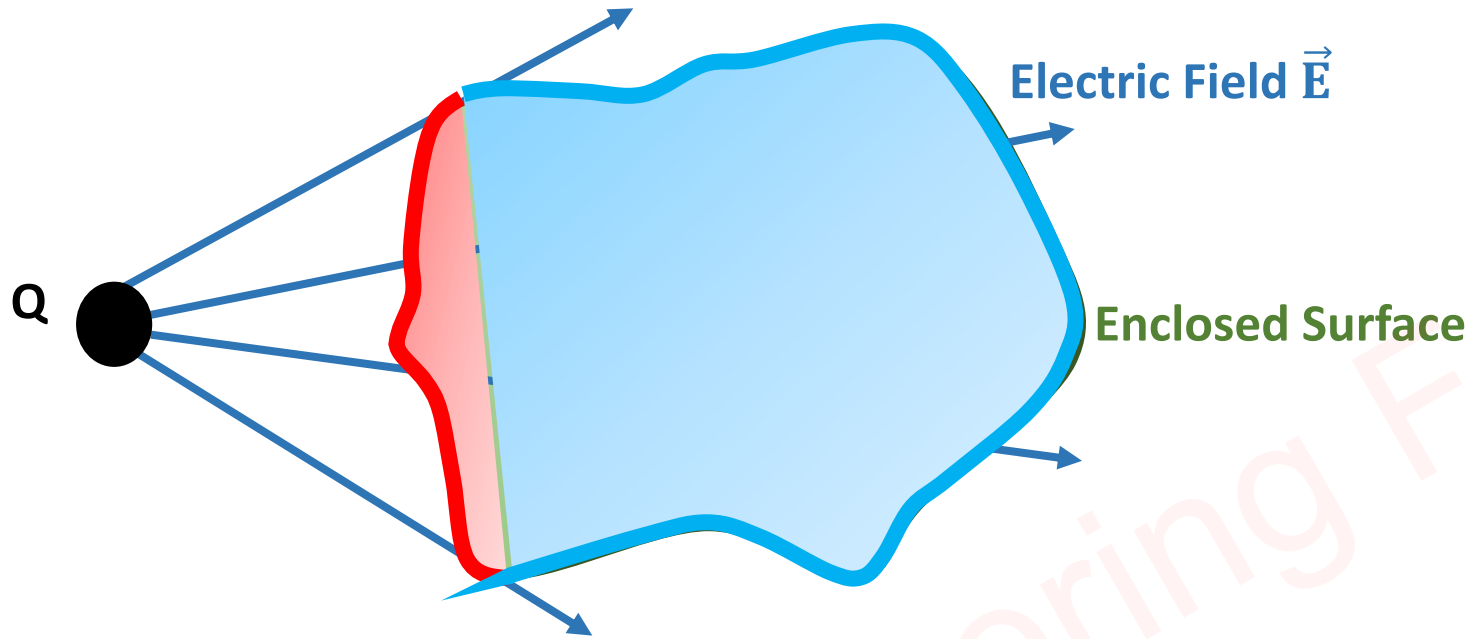


- If the charge is negative, then Electric Flux enters the enclosed surface.

$$\Phi = \int \vec{E} \cdot d\vec{S} = \frac{-Q}{\epsilon_0}$$

$$\Phi = \int \vec{D} \cdot d\vec{S} = -Q$$

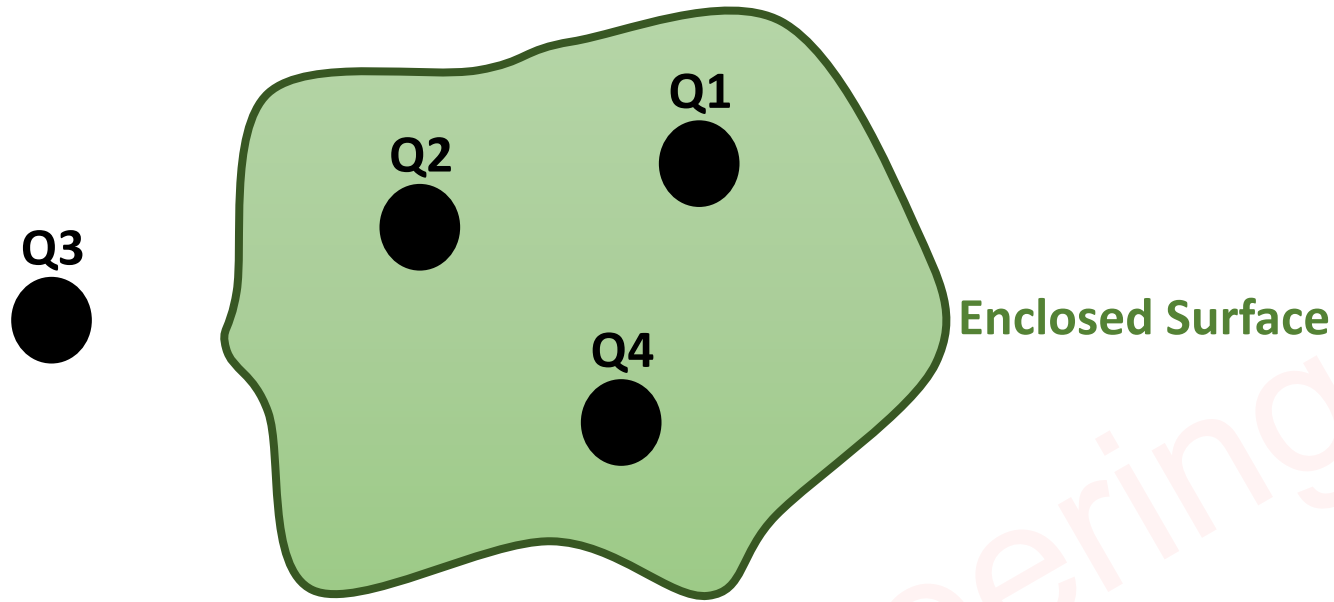
Case Studies of Gauss's Law of Electric Field



- Entering to the surface flux is Negative.
- Leaving to the surface flux is Positive.
- Hence, Total Flux is zero if the charge is located outside the surface.

Example of Gauss's Law of Electric Field

- Calculate the total Electric Flux ϕ for the given figure below.



$$\phi = \int \vec{E} \cdot d\vec{S} = \frac{[Q1 + Q2 + Q4]}{\epsilon_0}$$

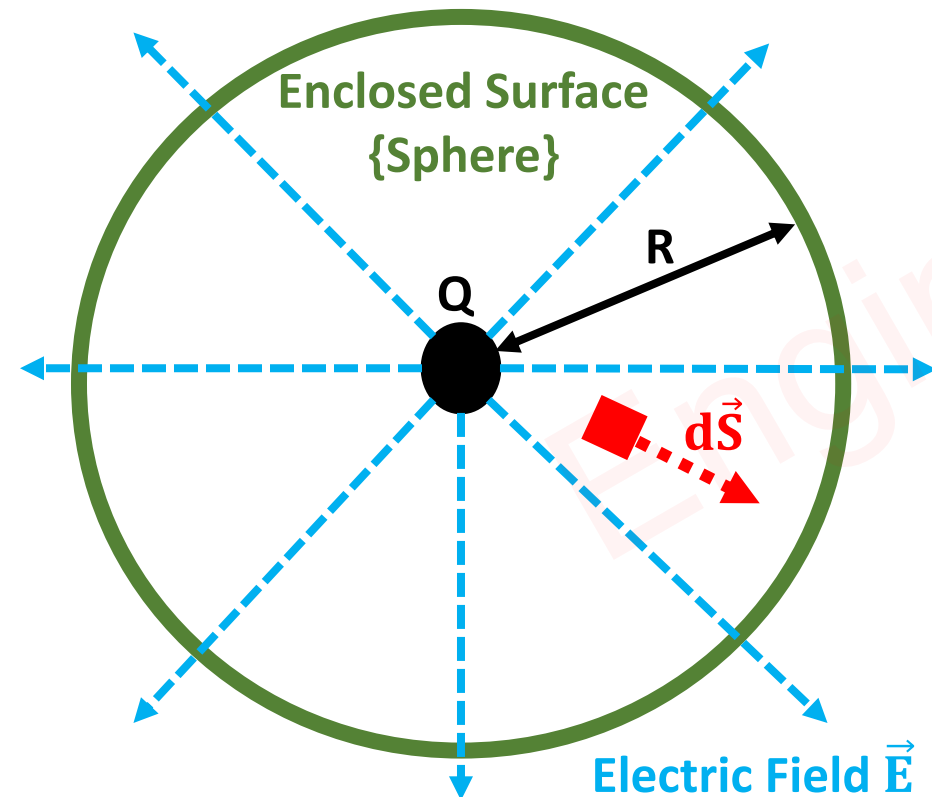
$$\phi = \int \vec{D} \cdot d\vec{S} = [Q1 + Q2 + Q4]$$

Applications of Gauss's Law

Electric Field due to point Charge

- Electric Field due to point charge Q at distance R is given by:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} \hat{R}$$



- As per Gauss's Law of Electric Flux is given by:

$$\Rightarrow \Phi = \oint \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0} \quad [\because d\vec{S} \parallel \vec{E} \Rightarrow \theta = 0]$$

$$\Rightarrow E(4\pi R^2) = \frac{Q}{\epsilon_0}$$

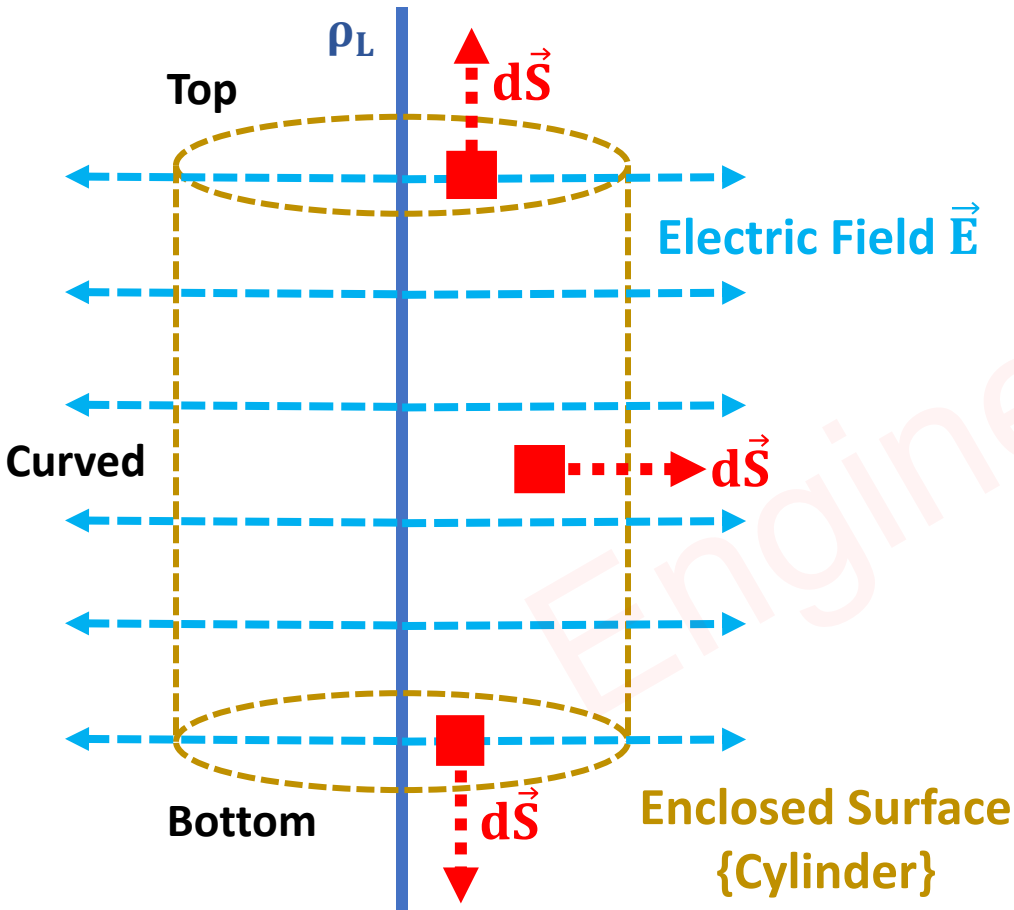
$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}$$

$$\Rightarrow \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} \hat{R}$$

Electric Field due to Line Charge

- Electric Field due to Line charge ρ_L at distance R is given by:

$$\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 R} \hat{R}$$



- As per Gauss's Law of Electric Flux is given by:

$$\Rightarrow \Phi = \oint \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0}$$

$$\Rightarrow \Phi = \int_{\text{Top}} \vec{E} \cdot d\vec{S} + \int_{\text{Bottom}} \vec{E} \cdot d\vec{S} + \int_{\text{Curved}} \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0}$$

$$\Rightarrow \Phi = \int_{\text{Curved}} \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0}$$

$$\Rightarrow E(2\pi R)L = \frac{Q}{\epsilon_0} \quad \left[\because \rho_L = \frac{Q}{L} \right]$$

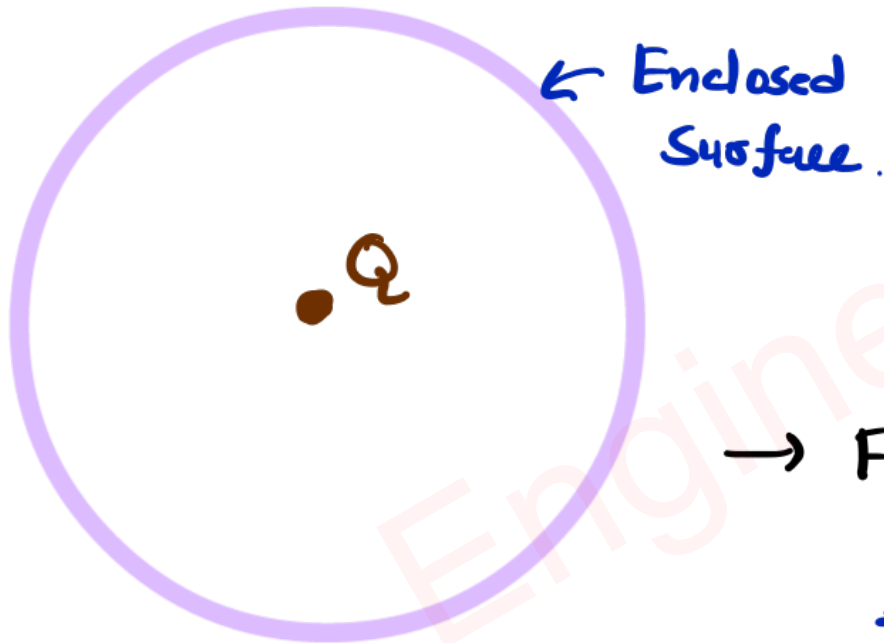
$$\Rightarrow E = \frac{\rho_L}{2\pi\epsilon_0 R}$$

$$\Rightarrow \vec{E} = \frac{\rho_L}{2\pi\epsilon_0 R} \hat{R}$$

Maxwell's 1st Equation

→ As per Gauss's Law of E field.

$$\Rightarrow \oint \vec{D} \cdot d\vec{S} = Q \text{ --- (1)}$$



→ If Charge Q is Volume Charge.

$$\Rightarrow Q = \int \rho_v dV \text{ --- (2)}$$

→ From eq.ⁿ (1) & (2)

$$\Rightarrow \oint \vec{D} \cdot d\vec{S} = \int \rho_v dV \text{ --- (A)}$$

→ Equation (A) is Integral form of Maxwell's 1st eq.ⁿ

→ As per Divergence Theorem

$$\Rightarrow \int \vec{F} \cdot d\vec{S} = \int \vec{\nabla} \cdot \vec{F} dV \text{ --- (3)}$$

→ From eqⁿ (3)

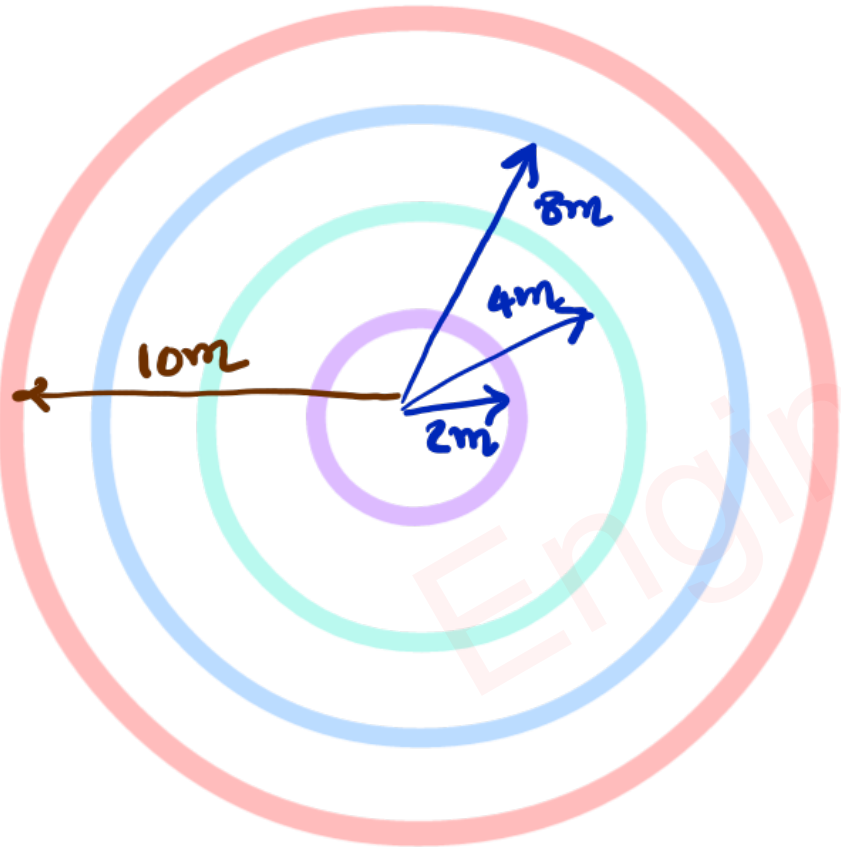
$$\Rightarrow \int \vec{D} \cdot d\vec{S} = \int \vec{\nabla} \cdot \vec{D} dV = \int \rho_v dV$$

$$\Rightarrow \vec{\nabla} \cdot \vec{D} = \rho_v \text{ --- (B)}$$

→ eqⁿ (B) is point form or differential form of Maxwell's 1st eqⁿ.

Examples of Gauss's Law

Question 1 – Concentric Spherical shells of radii 2m, 4m, and 8m carry uniform surface charge densities of 20 nC/m^2 , -4 nC/m^2 , and ρ_s , respectively. The value of ρ_s required to ensure that the electric flux density $D = 0$ at radius 10m is nC/m^2



⇒ As per Gauss law of Electric field

$$\Rightarrow \oint \vec{D} \cdot d\vec{S} = Q$$

$$\Rightarrow Q = 0$$

$$\Rightarrow Q_1 + Q_2 + Q_3 = 0$$

$$\Rightarrow (20 \text{ nC/m}^2) (\cancel{4\pi} 2^2) + (-4 \text{ nC/m}^2) (\cancel{4\pi} 4^2) + \rho_s (\cancel{4\pi} 8^2) = 0$$

$$\Rightarrow 80 - 64 + 64 \rho_s = 0$$

$$\Rightarrow \rho_s = \frac{-16}{64} = -\frac{1}{4} \text{ nC/m}^2$$

Question 2 – Which of the following equations represents the Gauss Law in a homogeneous isotropic medium?

☒ a. $\oint \mathbf{D} \cdot d\mathbf{s} = \int \rho_v dV$

b. $\vec{V} \times \vec{H} = \vec{D}$

c. $\vec{V} \cdot \vec{H} + \rho = 0$

d. $\vec{V} \cdot \vec{E} = \rho / \epsilon_0$

→ Integral Form (Gauss Law)

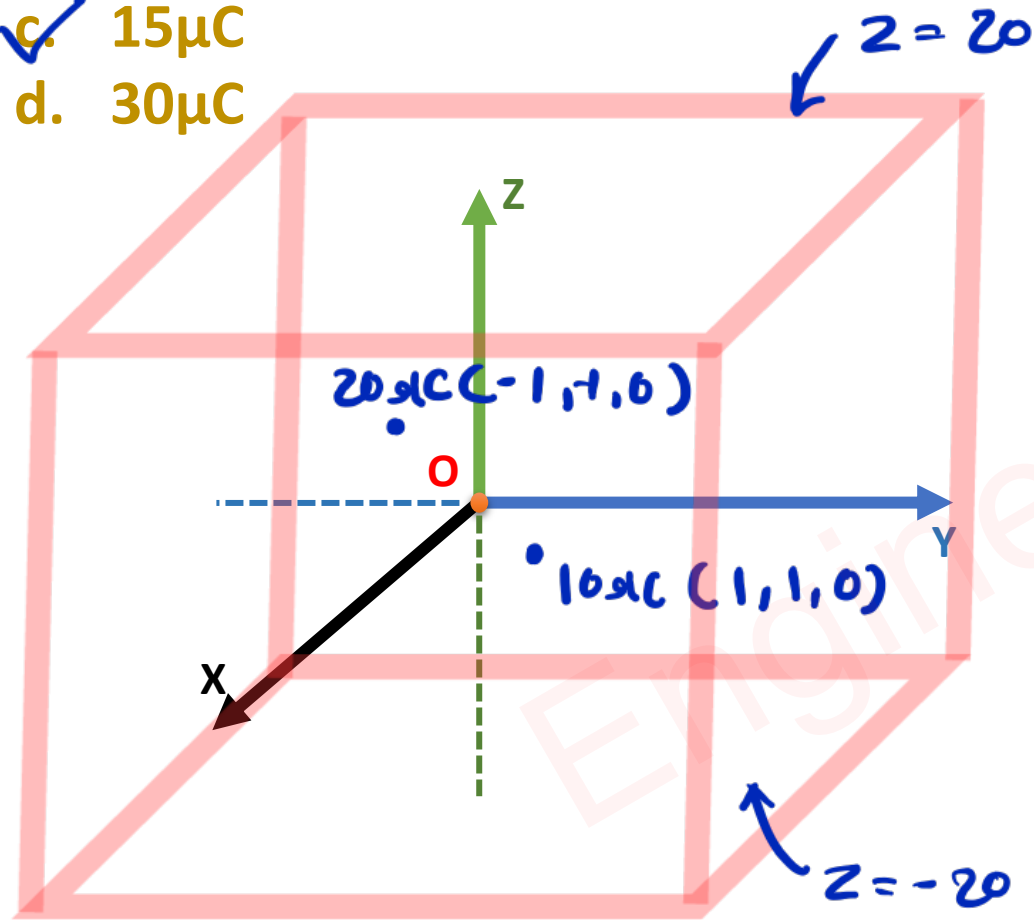
$$\Rightarrow \oint \vec{D} \cdot d\vec{s} = \int \rho_v dV \quad \text{or} \quad \oint \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} \int \rho_v dV$$

→ Point Form / Differential Form (Gauss Law)

$$\Rightarrow \vec{D} \cdot \vec{D} = \rho_v \quad \text{or} \quad \vec{D} \cdot \vec{E} = \frac{\rho_v}{\epsilon_0}$$

Question 3 – Two point charges $Q_1 = 10\mu\text{C}$ and $Q_2 = 20\mu\text{C}$ are placed at coordinates $(1,1,0)$ and $(-1,-1,0)$, respectively. The total electric flux passing through a plane $Z=20$ will be

- a. $7.5\mu\text{C}$
- b. $13.5\mu\text{C}$
- ✓ c. $15\mu\text{C}$
- d. $30\mu\text{C}$



⇒ As per Gauss Law of E field

$$\Rightarrow \phi = Q_{\text{enclosed}}$$

$$\Rightarrow \underbrace{\phi_F + \phi_B + \phi_R + \phi_L + \phi_{20} + \phi_{-20}}_{=0} = 30\mu\text{C}$$
$$\phi_{20} = \phi_{-20}$$

$$\Rightarrow 2\phi_{20} = 30\mu\text{C}$$

$$\Rightarrow \phi_{20} = 15\mu\text{C}$$

Question 4 – A hollow metallic sphere of radius r is kept at potential of 1 volt. The total electric flux coming out of the concentric spherical surface of radius $R(>r)$ is

- ✓ a. $4\pi\epsilon_0 r$
- b. $4\pi\epsilon_0 r^2$
- c. $4\pi\epsilon_0 R$
- d. $4\pi\epsilon_0 R^2$

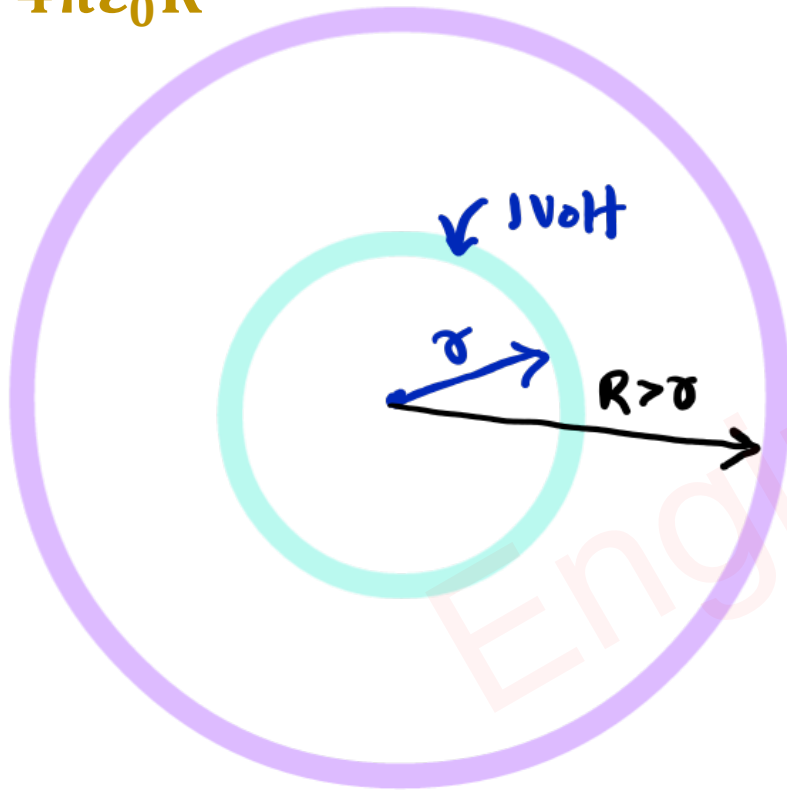
→ As per Gauss Law

$$\Rightarrow \phi = Q_{\text{enclosed}}$$

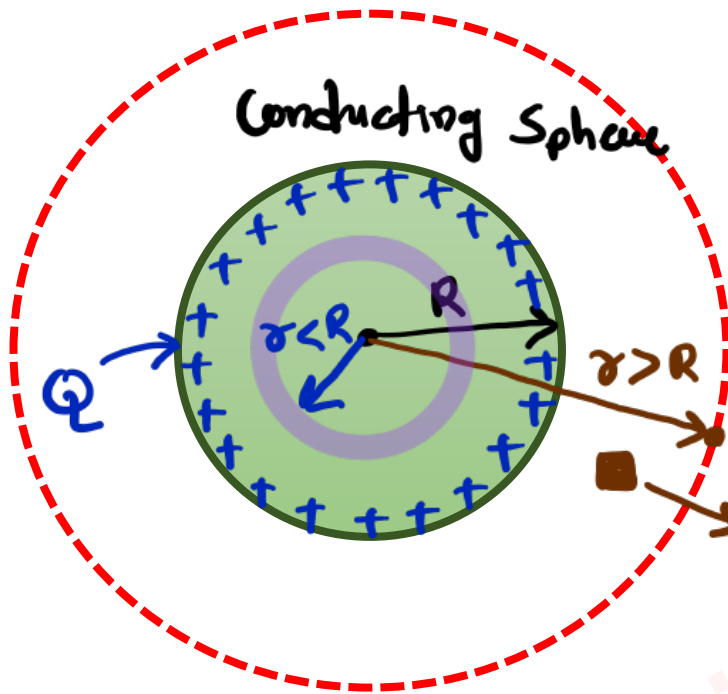
$$\Rightarrow V = k \frac{Q}{r}$$

$$\Rightarrow 1 = \left(\frac{1}{4\pi\epsilon_0} \right) \frac{Q}{r}$$

$$\Rightarrow \boxed{Q = 4\pi\epsilon_0 r}$$



Electric Field due to Conducting Sphere



→ Electric field

⇒ $r > R$ (Outside)

⇒ $r = R$ (On Surface)

⇒ $r < R$ (Inside)

→ Use-I, $r > R$ (Outside)

→ As per Gauss Law

$$\Rightarrow \phi = \oint \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0} \quad [\vec{E} \parallel d\vec{S} \Rightarrow \theta = 0]$$

$$\Rightarrow E (4\pi r^2) = \frac{Q}{\epsilon_0}$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

~ Case-II, $r = R$ (On Surface)

~ As per Gauss Law

$$\Rightarrow \phi = \oint \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$\Rightarrow E(4\pi R^2) = \frac{Q}{\epsilon_0}$$

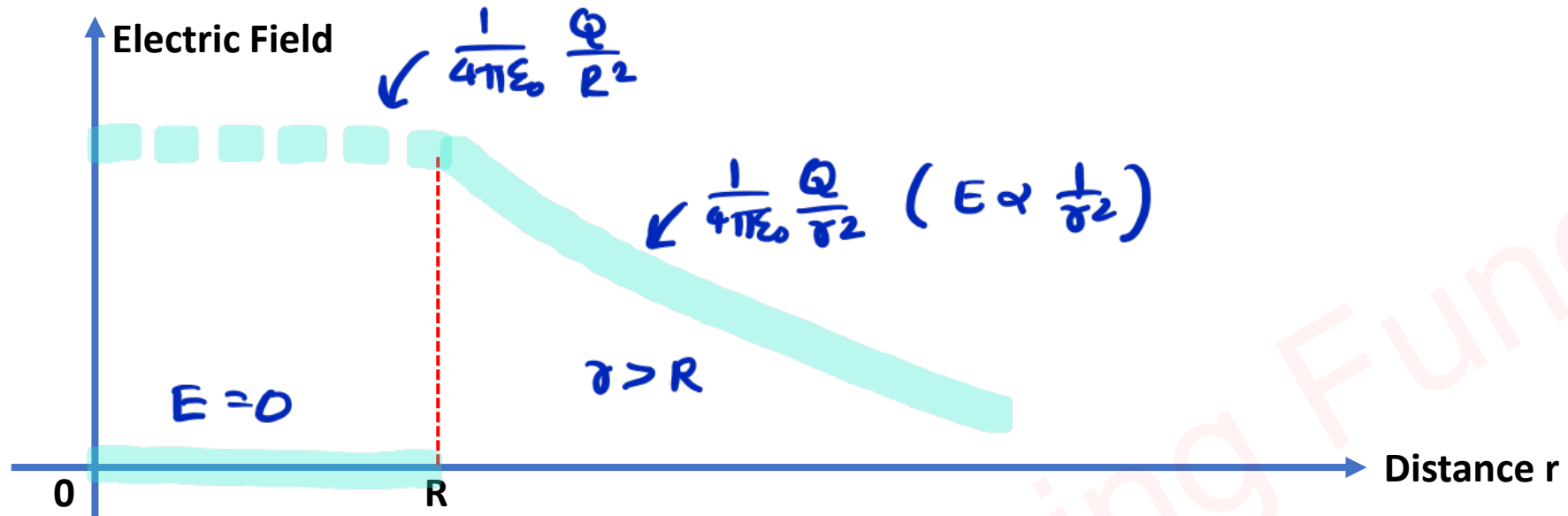
$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}$$

~ Case-III, $r < R$ (Inside Sphere)

~ As per Gauss Law.

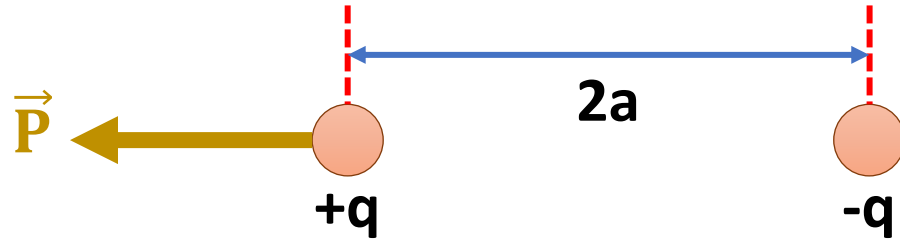
$$\Rightarrow \phi = \oint \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0} = 0$$

$$\Rightarrow E = 0$$



Electric Dipole & Dipole Moment

- Electric Dipole is a pair of charges with same magnitude and opposite polarity.



- Electric Dipole Moment is a product of charge and spacing between two charges for given Electric Dipole.

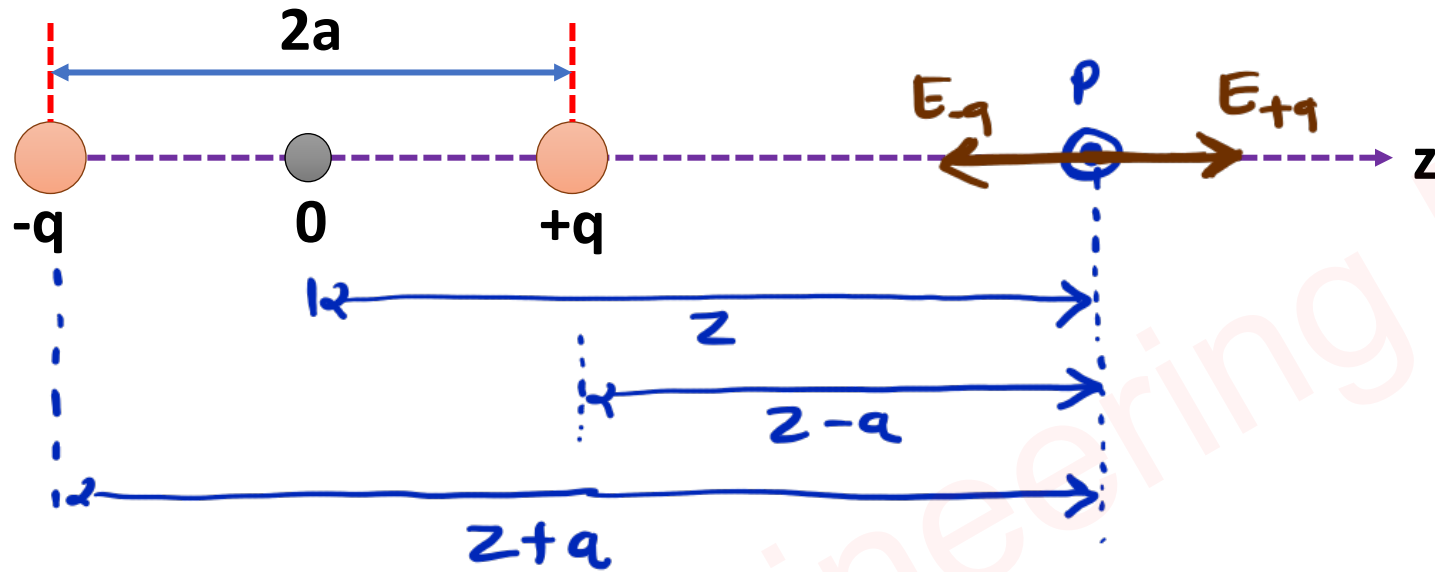
$$P = 2a \times q$$

- Direction of Electric Dipole Moment is from Negative charge to positive charge.

$$\vec{P} = 2\vec{a} \times q$$

Electric field due to Electric Dipole

Electric Field due to Dipole on the Axis of Dipole



∴ Hence, Total Electric field

$$\Rightarrow E = E_{+q} + E_{-q}$$

$$\Rightarrow E = k \frac{q}{(z-a)^2} - k \frac{q}{(z+a)^2}$$

$$\Rightarrow E = kq \left[\frac{1}{(z-a)^2} - \frac{1}{(z+a)^2} \right]$$

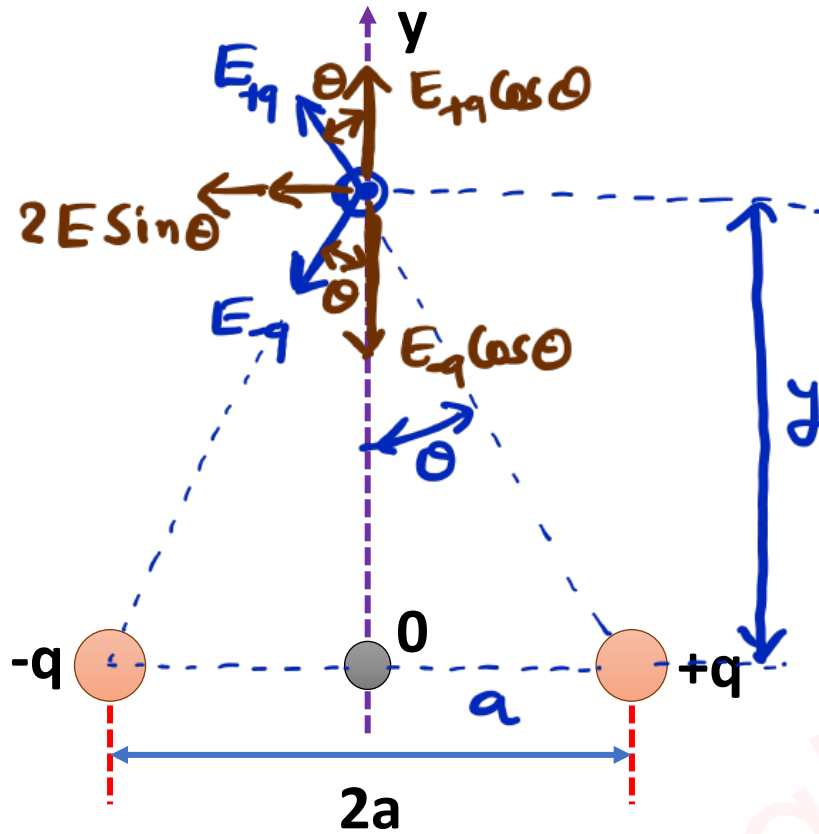
$$\Rightarrow E = kq \left[\frac{(z+a)^2 - (z-a)^2}{(z^2-a^2)^2} \right]$$

$$\Rightarrow E = kq \left[\frac{4za}{(z^2-a^2)^2} \right] \quad [z \gg a \Rightarrow (z^2-a^2)^2 = z^4]$$

$$\Rightarrow E = kq \left[\frac{4a}{z^3} \right] \quad [p = 2aq, \vec{p} = 2a\vec{1}q]$$

$$\Rightarrow E = \frac{2kp}{z^3} \quad \text{or} \quad \vec{E} = 2k \frac{\vec{p}}{z^3}$$

Electric Field due to Dipole on the Equator of Dipole



⇒ Total Electric field

$$\Rightarrow E = -2E \sin \theta$$

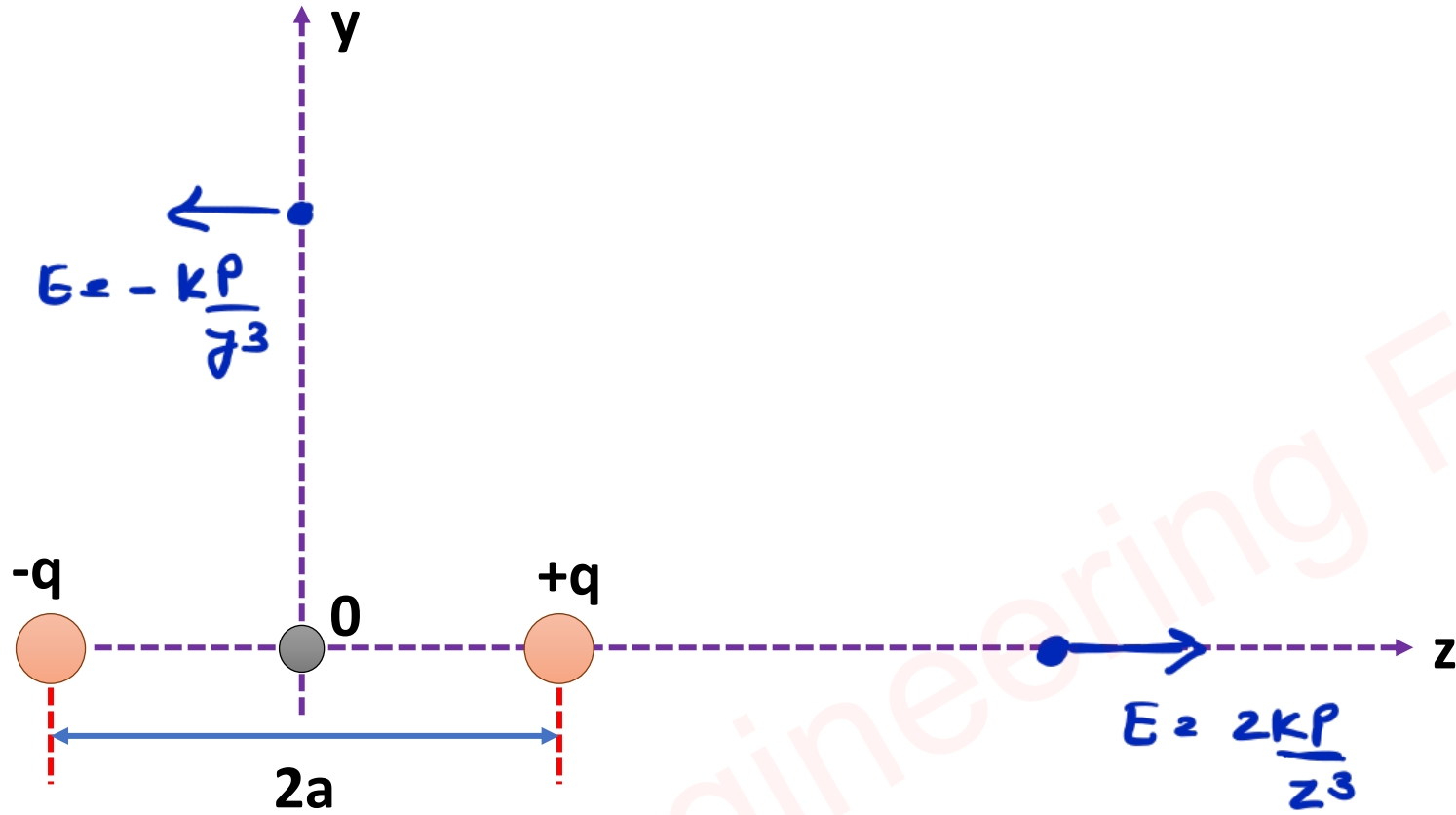
$$\Rightarrow E = -2 \left(k \frac{q}{(a^2 + y^2)} \right) \frac{a}{\sqrt{a^2 + y^2}}$$

$$\Rightarrow E = - \frac{2kqa}{(a^2 + y^2)^{3/2}} \quad [p = 2aq]$$

$$\Rightarrow E = - \frac{kp}{(a^2 + y^2)^{3/2}} \quad [y \gg a \Rightarrow (a^2 + y^2)^{3/2} = y^3]$$

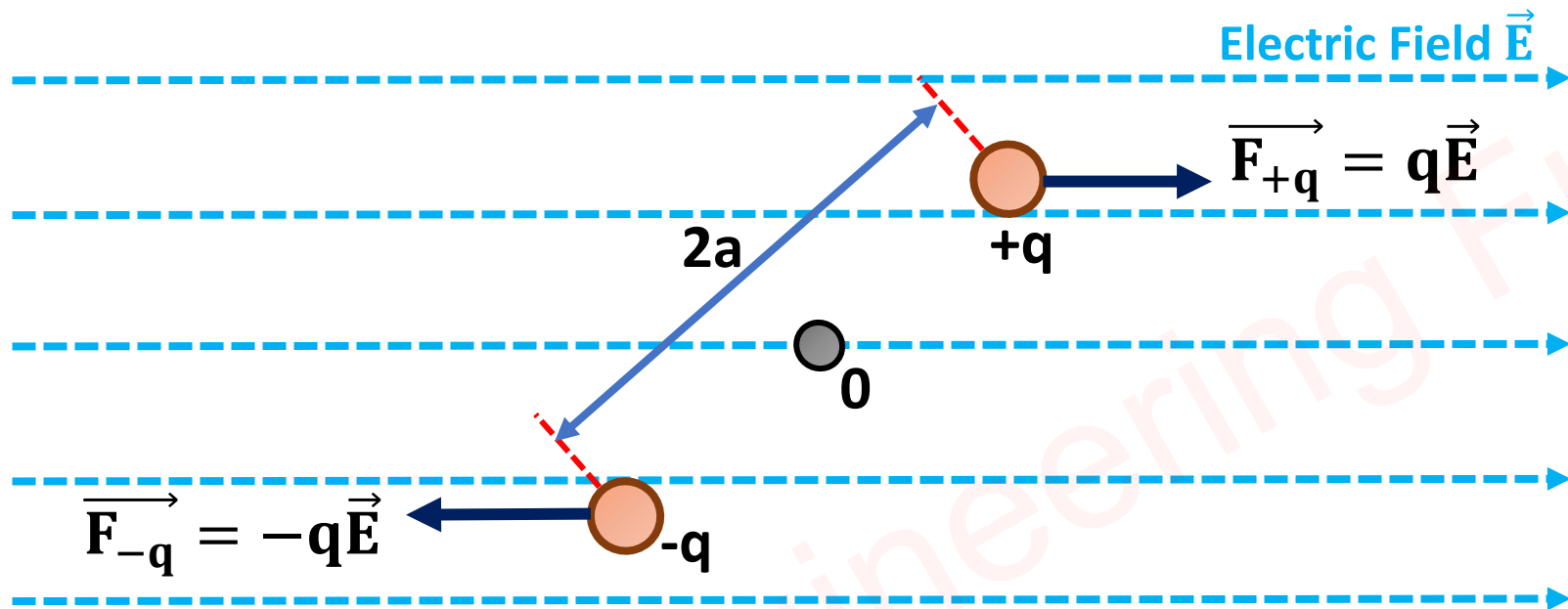
$$\Rightarrow E = -k \frac{p}{y^3} \quad \text{or} \quad \vec{E} = -k \frac{\vec{p}}{y^3}$$

Electric Field due to Dipole



Force & Torque on Electric Dipole

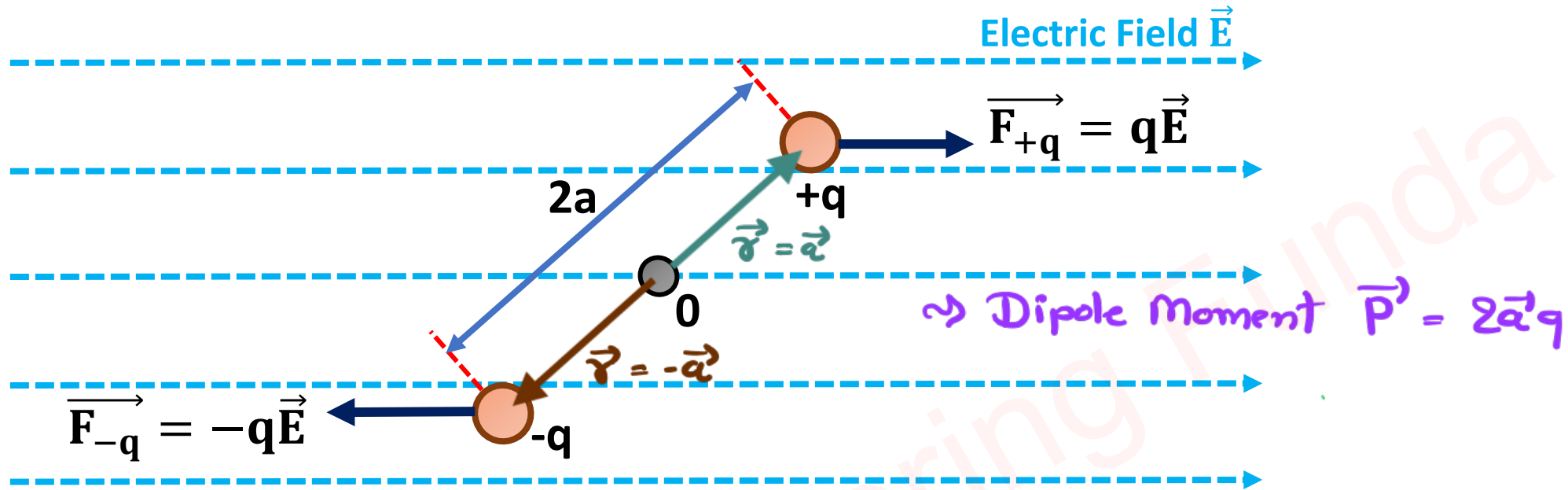
Force on Electric Dipole due to uniform Electric Field



- Total Force on the Dipole is given by:

$$\Rightarrow \vec{F} = \vec{F}_{+q} + \vec{F}_{-q} = q\vec{E} - q\vec{E} = 0$$

Torque on Electric Dipole due to uniform Electric Field



- Torque on Positive Charge is given by:

$$\Rightarrow \vec{\tau}_{+q} = \vec{r} \times \vec{F}$$

$$\Rightarrow \vec{\tau}_{+q} = \vec{a} \times q\vec{E}$$

- Torque on Negative Charge is given by:

$$\Rightarrow \vec{\tau}_{-q} = \vec{r} \times \vec{F}$$

$$\Rightarrow \vec{\tau}_{-q} = -\vec{a} \times -q\vec{E}$$

- Total Torque is given by:

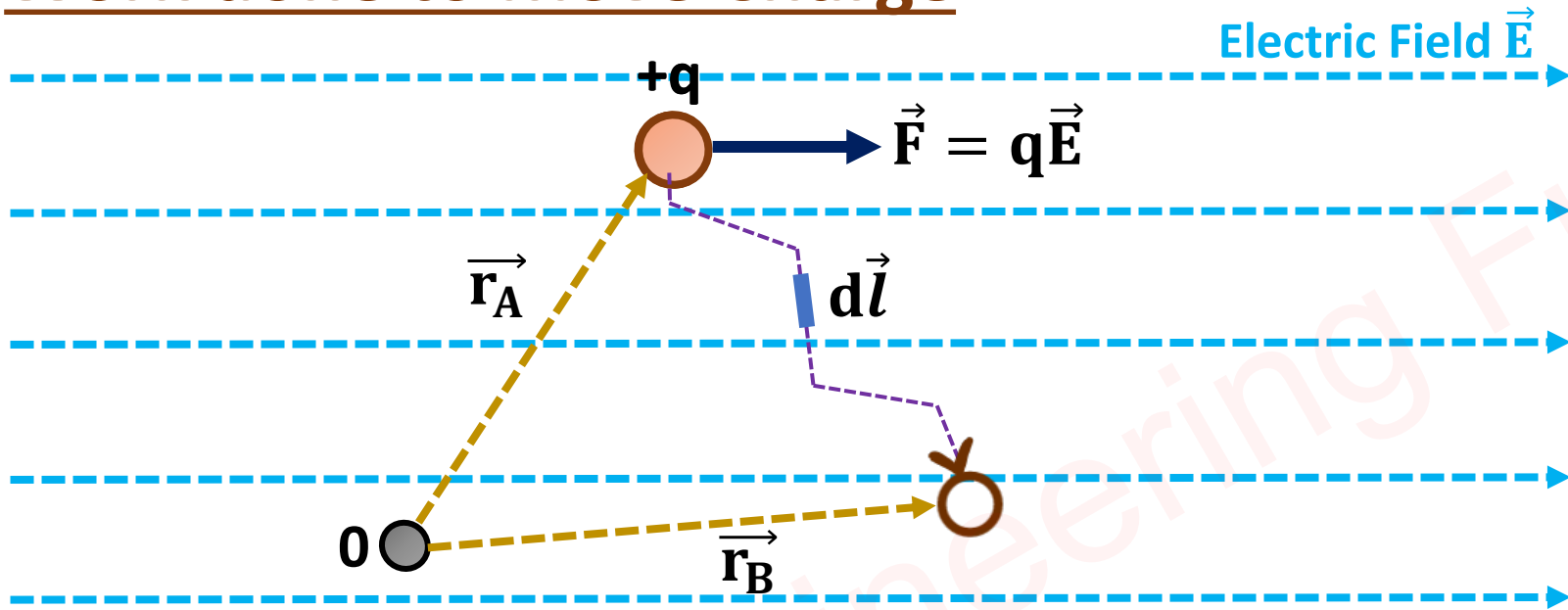
$$\Rightarrow \vec{\tau} = \vec{\tau}_{+q} + \vec{\tau}_{-q}$$

$$\Rightarrow \vec{\tau} = 2(\vec{a} \times q\vec{E})$$

$$\Rightarrow \vec{\tau} = \vec{P} \times \vec{E}$$

Electric Potential

Work done to Move Charge



- Force on charge $+q$ due to Electric field $\vec{E}$ is $\vec{F}$.

$$\Rightarrow \vec{F} = q\vec{E}$$

- For $d\vec{l}$ movement work done is dW .

$$\Rightarrow dW = \vec{F} \cdot d\vec{l}$$

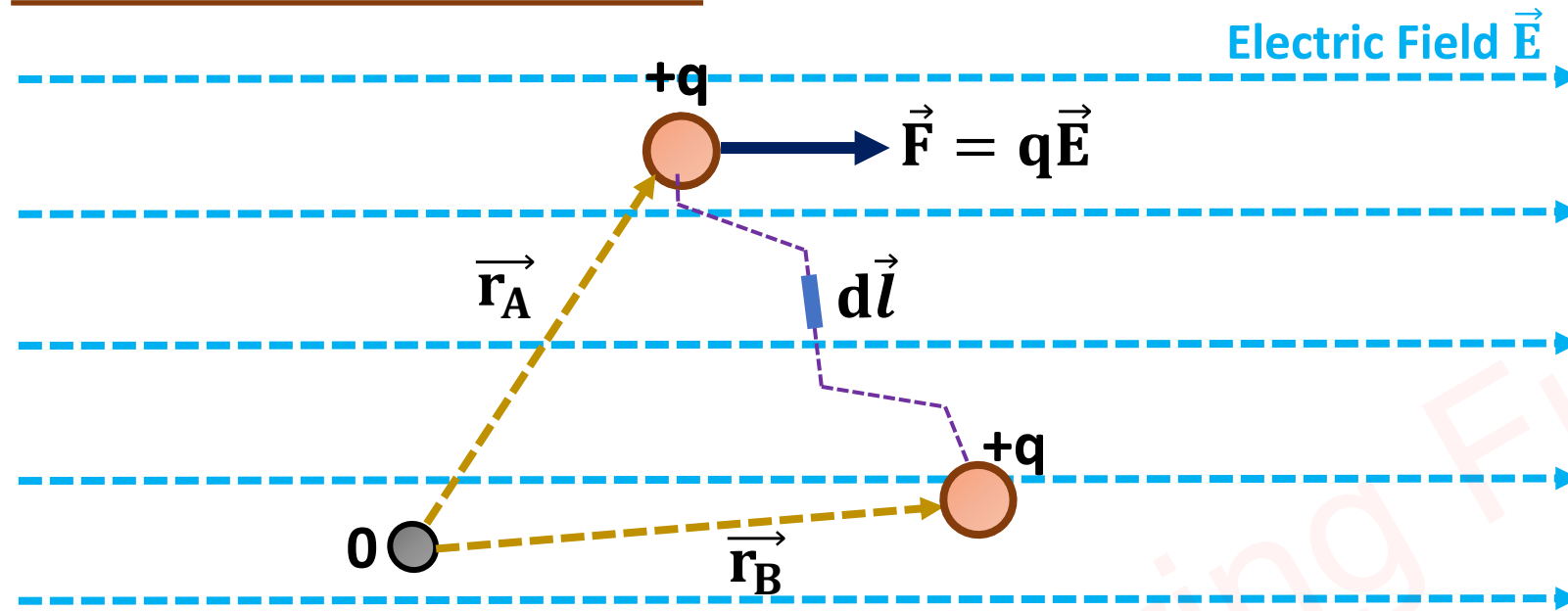
$$\Rightarrow dW = q\vec{E} \cdot d\vec{l}$$

- Hence, the Total Work done is W .

$$\Rightarrow W = \int dW = \int q\vec{E} \cdot d\vec{l}$$

$$\Rightarrow W = \int_{r_A}^{r_B} q\vec{E} \cdot d\vec{l}$$

Potential Difference



- Hence, the Total Work done is W .

$$\Rightarrow W = \int_{r_A}^{r_B} q\vec{E} \cdot d\vec{l}$$

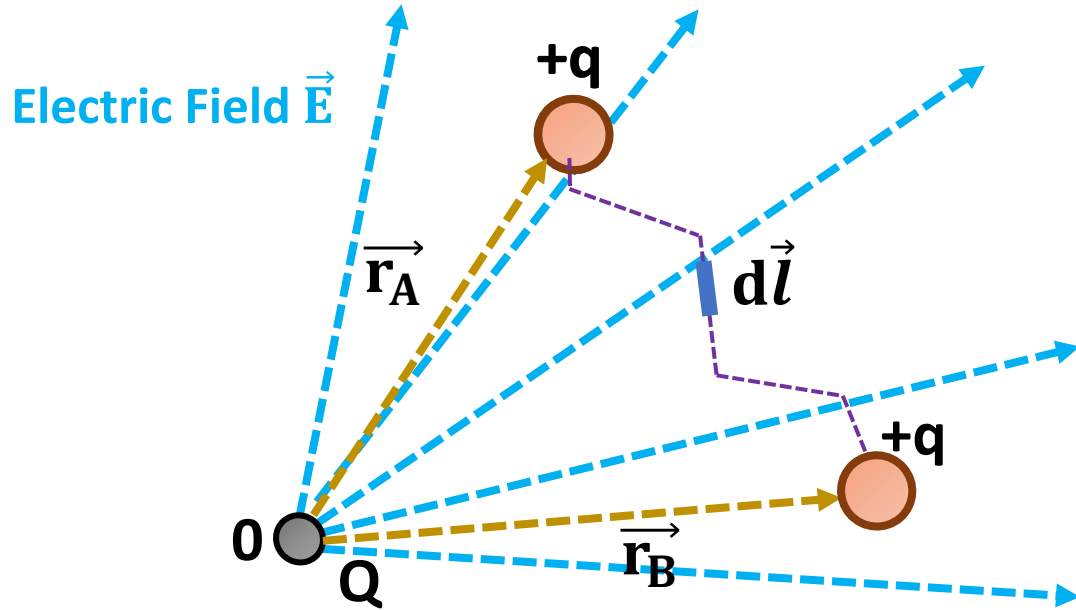
- Given Work done is done by Electric Field $\vec{E}$.

- Potential Difference is work done per unit charge.

$$\Rightarrow V_{AB} = \frac{W}{q} = - \int_{r_A}^{r_B} \vec{E} \cdot d\vec{l}$$

-Ve sign is placed due to work done by an external force.

Potential Difference due to charge Q



- At origin Q charge is placed. So, Electric Field due to charge Q is given by:

$$\Rightarrow \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} \hat{R}$$

- Hence, Potential Difference is given by;

$$\Rightarrow V_{AB} = \frac{W}{q} = - \int_{r_A}^{r_B} \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} \cdot dR$$

- Potential Difference is work done per unit charge.

$$\Rightarrow V_{AB} = \frac{W}{q} = - \int_{r_A}^{r_B} \vec{E} \cdot d\vec{l}$$

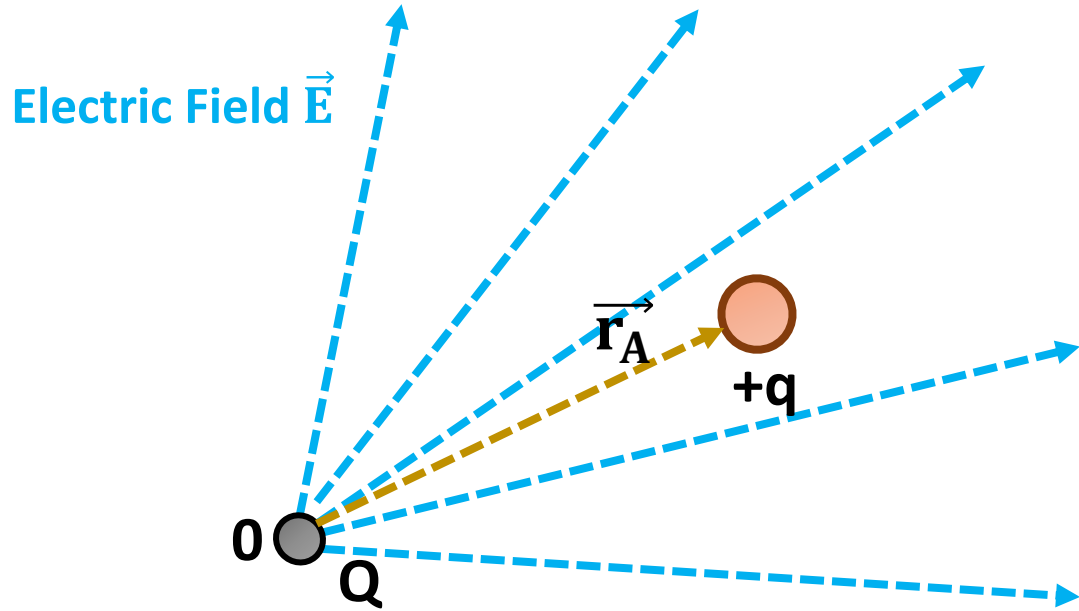
-Ve sign is placed due to work done by an external force.

$$\Rightarrow V_{AB} = V_B - V_A = - \frac{Q}{4\pi\epsilon_0} \int_{r_A}^{r_B} \frac{1}{R^2} \cdot dR$$

$$\Rightarrow V_{AB} = V_B - V_A = - \frac{Q}{4\pi\epsilon_0} \left[\frac{-1}{R} \right]_{r_A}^{r_B}$$

$$\Rightarrow V_{AB} = V_B - V_A = \frac{1}{4\pi\epsilon_0} \frac{Q}{r_B} - \frac{1}{4\pi\epsilon_0} \frac{Q}{r_A}$$

Potential at a given point due to charge Q



- If we bring the charge from infinity to given position $\vec{r}_A$, then work done per unit charge is potential at given position.

$$\Rightarrow V_A = - \int_{\infty}^{r_A} \vec{E} \cdot d\vec{l} = - \int_{\infty}^{r_A} \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} dR$$

$$\Rightarrow V_A = - \frac{Q}{4\pi\epsilon_0} \int_{\infty}^{r_A} \frac{1}{R^2} dR$$

- Potential Difference is work done per unit charge.

$$\Rightarrow V_{AB} = \frac{W}{q} = - \int_{r_A}^{r_B} \vec{E} \cdot d\vec{l}$$

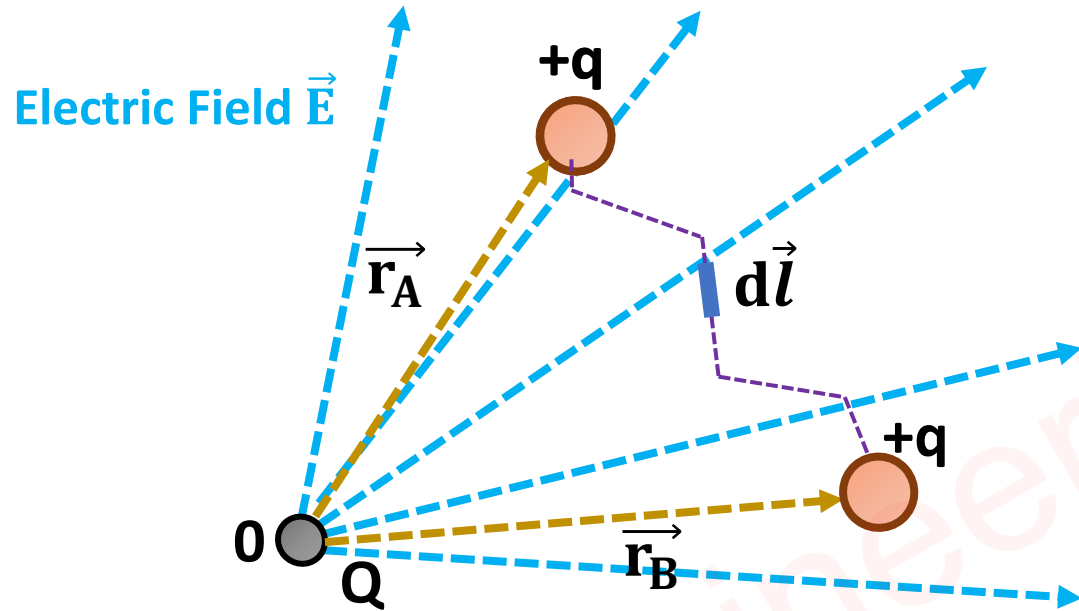
-Ve sign is placed due to work done by an external force.

$$\Rightarrow V_A = - \frac{Q}{4\pi\epsilon_0} \left[\frac{-1}{R} \right]_{\infty}^{r_A}$$

$$\Rightarrow V_A = \frac{1}{4\pi\epsilon_0} \frac{Q}{r_A}$$

Key Points regarding Potential

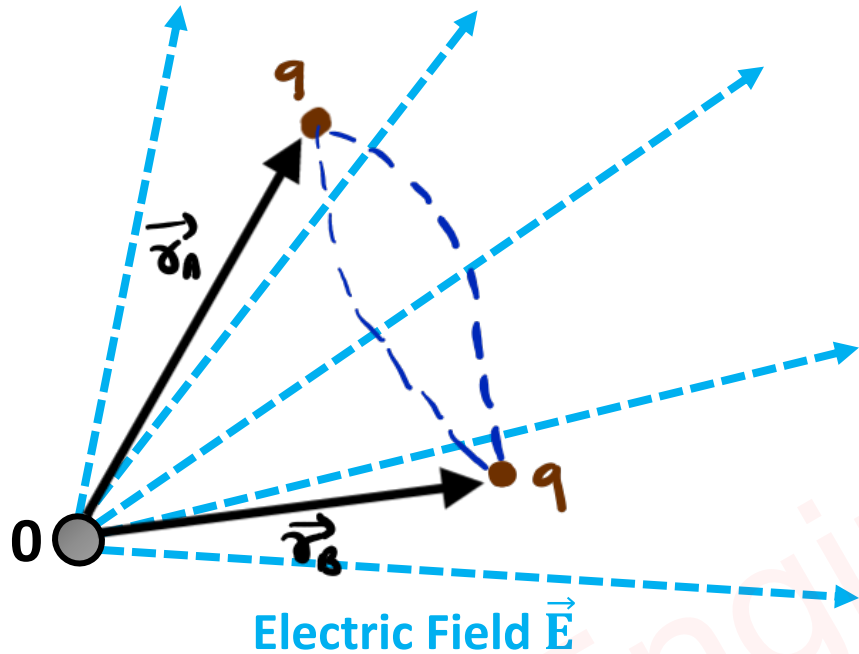
- $V_{AB} = -Ve$, {Work done happened by Electric Field}
- $V_{AB} = +Ve$, {Work done happened by External Force}



- Potential difference is independent of path.
- Unit of Potential is Voltage or J/C.
- Potential Difference can be calculated by:

$$\Rightarrow V_{AB} = V_B - V_A = - \int_{r_A}^{r_B} \vec{E} \cdot d\vec{l} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r_B} - \frac{1}{4\pi\epsilon_0} \frac{Q}{r_A}$$

Maxwell's 3rd Equation for Non-Time-Varying Electric Field



→ Potential difference

$$\Rightarrow V_{AB} = V_B - V_A = \frac{W}{q} = - \int_{r_A}^{r_B} \vec{E} \cdot d\vec{r}$$

→ For closed path

$$\Rightarrow \Delta V = - \oint \vec{E} \cdot d\vec{r}$$

$$= V_{AB} + V_{BA}$$

$$= - \int_{r_A}^{r_B} \vec{E} \cdot d\vec{r} + \left(- \int_{r_B}^{r_A} \vec{E} \cdot d\vec{r} \right)$$

$$= \cancel{V_B - V_A} + \cancel{V_A - V_B}$$

$$= 0$$

$$\Rightarrow \oint \vec{E} \cdot d\vec{l} = 0 \quad \text{--- (A)}$$

$\Rightarrow$ eq.ⁿ (A) is Maxwell's 3rd eq.ⁿ in Integral form for Non time varying field.

$\Rightarrow$ Workdone is given by

$$\Rightarrow \frac{W}{q} = \Delta V = - \oint \vec{E} \cdot d\vec{l} = 0$$

$$\Rightarrow W = 0$$

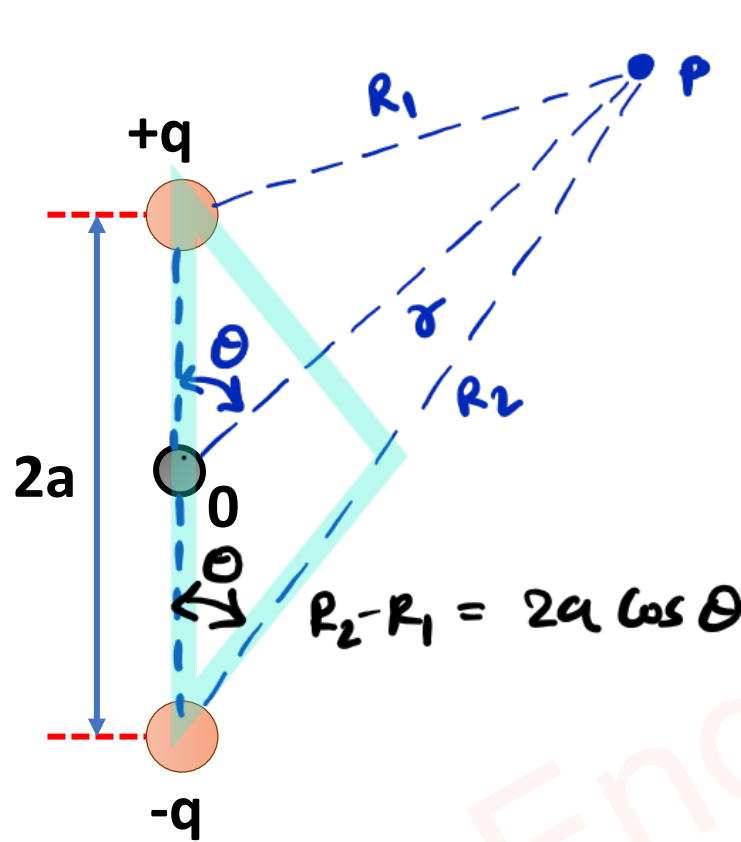
$\Rightarrow$ As per Stokes theorem

$$\Rightarrow \oint \vec{E} \cdot d\vec{l} = \int (\nabla \times \vec{E}) \cdot d\vec{S} = 0$$

$$\Rightarrow \nabla \times \vec{E} = 0 \quad \text{--- (B)}$$

$\Rightarrow$ eq.ⁿ (B) is Maxwell's 3rd eq.ⁿ in differential form / point form for Non time varying field.

Potential Due to Dipole



→ Total Potential at point P

$$\Rightarrow V = V_{+q} + V_{-q}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{R_1} + \frac{1}{4\pi\epsilon_0} \frac{-q}{R_2}$$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{R_2 - R_1}{R_1 R_2} \right]$$

→ If $r \gg 2a$, $R_1 R_2 = r^2$, $R_2 - R_1 = 2a \cos \theta$

$$\Rightarrow V = \frac{q}{4\pi\epsilon_0} \left[\frac{2a \cos \theta}{r^2} \right]$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} = k = 9 \times 10^9 \text{ SI} , \text{ Dipole Moment } p = 2aq$$

$$\Rightarrow V = \frac{k p}{r^2} \cos \theta$$

Engineering Funda

Examples of Electric Potential

Question 1 – Three point charges $-4\mu\text{C}$, $5\mu\text{C}$ and $3\mu\text{C}$ are located at $(2, -1, 3)$, $(0, 4, -2)$, and $(0, 0, 0)$, respectively. Find the potential at $(-1, 5, 2)$.

$$\begin{aligned}\sim) q_1 &= -4\mu\text{C} \longrightarrow \vec{r}_1 = (2, -1, 3) \text{ m} \\ q_2 &= 5\mu\text{C} \longrightarrow \vec{r}_2 = (0, 4, -2) \text{ m} \\ q_3 &= 3\mu\text{C} \longrightarrow \vec{r}_3 = (0, 0, 0) \text{ m} \\ V &= ? \longrightarrow \vec{r} = (-1, 5, 2) \text{ m}\end{aligned}$$

$$\longrightarrow V = k \frac{q}{r}$$

$$\sim) r_1 = \sqrt{(-3)^2 + 6^2 + (-1)^2} = \sqrt{46}$$

$$r_2 = \sqrt{(-1)^2 + 1^2 + 4^2} = \sqrt{18}$$

$$r_3 = \sqrt{(-1)^2 + 6^2 + 2^2} = \sqrt{30}$$

$$\sim) V = k \frac{q_1}{r_1} + k \frac{q_2}{r_2} + k \frac{q_3}{r_3} = k \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \right)$$

$$= 1 \times 10^9 \left(\frac{-4 \times 10^{-6}}{\sqrt{46}} + \frac{5 \times 10^{-6}}{\sqrt{18}} + \frac{3 \times 10^{-6}}{\sqrt{30}} \right)$$

$$= 9 \times 10^9 \times 10^{-6} \left(\frac{-4}{\sqrt{46}} + \frac{5}{\sqrt{18}} + \frac{3}{\sqrt{30}} \right)$$

$$V = 10.22 \text{ kV}$$

Engineering Funda

Question 2 – In Free space, $V = x^2y(z + 3)$ Volt. Find

a. Electric Field at (3, 4, -6)

b. The charge withing the cube, $0 < xyz < 1$

$$\leadsto \underbrace{V}_{\text{Scalar}} = - \int \vec{E} \cdot d\vec{l} \quad \leadsto \underbrace{\vec{E}}_{\text{Vector}} = - \vec{\nabla} V$$

$$\begin{aligned} \leadsto \vec{E} &= - \vec{\nabla} V \\ &= - \left(\frac{\partial V}{\partial x} \hat{i}_x + \frac{\partial V}{\partial y} \hat{i}_y + \frac{\partial V}{\partial z} \hat{i}_z \right) \\ &= - \left(2xy(z+3) \hat{i}_x + x^2(z+3) \hat{i}_y + x^2y \hat{i}_z \right) \end{aligned}$$

– At (3, 4, -6) m

$$\vec{E} = 72 \hat{i}_x + 27 \hat{i}_y - 36 \hat{i}_z \quad \text{V/m}$$

$$\leadsto |\vec{E}| = \sqrt{72^2 + 27^2 + (-36)^2} = 84.9 \text{ V/m}$$

$$\leadsto \vec{D} = \epsilon_0 \vec{E} \rightarrow \mathcal{J}_v = \vec{\nabla} \cdot \vec{D} \rightarrow Q = \int \mathcal{J}_v dV$$

$$\begin{aligned} \leadsto \vec{D} &= \epsilon_0 \vec{E} \\ &= \epsilon_0 [-(2xy(z+3))\hat{i}_x + x^2(z+3)\hat{i}_y + x^2y\hat{i}_z] \end{aligned}$$

$$\begin{aligned} \leadsto \mathcal{J}_v &= \vec{\nabla} \cdot \vec{D} \\ &= \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \\ &= -\epsilon_0 (2y(z+3)) \end{aligned}$$

$$\begin{aligned} \leadsto Q &= \int \mathcal{J}_v dV \\ &= \int_0^1 \int_0^1 \int_0^1 (-\epsilon_0 (2y(z+3))) dx dy dz \\ &= -\epsilon_0 [x]_0^1 [2y^2/2]_0^1 [z^2/2 + 3z]_0^1 \\ &= -7/2 \epsilon_0 = -7/2 \times 8.854 \times 10^{-12} = -30.9 \text{ pC} \end{aligned}$$

Examples of Potential from E Field

Question 1 – An electrostatic potential is given by $\Phi = 2x\sqrt{y}$ volts in the rectangular coordinate system. The magnitude of the electric field at $x=1\text{m}$ and $y=1\text{m}$ is V/m

$$\leadsto \underbrace{V}_{\text{Scalar}} = - \int \vec{E} \cdot d\vec{l} \quad \leadsto \underbrace{\vec{E}}_{\text{Vector}} = - \vec{\nabla} V$$

$$\begin{aligned} \leadsto \vec{E} &= - \vec{\nabla} \Phi \\ &= - \left(\frac{\partial \Phi}{\partial x} \hat{i}_x + \frac{\partial \Phi}{\partial y} \hat{i}_y + \frac{\partial \Phi}{\partial z} \hat{i}_z \right) \\ &= - \left(2\sqrt{y} \hat{i}_x + 2x \left(\frac{1}{2} y^{-1/2} \right) \hat{i}_y \right) \\ &= - 2\sqrt{y} \hat{i}_x - \frac{x}{\sqrt{y}} \hat{i}_y \end{aligned}$$

$$\begin{aligned} \text{At } x=1\text{m} \text{ \& } y=1\text{m} \\ &= (-2) \hat{i}_x - \hat{i}_y \text{ V/m} \end{aligned}$$

$$\begin{aligned} \rightarrow |\vec{E}| &= \sqrt{(-2)^2 + (-1)^2} \\ &= \sqrt{5} \text{ V/m} \end{aligned}$$

Question 2 – Given the potential function in free space to be, $V = 50x^2 + 50y^2 + 50z^2$ V. Find magnitude of E field at (1, -1, 1) V/m.

$$\Rightarrow \vec{E} = -\vec{\nabla} V$$

$$= - \left(\frac{\partial V}{\partial x} \hat{i}_x + \frac{\partial V}{\partial y} \hat{i}_y + \frac{\partial V}{\partial z} \hat{i}_z \right)$$

$$= - \left(100x \hat{i}_x + 100y \hat{i}_y + 100z \hat{i}_z \right)$$

– At (1, -1, 1) m

$$= -100 \hat{i}_x + 100 \hat{i}_y - 100 \hat{i}_z \text{ V/m}$$

$$\rightarrow |\vec{E}| = \sqrt{(-100)^2 + (100)^2 + (-100)^2}$$

$$= 100\sqrt{3} \text{ V/m}$$

Examples of Potential & Electric Field

Question 1 – An infinite number of charges, each equal to 'q' are placed at $x=1, 2, 4, 8, 16, \dots$. Find the potential and Electric Field at $x=0$.



$$\begin{aligned} \sim) V &= k \frac{q}{r_1} + k \frac{q}{r_2} + k \frac{q}{r_3} + k \frac{q}{r_4} + \dots \\ &= kq \left[\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \right] \\ &= kq \left[\frac{1}{1 - 1/2} \right] = 2kq = 2 \left(\frac{1}{4\pi\epsilon_0} \right) q = \frac{q}{2\pi\epsilon_0} \end{aligned}$$

$$\begin{aligned} \sim) E &= k \frac{q}{r_1^2} + k \frac{q}{r_2^2} + k \frac{q}{r_3^2} + k \frac{q}{r_4^2} + \dots \\ &= kq \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{8^2} + \dots \right] \\ &= kq \left[\frac{1}{1 - 1/2^2} \right] = \frac{4}{3} kq = \frac{4}{3} \left(\frac{1}{4\pi\epsilon_0} \right) q = \frac{q}{3\pi\epsilon_0} \end{aligned}$$

Question 2 – Two electric charges q and $-2q$ are placed at $(0, 0)$ and $(6, 0)$ on XY plane. The equation for the zero equipotential curve in XY plane is (x, y)

a. $x = -2$

b. $y = 2$

c. $x^2 + y^2 = 2$

☒ d. $(x + 2)^2 + y^2 = 16$

$$\rightarrow V = 0 = V_q + V_{-2q}$$

$$\Rightarrow k \frac{q_1}{r_1} + k \frac{q_2}{r_2} = 0$$

$$\Rightarrow k \frac{1}{\sqrt{x^2 + y^2}} + k \frac{(-2)}{\sqrt{(x-6)^2 + y^2}} = 0$$

$$\Rightarrow \frac{1}{\sqrt{x^2 + y^2}} = \frac{2}{\sqrt{(x-6)^2 + y^2}}$$

$$\Rightarrow 4(x^2 + y^2) = (x-6)^2 + y^2$$

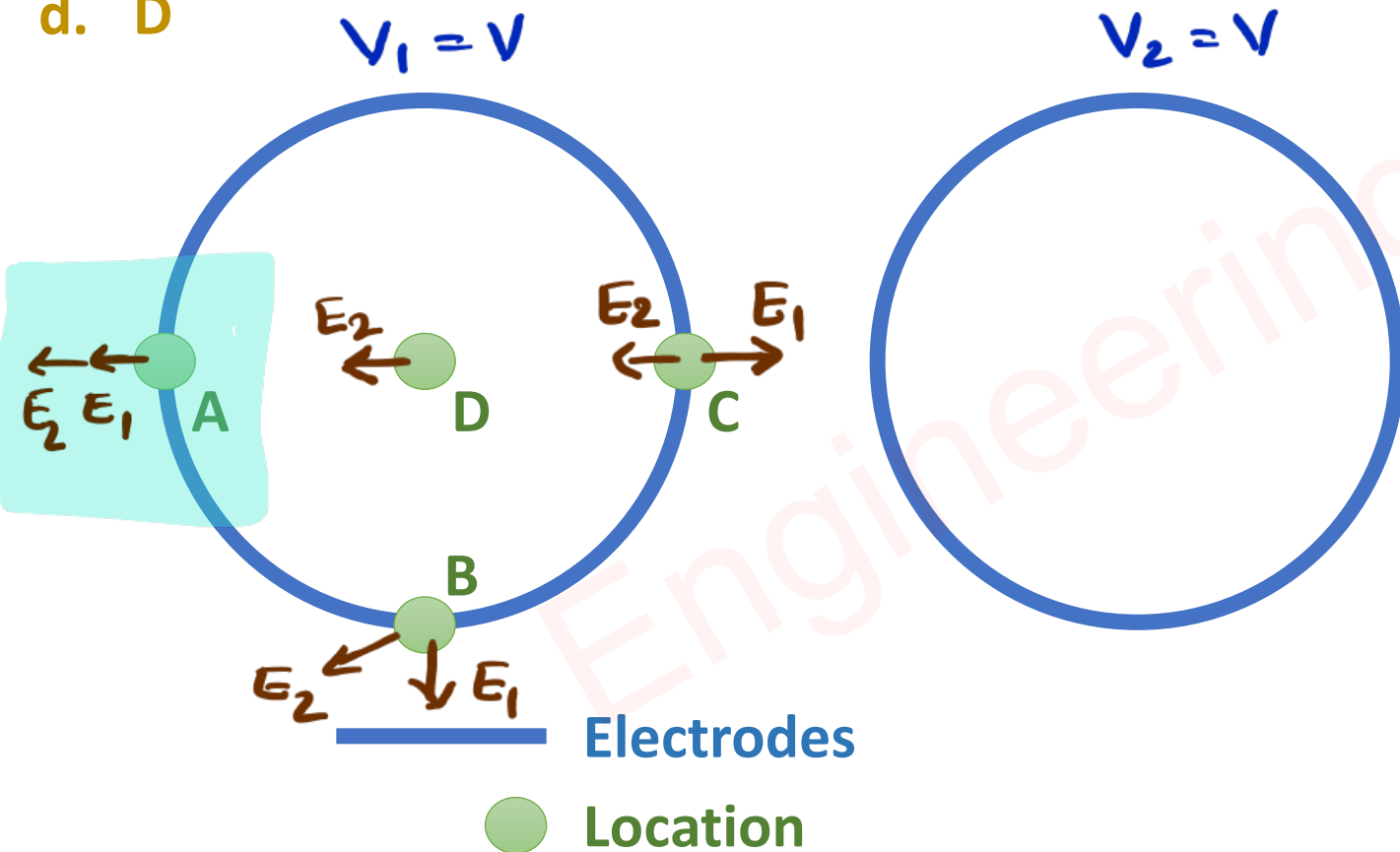
$$\Rightarrow 3x^2 + 12x - 36 + 3y^2 = 0$$

$$\Rightarrow x^2 + 4x - 12 + y^2 = 0$$

$$\Rightarrow (x+2)^2 + y^2 = 16$$

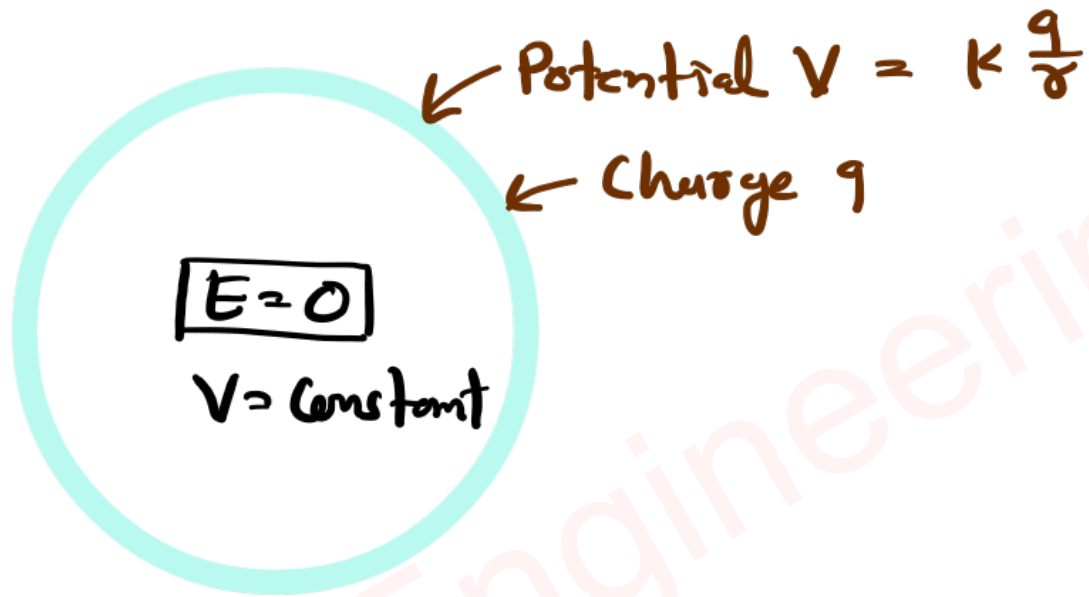
Question 3 – The electrodes, whose cross-sectional view is shown in figure are at the same potential. The maximum E field is at

- ✓ a. A
- b. B
- c. C
- d. D



Examples of Potential

Question 1 – A hollow conductor is at a potential V . The potential at any point inside the hollow is V



$$2) \vec{E} = -\vec{\nabla} V = 0$$

$$\Rightarrow \vec{\nabla} V = 0$$

$$\Rightarrow V = \text{constant}$$

Question 2 – If V, W and q stands for voltage, energy and charge, then V can be expressed as

- a. $V = \frac{dW}{dq}$
- ✓ b. $dV = \frac{dW}{dq}$
- c. $V = \frac{dq}{dW}$
- d. $dV = \frac{dq}{dW}$

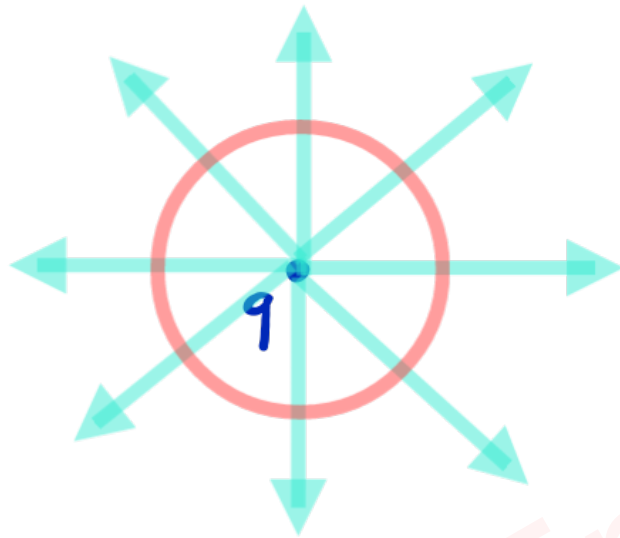
$$\Rightarrow dW = dq dV$$

$$\Rightarrow dV = \frac{dW}{dq}$$

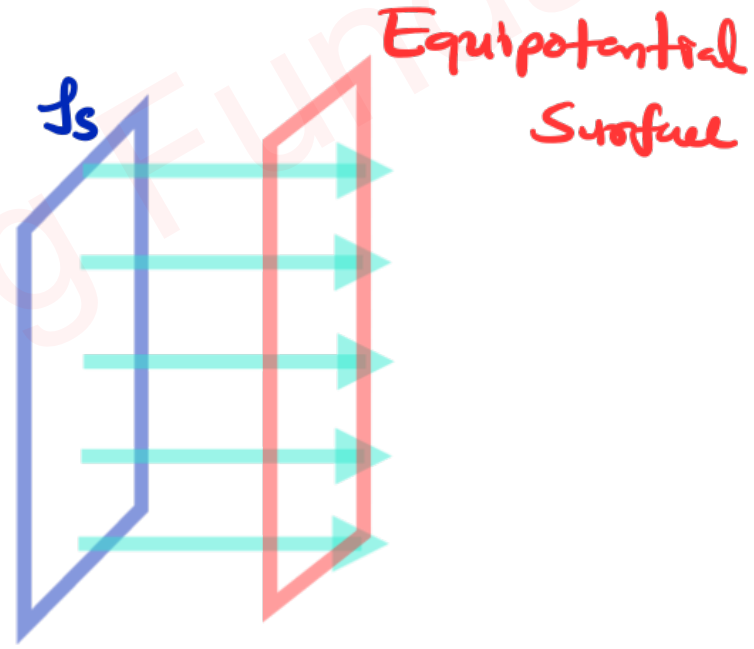
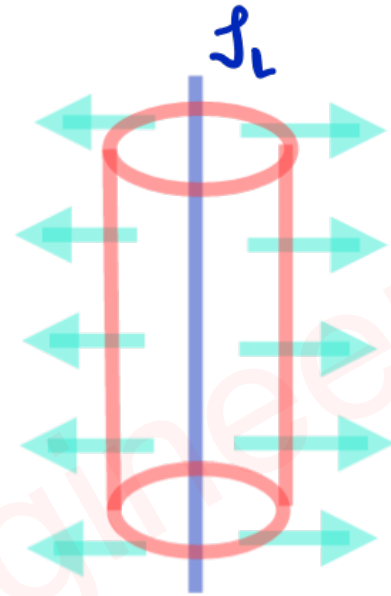
Engineering Funda

Question 3 – In a uniform electric field, field lines, and equipotential surface are

- a. Parallel to each other
- b. Intersects at 45 Degree
- c. Intersects at 30 Degree
- ✓ d. Orthogonal to each other

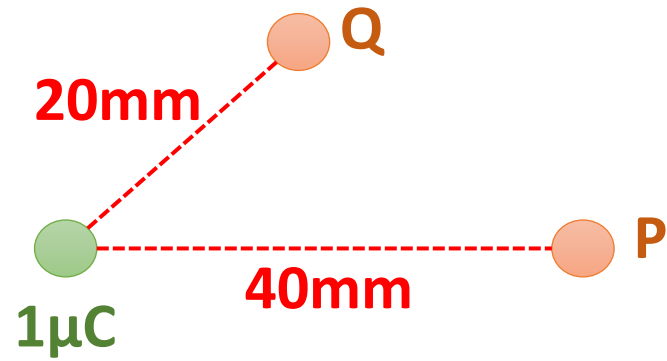


Electric field



Equipotential Surface

Question 4 – A point charge of $1\mu\text{C}$ is placed in space as shown figure. Find V_{QP} .

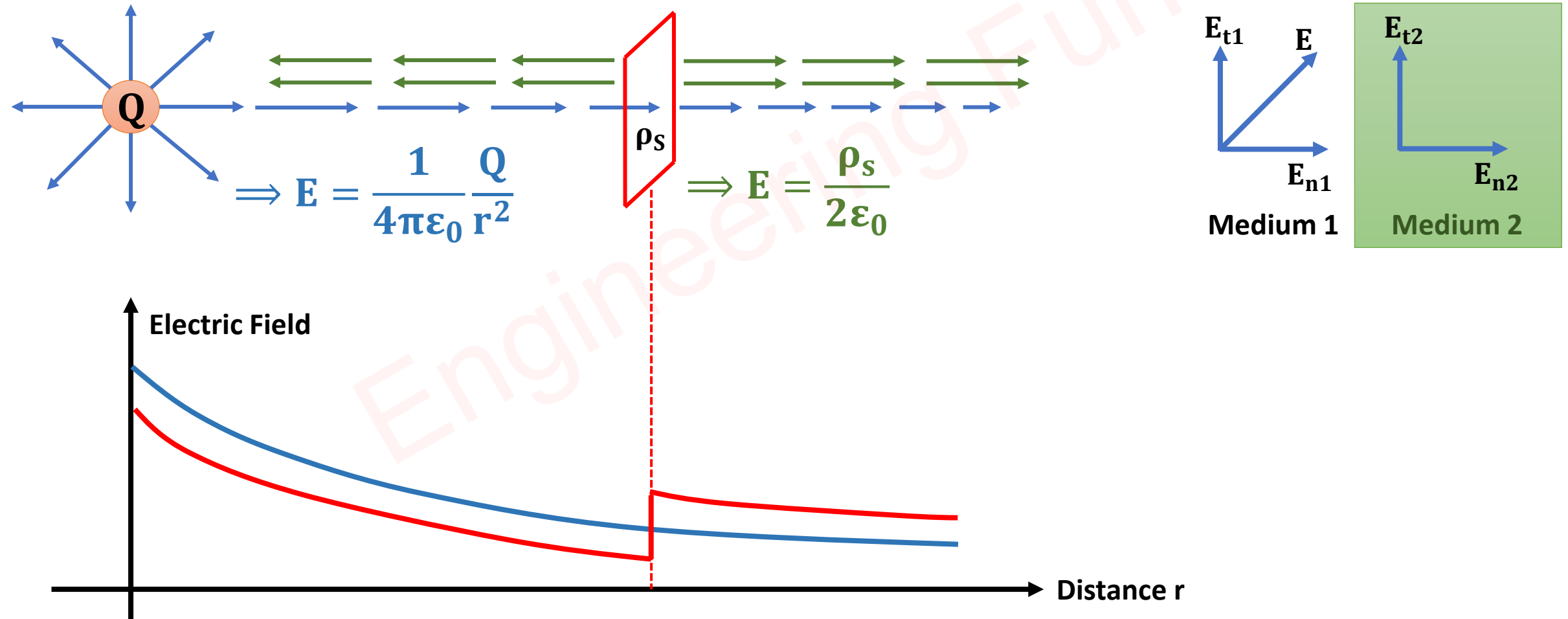


$$\begin{aligned}\rightarrow V_{QP} &= V_P - V_Q \\ &= k \frac{Q}{r_P} - k \frac{Q}{r_Q} \\ &= 9 \times 10^9 \times 10^{-6} \left(\frac{1}{40 \times 10^{-3}} - \frac{1}{20 \times 10^{-3}} \right) \\ &= -225 \text{ kV}\end{aligned}$$

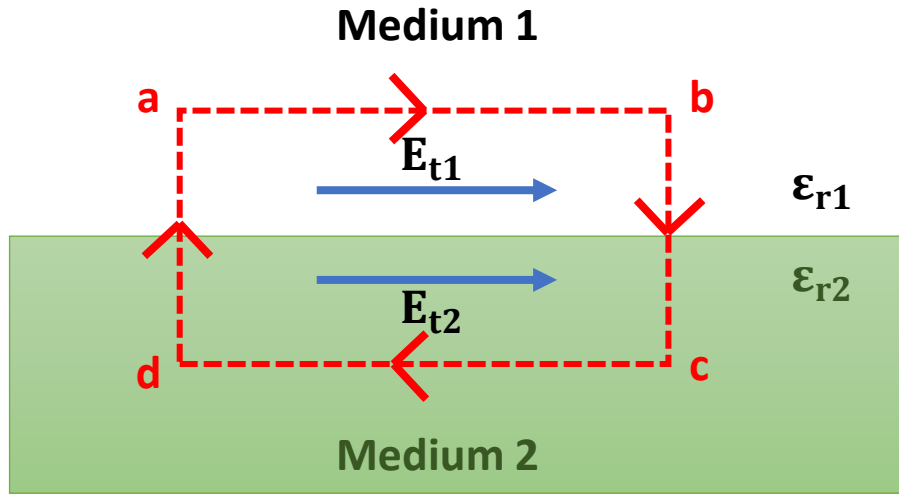
Boundary Conditions for Electric Field

Why should we identify Boundary Conditions for the Electric Field?

- Boundary Conditions explain the continuity and discontinuity of the Electric Field when the Electric Field enters from one medium to another medium.



Boundary Conditions of Electric Field for Tangential Components



- As per Maxwell's 3rd equation for Non-Time Varying Field:

$$\Rightarrow \oint \vec{E} \cdot d\vec{l} = 0$$

$$\Rightarrow \int_a^b \vec{E} \cdot d\vec{l} + \int_b^c \vec{E} \cdot d\vec{l} + \int_c^d \vec{E} \cdot d\vec{l} + \int_d^a \vec{E} \cdot d\vec{l} = 0$$

$$\Rightarrow E_{t1}AB + 0 - E_{t2}CD + 0 = 0 \quad \{\because AB = CD\}$$

$$\Rightarrow E_{t1} = E_{t2}$$

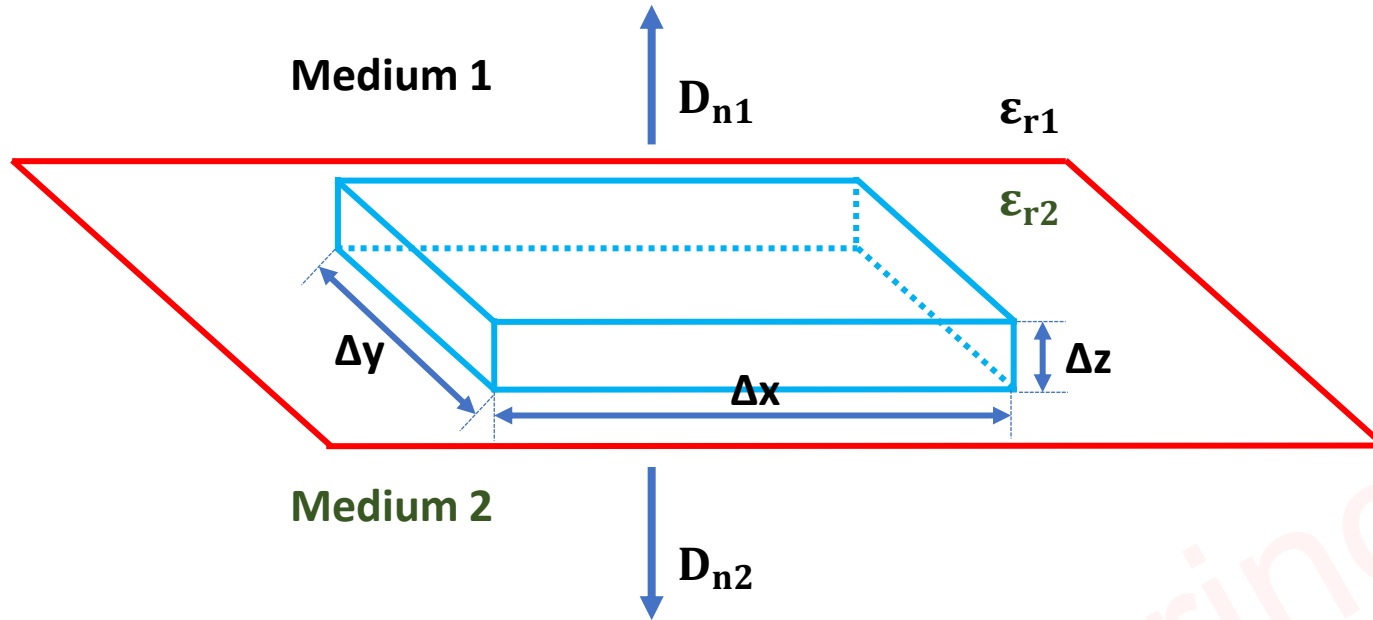
- Hence, the Electric field is continuous for the tangential component.
- The Electric Flux density is given by:

$$\Rightarrow \vec{D} = \epsilon_0 \epsilon_r \vec{E}$$

$$\Rightarrow \frac{D_{t1}}{\epsilon_0 \epsilon_{r1}} = \frac{D_{t2}}{\epsilon_0 \epsilon_{r2}}$$

$$\Rightarrow \frac{D_{t1}}{D_{t2}} = \frac{\epsilon_{r1}}{\epsilon_{r2}}$$

Boundary Conditions of Electric Field for Normal Components



- As per Maxwell's 1st equation for electric Field:

$$\Rightarrow \oint \vec{D} \cdot d\vec{S} = Q$$

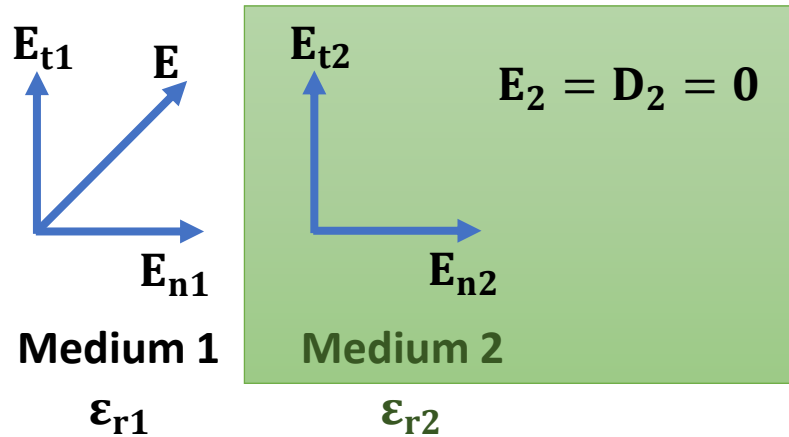
$$\Rightarrow \int_{\text{Top}} \vec{D} \cdot d\vec{S} + \int_{\text{Bottom}} \vec{D} \cdot d\vec{S} + \int_{\text{Left}} \vec{D} \cdot d\vec{S} + \int_{\text{Right}} \vec{D} \cdot d\vec{S} + \int_{\text{Front}} \vec{D} \cdot d\vec{S} + \int_{\text{Back}} \vec{D} \cdot d\vec{S} = Q$$

$$\Rightarrow D_{n1}\Delta x\Delta y - D_{n2}\Delta x\Delta y = Q$$

$$\Rightarrow D_{n1} - D_{n2} = Q/\Delta x\Delta y \quad \{\because \rho_s = Q/\Delta x\Delta y\}$$

$$\Rightarrow D_{n1} - D_{n2} = \rho_s$$

Boundary Conditions of Electric Field for Conductor in Free Space



- For the perfect conductor, the Electric Field Inside the perfect conductor is Zero.
- For the perfect conductor, the Tangential Component of the Electric Field is given by:

$$\Rightarrow E_{t1} = E_{t2} = 0$$

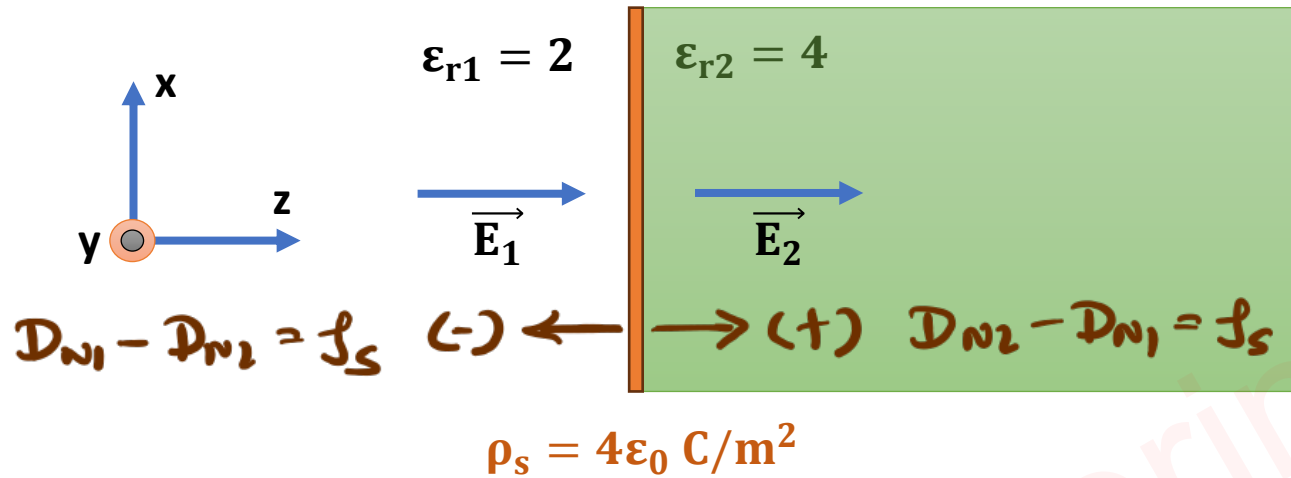
- For the perfect conductor, the Normal Component of the Electric Field is given by:

$$\Rightarrow D_{n1} - D_{n2} = \rho_s$$

$$\Rightarrow D_{n1} = \rho_s$$

Example of Boundary Conditions for Electric Field

Question 1 – $\vec{E}_1 = 2\hat{i}_x + 4\hat{i}_y + 6\hat{i}_z$ V/m, Find $\vec{E}_2$ using Boundary Conditions.



∴ Tangential Components

$$\Rightarrow E_{t1} = E_{t2}$$

∴ Normal Components

$$\Rightarrow D_{n1} - D_{n2} = I_s$$

or

$$\Rightarrow D_{n2} - D_{n1} = I_s$$

$$\rightarrow \vec{E}_1 = 2\hat{i}_x + 4\hat{i}_y + 6\hat{i}_z \text{ V/m}$$

$$\vec{E}_2 = 2\hat{i}_x + 4\hat{i}_y + E_{n2}\hat{i}_z = 2\hat{i}_x + 4\hat{i}_y + 4\hat{i}_z \text{ V/m}$$

$$\rightarrow D_{n2} - D_{n1} = I_s$$

$$\Rightarrow D_{n2} = I_s + D_{n1}$$

$$= 4\epsilon_0 + \epsilon_0 \epsilon_{r1} E_{n1}$$

$$= 4\epsilon_0 + \epsilon_0 (2)(6) = 16\epsilon_0$$

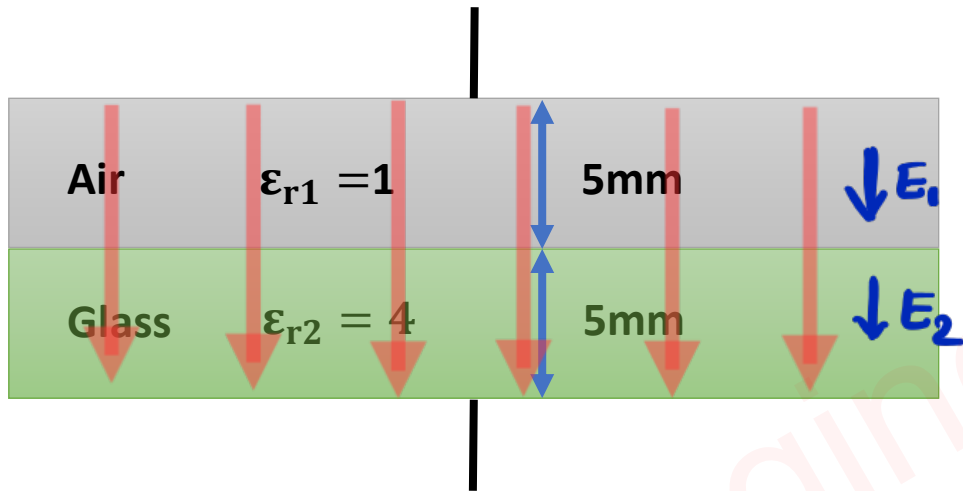
$$\rightarrow E_{n2} = \frac{D_{n2}}{\epsilon_0 \epsilon_{r2}}$$

$$= \frac{4 \cancel{\epsilon_0}}{\cancel{\epsilon_0} 4}$$

$$= 4 \text{ V/m}$$

Examples of Boundary Conditions for Electric Field

Question 2 – As shown below, a parallel plate capacitor is partially filled with glass of dielectric constant 4. The dielectric strength of Air and glass are 30 kV/cm and 300 kV/cm, respectively. The maximum voltage can be applied across the capacitor without any breakdown is



⇒ Normal Components of boundary cond.ⁿ

$$\Rightarrow D_{N2} - D_{N1} = \rho_s \quad (\rho_s = 0)$$

$$\Rightarrow D_{N2} - D_{N1} = 0$$

$$\Rightarrow D_{N2} = D_{N1}$$

$$\Rightarrow \epsilon_0 \epsilon_{r2} E_2 = \epsilon_0 \epsilon_{r1} E_1$$

$$\Rightarrow E_1 = 4 E_2$$

$$\begin{aligned} \Rightarrow E_1 &= 30 \text{ kV/cm} \\ &= 30 \times \frac{10^3}{10^{-2}} \frac{\text{V}}{\text{m}} \\ &= 3 \times 10^6 \text{ V/m} \end{aligned}$$

$$\begin{aligned} \rightarrow E_2 &= E_1 / 4 \\ &= 0.75 \times 10^6 \text{ V/m} \end{aligned}$$

$$\Rightarrow V = V_1 + V_2$$

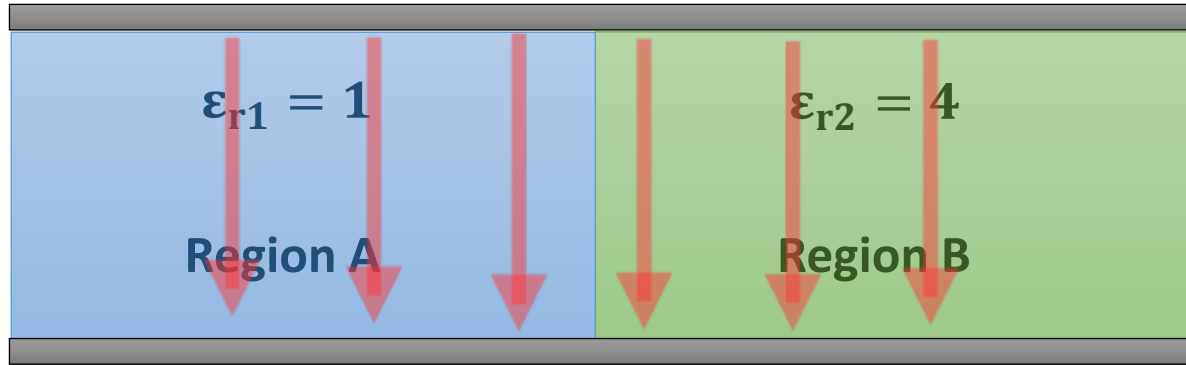
$$= E_1 d_1 + E_2 d_2$$

$$= 3 \times 10^6 \times 5 \times 10^{-3} + 0.75 \times 10^6 \times 5 \times 10^{-3}$$

$$= 18.75 \times 10^3 \text{ Volt}$$

Engineering Funda

Question 3 – A parallel plate capacitor is filled with dielectrics as shown figure. If in region A, E field is 4kV/cm then the E field in region B is

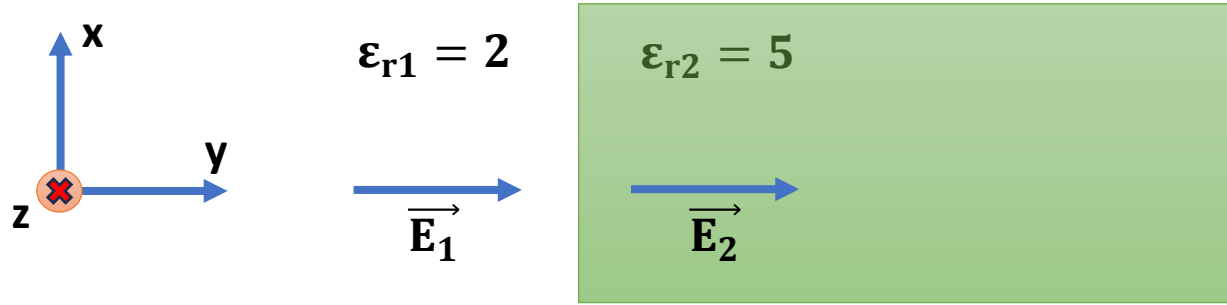


∴ Tangential Component boundary Cond.ⁿ

$$\Rightarrow E_{t1} = E_{t2}$$

$$\begin{aligned}\Rightarrow E_{t2} &= 4 \text{ kV/cm} \\ &= 4 \times \frac{10^3}{10^{-2}} \frac{\text{V}}{\text{m}} \\ &= 4 \times 10^5 \text{ V/m}\end{aligned}$$

Question 4 – $\vec{E}_1 = 3\hat{i}_x + 4\hat{i}_y + 2\hat{i}_z$ V/m, Find $\vec{E}_2$ using Boundary Conditions.



~ For tangential components

$$\Rightarrow E_{t1} = E_{t2}$$

→ For Normal components

$$\Rightarrow D_{n2} - D_{n1} = \rho_s \quad (\rho_s = 0)$$

$$\Rightarrow D_{n2} = D_{n1}$$

$$\Rightarrow \epsilon_0 \epsilon_{r2} E_{n2} = \epsilon_0 \epsilon_{r1} E_{n1}$$

$$\Rightarrow E_{n2} = 0.4 E_{n1}$$

$$= 0.4 \times 4$$

$$= 1.6 \text{ V/m}$$

$$\sim \vec{E}_1 = 3\hat{i}_x + 4\hat{i}_y + 2\hat{i}_z \text{ V/m}$$

$$\sim \vec{E}_2 = 3\hat{i}_x + 1.6\hat{i}_y + 2\hat{i}_z \text{ V/m}$$